Lecture 1

Probability Essentials

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Review of probability

- In Frequentist or classical statistics, we assume that the data $\mathbf{Y}$ follows a model that involves some unknown fixed parameter $\boldsymbol{\theta}$.
- In Bayesian statistics, we also assume a probability distribution (prior) for $\boldsymbol{\theta}$.
- The most important thing in Bayesian statistics is computing the posterior, i.e., the distribution of the parameters $\boldsymbol{\theta}$ after observing the data $\mathbf{Y}$.
- This is the conditional distribution of $\boldsymbol{\theta}$ given $\mathbf{Y}$.
- Therefore, we need to review the probability concepts that lead to the conditional distribution of one variable conditioned on another.

  1. Probability mass (PMF) and density (PDF) functions (Lecture 1)
  2. Joint distributions (Lecture 2)
  3. Marginal and conditional distributions (Lecture 2)

Random variables

- A random variable is generally denoted by X (capital script).
- We want to compute the probability of X taking a specific value x (lowercase script).
- This is denoted by $\text{Prob}(X = x)$.
- We also might want to compute the probability of X being in a set $\mathcal{A}$.
- This is denoted by $\text{Prob}(X \in \mathcal{A})$.
- The set of possible value that X can take on is called its support ($\mathcal{S}$).

Random variables - example

Example 1: X is the outcome of rolling a die.

- The support is $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. Let $\mathcal{A} = \{2, 3\}$.
- $\text{Prob}(X = 5) = 1/6$, and $\text{Prob}(X \in \mathcal{A}) = 2/6 = 1/3$.

Example 2: X is the weight of a newborn baby.

- The support is $\mathcal{S} = (0, \infty)$.
- $\text{Prob}(X \in (0, \infty)) = 1$.

Example 3: X is the amount of rainfall measured at a weather station.

- The support is $\mathcal{S} = [0, \infty)$.
- $\text{Prob}(X \in [0, \infty)) = 1$ and $\text{Prob}(X = 0) \in [0, 1]$.

What is probability?

Academician A. N. Kolmogorov, S. N. Bose, P. C. Mahalanobis at a Special Convocation held to confer Honorary degree of Doctor of Science of the I.S.I. to Kolmogorov on 28 April, 1962

Figure: Copyright: ISI Kolkata archive

What is probability?

Objective (associated with a Frequentist)

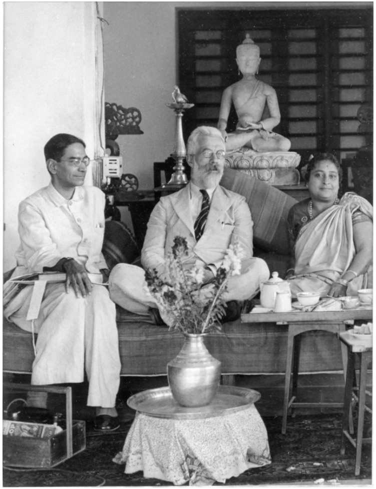
- $\text{Prob}(X = x)$ as a purely mathematical statement.
- If we repeatedly sampled X, the the proportion of draws equal to x converges to $\text{Prob}(X = x)$.

Subjective (associated with Bayesian)

- $\text{Prob}(X = x)$ represents an individual's degree of belief.
- Often quantified as the amount an individual would be willing to wager that X will be x.

A Bayesian analysis makes use of both of these concepts.

What is uncertainty?



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What is uncertainty?

Aleatoric uncertainty (likelihood)

- Uncontrollable randomness in the experiment.
- For example, the results of a fair coin flip can never be predicted with certainty.

Epistemic uncertainty (prior/posterior)

- Uncertainty about a quantity that could theoretically be known.
- For example, if we flipped a coin infinitely-many times we could know the true probability of a head.

A Bayesian analysis makes use of both of these concepts.

Probability versus statistics

Probability is the forward problem

- We assume we know how the data are being generated and compute the probability of events.
- For example, what is the probability of flipping 5 straight heads if the coin is fair?

Statistics is the inverse problem

- We use data to learn about the data-generating mechanism.
- For example, if we flipped five straight head, can we conclude the coin is biased?

Any statistical analysis obviously relies on probability.

Univariate distributions

- We often distinguish between **discrete** and **continuous** random variables (let us forget the discontinuous case for the sake of simplicity).

- The random variable X is discrete if its support $\mathcal{S}$ is countable

- Examples:

  $X \in \{0, 1, 2, 3\}$ is the number of successes in 3 trials.

  $X \in \{0, 1, 2, ...\}$ is the number users that visit a website.

Univariate distributions

- We often distinguish between **discrete** and **continuous** random variables.
- The random variable X is continuous if its support $\mathcal{S}$ is uncountable.
- Examples with $\mathcal{S} = (0, \infty)$:

  $X > 0$ is weight of a baby.

  $X > 0$ is the wind speed.
- Examples with $\mathcal{S} = \mathbb{R}$:

  X is the recorded temperature at a weather station.

  X is the surge in the share price of a company.

Discrete univariate distributions

- If X is discrete we describe its distribution with its **probability mass function** (PMF).
- The PMF is $f(x) = \text{Prob}(X = x)$.
- The domain ($\mathcal{S}$) of X is the set of x with $f(x) > 0$.
- We must have $f(x) \geq 0$ and $\sum_{x \in \mathcal{S}} f(x) = 1$.
- The mean is $\text{E}(X) = \sum_{x \in \mathcal{S}} xf(x)$.
- The variance is $\text{V}(X) = \sum_{x \in \mathcal{S}} [x - E(X)]^2 f(x)$.

Parametric families of distributions

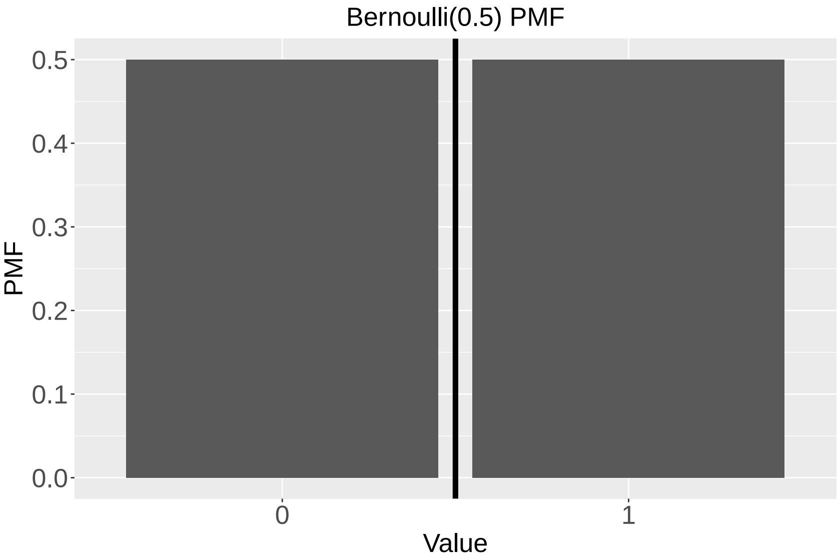
- A statistical analysis typically proceeds by selecting a PMF that seems to match the distribution of a sample.
- We rarely know the PMF exactly, but we assume it is from a parametric family of distributions.
- For example, Binomial(10,0.5) and Binomial(4,0.1) are different but both from the binomial family.
- A family of distributions have the same equation for the PMF but differ by some unknown parameters θ.
- We must estimate these parameters.

Example: $X \sim \text{Bernoulli}(\theta)$

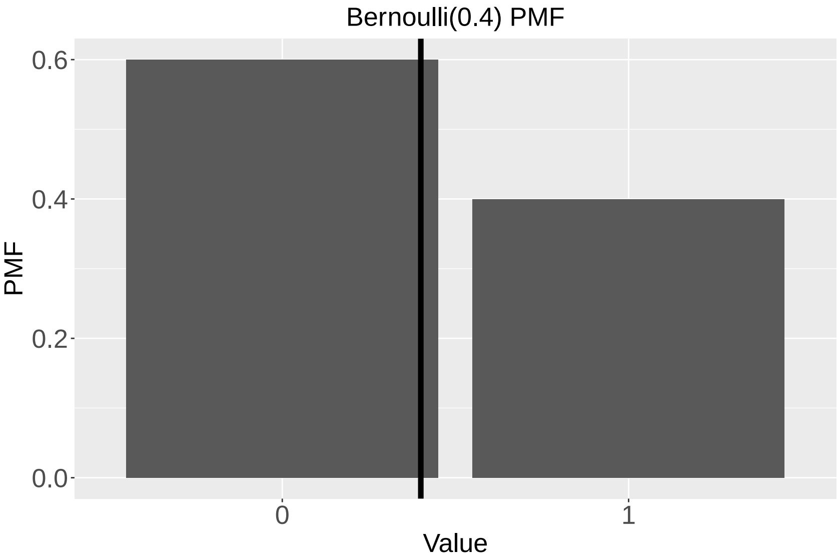
- Example: X is a success (1) or failure (0).
- Domain: $X \in \{0, 1\}$ (i.e., X is binary).
- PMF: $P(X = 0) = 1 - \theta$ and $P(X = 1) = \theta$.
- Parameter: $\theta \in [0, 1]$ is the success probability.
- Mean: $\text{E}(X) = \sum_x xf(x) = 0(1 - \theta) + 1\theta = \theta$.
- Variance:

$$\text{V}(X) = \sum_x (x - \theta)^2 f(x) = (0 - \theta)^2(1 - \theta) + (1 - \theta)^2\theta = \theta(1 - \theta).$$

Example: $X \sim \text{Bernoulli}(0.5)$



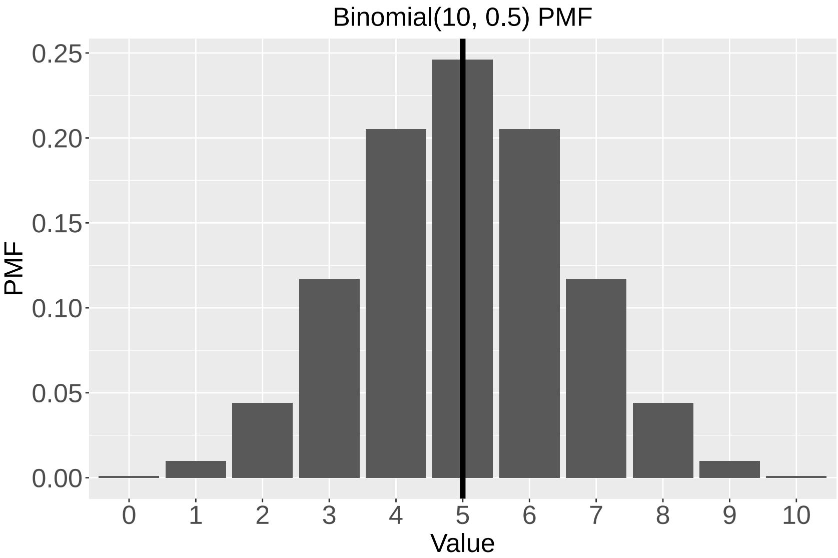
Example: $X \sim \text{Bernoulli}(0.4)$



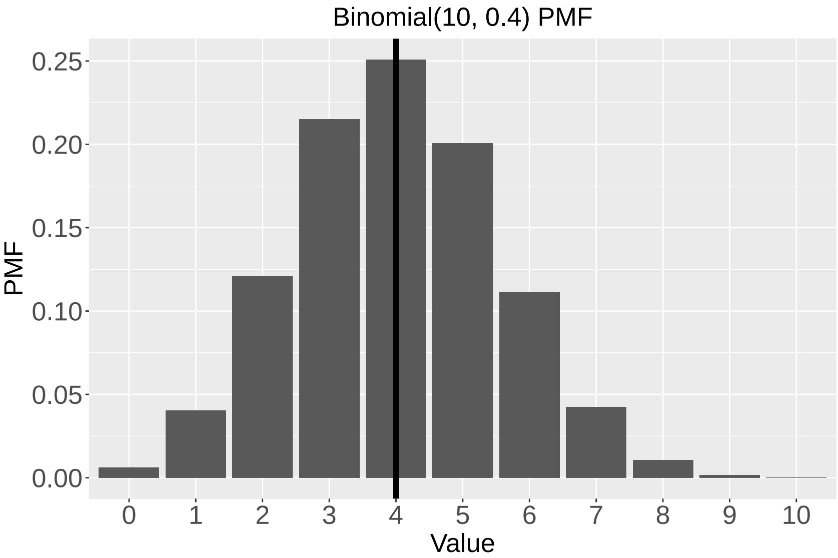
Example: $X \sim \text{Binomial}(N, \theta)$

- Example: X is a number of successes in N trials.
- Construction: $X = Y_1 + \ldots + Y_N$ where $Y_1, \ldots, Y_N \overset{IID}{\sim} \text{Bernoulli}(\theta)$.
- Domain: $X \in \{0, 1, ..., N\}$.
- PMF: $P(X = x) = \binom{N}{x}\theta^x(1-\theta)^{N-x}$.
- Parameter: $\theta \in [0, 1]$ is the success probability of each trial.
- Mean: $\mathrm{E}(X) = \sum_{x=0}^{N} xf(x) = n\theta$.
- Variance: $\mathrm{V}(X) = n\theta(1-\theta)$.

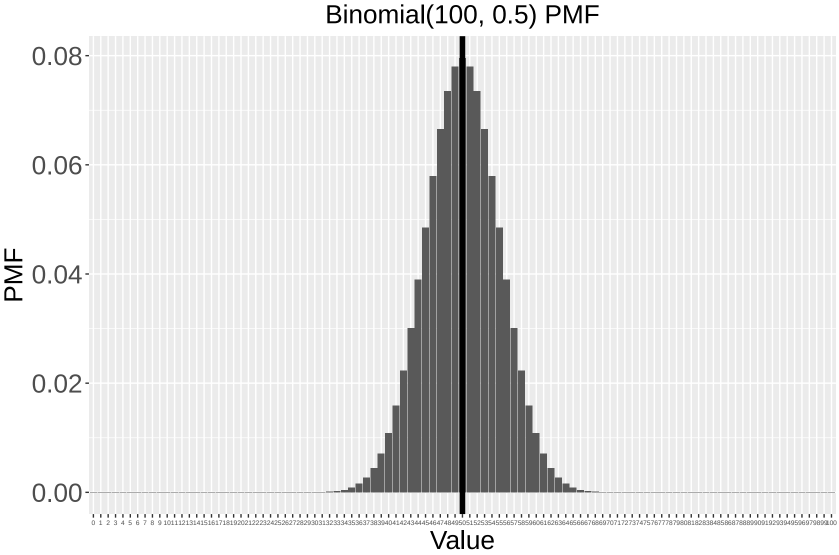
Example: $X \sim \text{Binomial}(10, 0.5)$



Example: $X \sim \text{Binomial}(10, 0.4)$



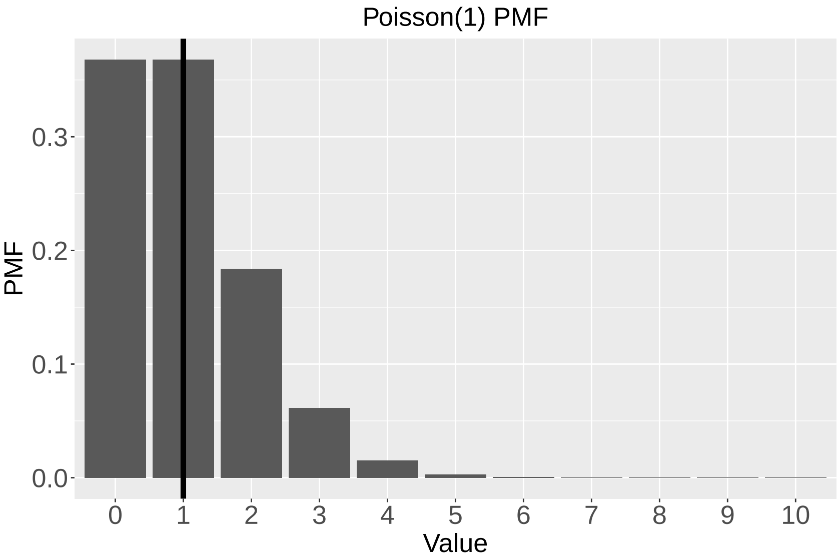
Example: $X \sim \text{Binomial}(100, 0.5)$



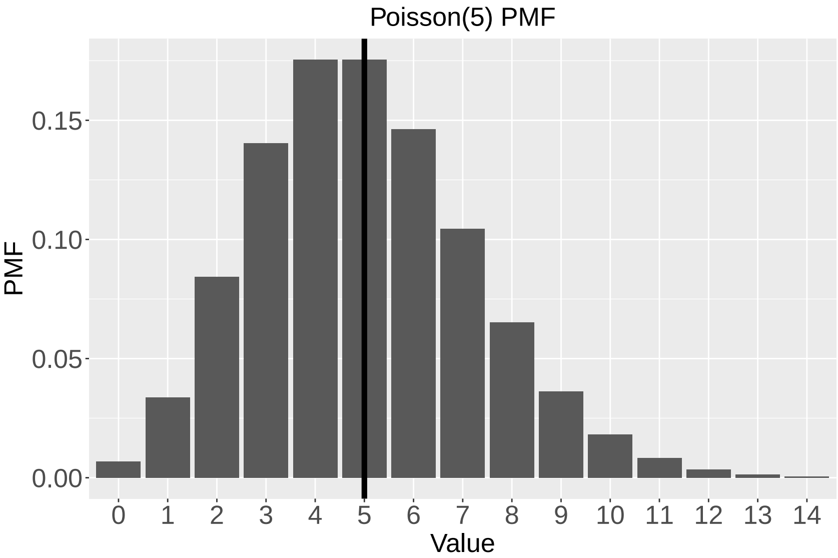
Example: $X \sim \text{Poisson}(N\theta)$

- Example: X is the number events that occur in N units of time
- Often the distribution is presented with $N = 1$.
- Domain: $X \in \{0, 1, 2, ...\}$.
- PMF: $P(X = x) = \frac{\exp(-N\theta)(N\theta)^x}{x!}$.
- Parameter: θ is the expected number of events per unit of time
- Mean: $\mathrm{E}(X) = N\theta$.
- Variance: $\mathrm{V}(X) = N\theta$.

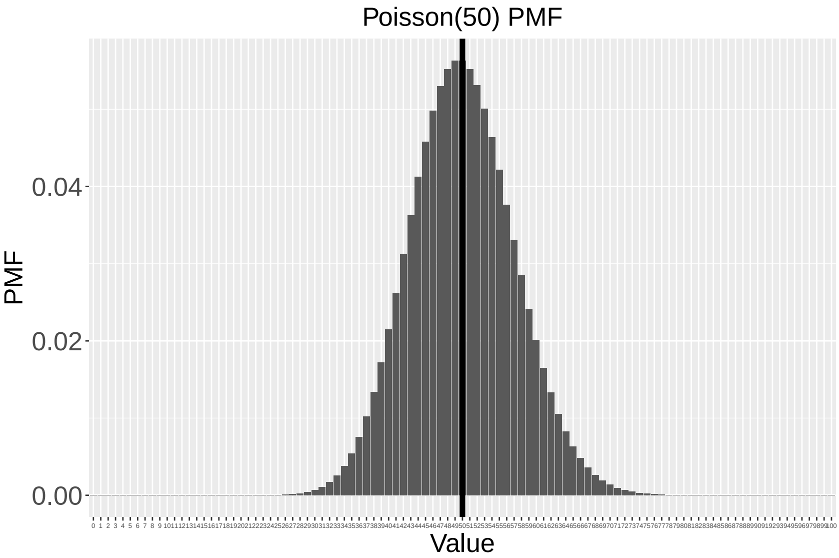
Example: $X \sim \text{Poisson}(1)$



Example: $X \sim \text{Poisson}(5)$



Example: $X \sim$ Poisson(50)



Continuous univariate distributions

- If X is continuous we describe its distribution with the cumulative distribution function $F(x) \in [0, 1]$ or by probability density function (PDF) $f(x) \geq 0$.
- Here, $F(x) = \int_{-\infty}^{x} f(y)dy$ is a continuous function.
- Therefore, $P(X = x) = F(x) - \lim_{\delta \to 0} F(x - \delta) = 0$.
- Thus, the PMF is meaningless here.

Continuous univariate distributions

- Probabilities are computed as areas under the PDF curve

$$\text{Prob}(l < X < u) = \int_l^u f(x)dx.$$

- Therefore, $f(x)$ must satisfy $f(x) \geq 0$ and

$$\text{Prob}(-\infty < X < \infty) = \int_{-\infty}^{\infty} f(x)dx = 1.$$

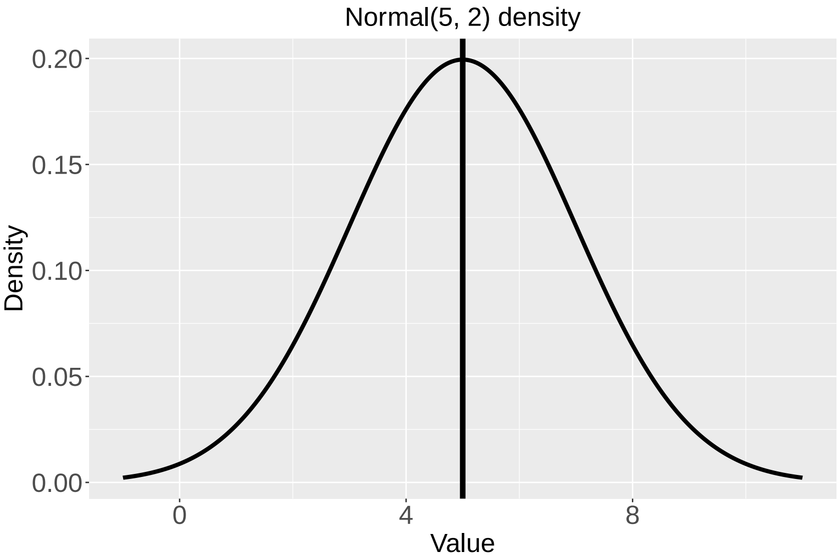
Continuous univariate distributions

- The domain is the set of x values with $f(x) > 0$.
- The mean and the variance are defined similarly to the discrete case but with the sums replaced by integrals.
- The mean is
$$\mathrm{E}(X) = \int xf(x)dx.$$
- The variance is
$$\mathrm{V}(X) = \int [x - E(X)]^2 f(x)dx.$$

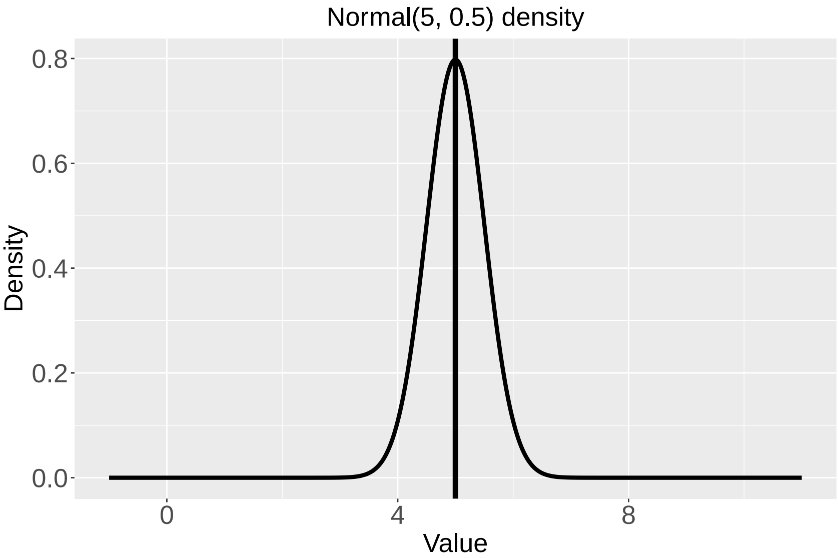
Example: $X \sim \text{Normal}(\mu, \sigma^2)$

- Example: X is the average temperature at a weather station on a certain day.
- Domain: $X \in (-\infty, \infty)$.
- PDF: $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right]$.
- Parameters: μ is the mean, $\sigma^2 > 0$ is the variance.
- Mean: $\mathrm{E}(X) = \mu$.
- Variance: $\mathrm{V}(X) = \sigma^2$.

Example: $X \sim \text{Normal}(5, 2^2)$



Example: $X \sim \text{Normal}(5, 0.5^2)$

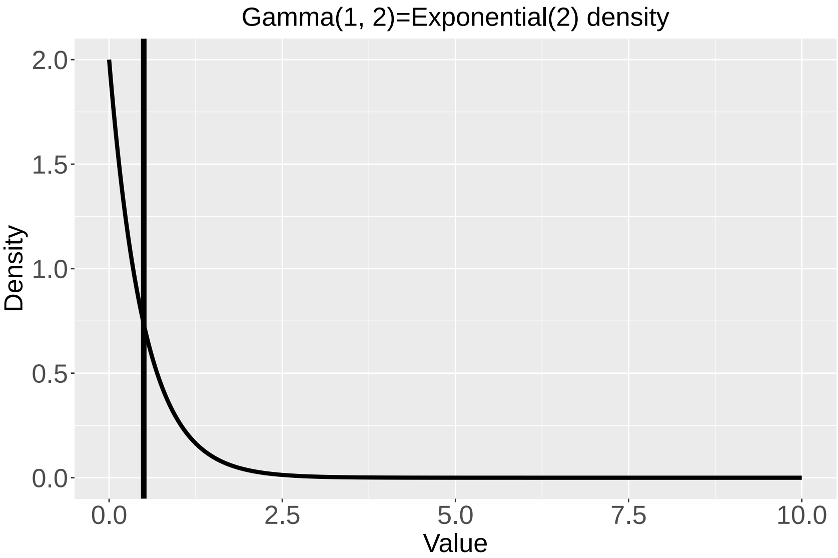


Example: $X \sim \text{Gamma}(a, b)$

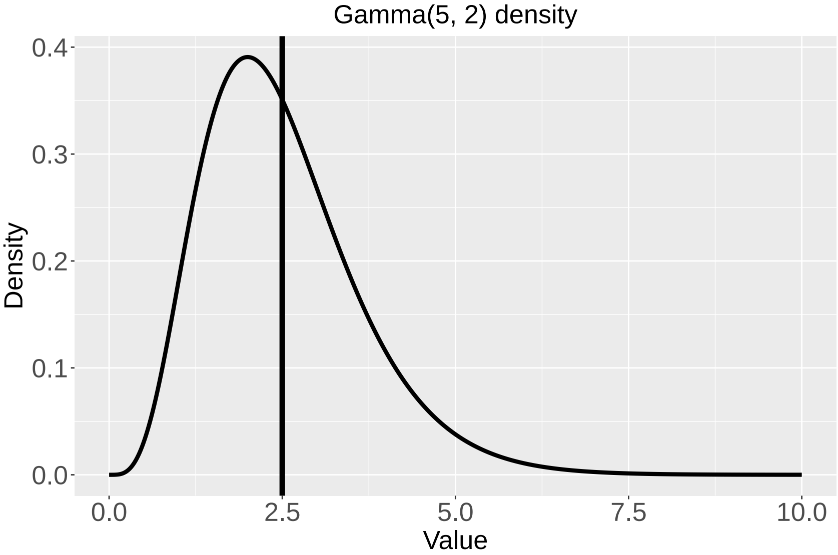
- Example: X is a height of a baby.
- Domain: $X \in (0, \infty)$.
- PDF: $f(x) = \frac{b^a}{\Gamma(a)} x^{a-1} \exp(-bx)$.
- Parameters: $a > 0$ is the shape, $b > 0$ is the rate.
- Mean: $\text{E}(X) = \frac{a}{b}$.
- Variance: $\text{V}(X) = \frac{a}{b^2}$.
- Be careful: Sometimes the PDF is given as

$$f(x) = \frac{1}{\Gamma(a) b^a} x^{a-1} \exp(-x/b).$$

Example: $X \sim \text{Gamma}(1, 2)$



Example: $X \sim \text{Gamma}(5, 2)$

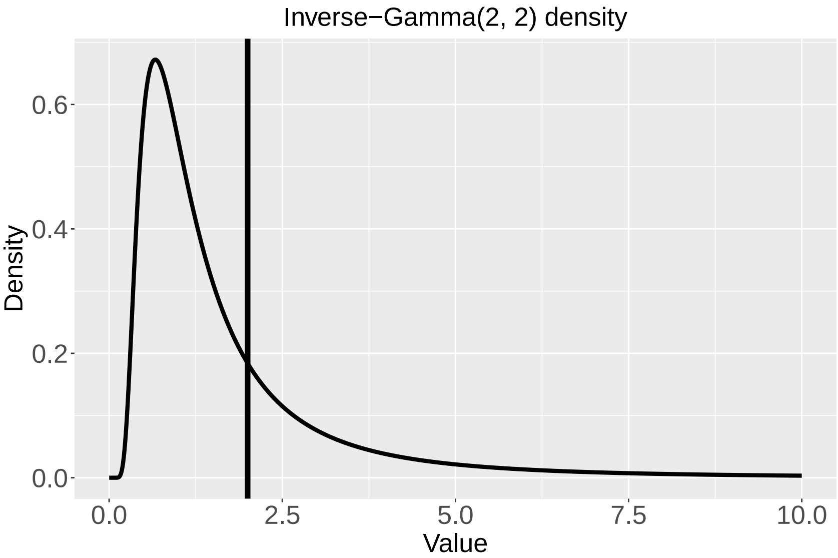


Example: $X \sim$ Inverse-Gamma(a, b)

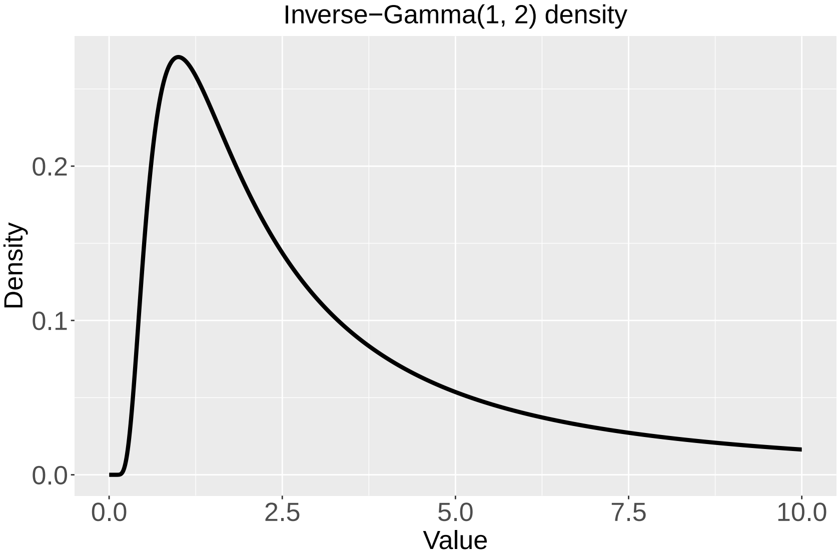
- If $Y \sim$ Gamma(a, b) and $X = 1/Y$, then $X \sim$ Inverse-Gamma(a, b).
- Domain: $X \in (0, \infty)$.
- PDF: $f(x) = \frac{b^a}{\Gamma(a)} x^{-a-1} \exp(-b/x)$.
- Parameters: $a > 0$ is the shape, $b > 0$ is the rate.
- Mean: $\mathrm{E}(X) = \frac{b}{a-1}$ if $a > 1$.
- Variance: $\mathrm{V}(X) = \frac{b^2}{(a-1)^2(a-2)}$ if $a > 2$.
- Be careful: Sometimes the PDF is given as

$$f(x) = \frac{1}{\Gamma(a) b^a} x^{-a-1} \exp[-1/(bx)].$$

Example: $X \sim$ Inverse-Gamma$(2, 2)$



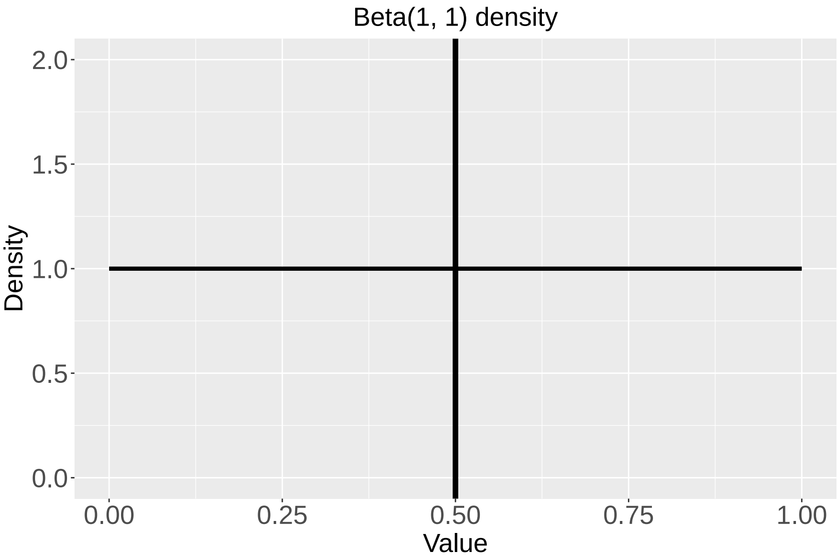
Example: $X \sim \text{Inverse-Gamma}(1, 2)$



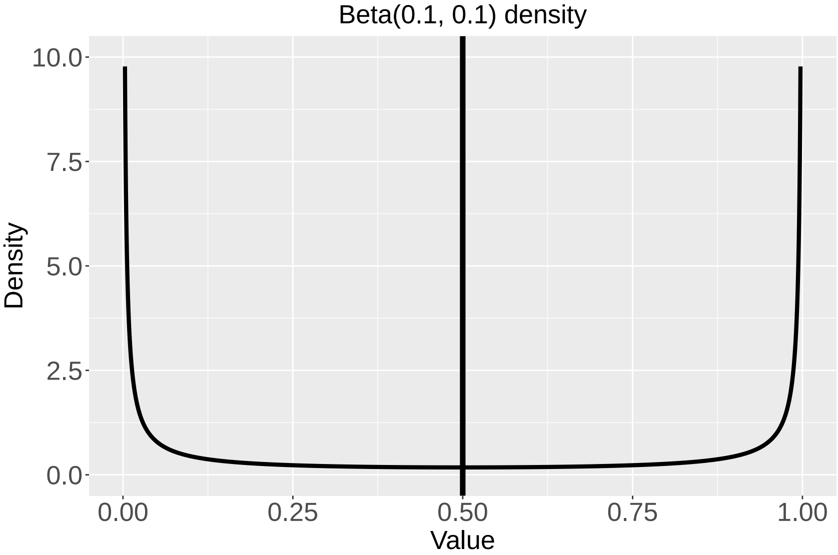
Example: $X \sim \text{Beta}(a, b)$

- Example: X is a probability.
- Domain: $X \in [0, 1]$.
- PDF: $f(x) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} x^{a-1}(1-x)^{b-1}$.
- Parameters: $a > 0$ and $b > 0$.
- Mean: $\text{E}(X) = \frac{a}{a+b}$.
- Variance: $\text{V}(X) = \frac{ab}{(a+b)^2(a+b+1)}$.

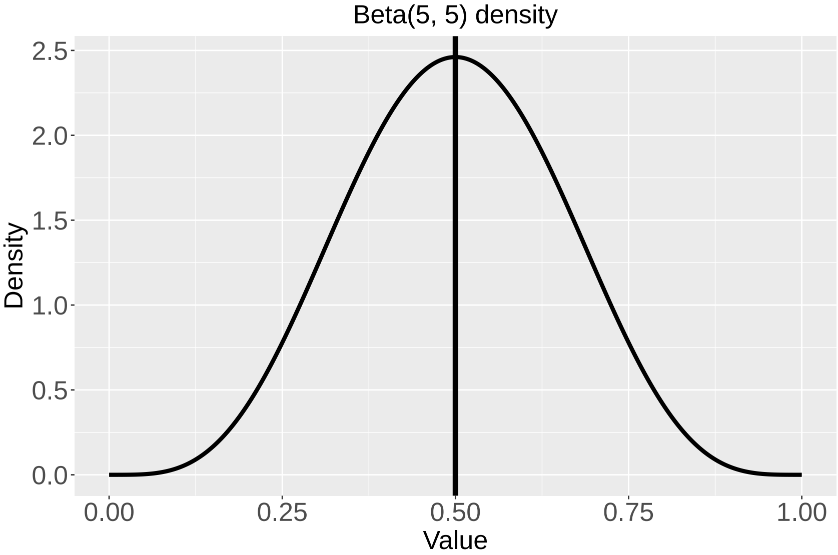
Example: $X \sim \text{Beta}(1, 1)$



Example: $X \sim \text{Beta}(0.1, 0.1)$



Example: $X \sim \text{Beta}(5, 5)$



Thank you!

Marginal distribution

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right)$$

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2}\cdot\frac{x_1^2+x_2^2-2\rho x_1 x_2}{1-\rho^2}}$$

$$f_{X_1}(x_1) = \int_{-\infty}^{+\infty} f(x_1, x_2)\, dx_2$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2}\cdot\frac{x_1^2+x_2^2-2\rho x_1 x_2}{1-\rho^2}}\, dx_2$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \cdot e^{-\frac{1}{2}\cdot\frac{x_1^2}{1-\rho^2}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2}\cdot\frac{x_2^2-2\rho x_1 x_2}{1-\rho^2}}\, dx_2$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2}\cdot\frac{x_1^2}{1-\rho^2}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2}\cdot\frac{x_2^2-2\rho x_1 x_2+(\rho x_1)^2-(\rho x_1)^2}{1-\rho^2}}\, dx_2$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2}\cdot\frac{x_1^2}{1-\rho^2}} \times e^{\frac{1}{2}\cdot\frac{(\rho x_1)^2}{1-\rho^2}}$$

$$\times \sqrt{2\pi}\sqrt{1-\rho^2} \underbrace{\int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2}\cdot\frac{(x_2-\rho x_1)^2}{1-\rho^2}}\, dx_2}_{\rightarrow 1}$$

$$= \frac{1}{2\pi\cancel{\sqrt{1-\rho^2}}} \times e^{-\frac{1}{2}\cdot\frac{x_1^2-\rho^2 x_1^2}{1-\rho^2}} \times \sqrt{2\pi} \times \cancel{\sqrt{1-\rho^2}}$$

$$= \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\cdot x_1^2 \cancel{(1-\rho^2)}/\cancel{(1-\rho^2)}} = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x_1^2}$$

So, the marginal distribution of X_1 is $N(0,1)$

Conditional distribution:

The joint distribution is

$$f(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2}\cdot\frac{x_1^2+x_2^2-2\rho x_1 x_2}{1-\rho^2}}$$

$$f_{X_1}(x_1) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x_1^2}$$

So, $$f_{X_2|X_1}(x_2|x_1) = \frac{f_{X_1 X_2}(x_1, x_2)}{f_{X_1}(x_1)}$$

$$= \frac{\frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2}\cdot\frac{x_1^2+x_2^2-2\rho x_1 x_2}{1-\rho^2}}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x_1^2}}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2}\left[\frac{x_1^2+x_2^2-2\rho x_1 x_2}{1-\rho^2} - x_1^2\right]}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2}\frac{x_1^2+x_2^2-2\rho x_1 x_2-(1-\rho^2)x_1^2}{1-\rho^2}}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} e^{-\frac{1}{2}\frac{\rho^2 x_1^2+x_2^2-2\rho x_1 x_2}{1-\rho^2}}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \cdot e^{-\frac{1}{2}\frac{(x_2-\rho x_1)^2}{1-\rho^2}}$$

So, $X_2 | X_1 = x_1 \sim N(\rho x_1, 1-\rho^2)$

Lecture 2

Probability Essentials

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Review of probability (Recap)

- In Frequentist or classical statistics, we assume that the data **Y** follows a model that involves some unknown fixed parameter $\boldsymbol{\theta}$.
- In Bayesian statistics, we also assume a probability distribution (prior) for $\boldsymbol{\theta}$.
- The most important thing in Bayesian statistics is computing the posterior, i.e., the distribution of the parameters $\boldsymbol{\theta}$ after observing the data **Y**.
- This is the conditional distribution of $\boldsymbol{\theta}$ given **Y**.
- Therefore, we need to review the probability concepts that lead to the conditional distribution of one variable conditioned on another.

  1. Probability mass (PMF) and density (PDF) functions (Lecture 1)
  2. Joint distributions (Lecture 2)
  3. Marginal and conditional distributions (Lecture 2)

Joint distributions

- $\mathbf{X} = (X_1, ..., X_p)$ is a random vector (vectors and matrices should be in bold).
- For convenience, let's consider only $p = 2$ random variables X and Y.
- (X, Y) is discrete if it can take on a countable number of values, such as

  X = number of students who were tested for Covid at IIT-K and
  Y = number of students who tested positive.
- (X, Y) is continuous if it can take on an uncountable number of values, such as

  X = height of a person and Y = weight of a person.

Discrete random variables

- The joint PMF is
$$f(x, y) = \text{Prob}(X = x, Y = y)$$
- Example: patients are randomly assigned a dose and followed to determine whether they develop a tumor.
- $X \in \{10, 20, 30\}$ is the dose; $Y \in \{0, 1\}$ is 1 if a tumor develops and 0 otherwise
- The joint PMF is

	X		
Y	10	20	30
0	0.469	0.124	0.049
1	0.231	0.076	0.051

Discrete random variables

- The **marginal PMF** for X is

$$f_X(x) = \text{Prob}(X = x) = \sum_y f(x, y).$$

- The **marginal PMF** for Y is

$$f_Y(y) = \text{Prob}(Y = y) = \sum_x f(x, y).$$

- The marginal distribution is the same as univariate distribution as if we ignored the other variable.

Discrete random variables

- Example: X = dose and Y = tumor status.
- Find the marginal PMFs of X and Y.
- The joint PMF:

	X		
Y	10	20	30
0	0.469	0.124	0.049
1	0.231	0.076	0.051

Discrete random variables

- Example: X = dose and Y = tumor status.
- Find the marginal PMFs of X and Y.
- The joint PMF:

	X		
Y	10	20	30
0	0.469	0.124	0.049
1	0.231	0.076	0.051

- $$\begin{aligned} & \text{Prob}(Y = 0) \\ = \; & \text{Prob}(X = 10, Y = 0) + \text{Prob}(X = 20, Y = 0) + \text{Prob}(X = 30, Y = 0) \\ = \; & 0.469 + 0.124 + 0.049. \end{aligned}$$

Discrete random variables

- The **Conditional PMF** of Y given X is

$$f(y|x) = \text{Prob}(Y = y|X = x) = \frac{\text{Prob}(X = x, Y = y)}{\text{Prob}(X = x)} = \frac{f(x, y)}{f_X(x)}.$$

- Here x is treated as a fixed number, and so $f(y, x)$ is only a function of y.
- However, we can't use $f(x, y)$ as the PMF for Y because

$$\sum_y f(x, y) = f_X(x) \neq 1$$

- Dividing by $f_X(x)$ makes $f(y|x)$ valid

$$\sum_y f(y|x) = \sum_y \frac{f(y, x)}{f_X(x)} = \frac{\sum_y f(y, x)}{f_X(x)} = \frac{f_X(x)}{f_X(x)} = 1$$

Discrete random variables

- Example: X = dose and Y = tumor status.
- Find $f(y|x)$.
- The joint PMF:

	X		
Y	10	20	30
0	0.469	0.124	0.049
1	0.231	0.076	0.051

Discrete random variables

- Example: X = dose and Y = tumor status.
- Find $f(y|x)$.
- The joint PMF:

	X		
Y	10	20	30
0	0.469	0.124	0.049
1	0.231	0.076	0.051

- $$\begin{aligned} & \text{Prob}(Y = 0|X = 10) \\ = \; & \text{Prob}(X = 10, Y = 0)/\text{Prob}(X = 10) \\ = \; & 0.469/(0.469 + 0.231). \end{aligned}$$

Monte Hall problem

- `https://en.wikipedia.org/wiki/Monty_Hall_problem`
- Suppose you're on a game show, and you're given the choice of three doors.
- Behind one door is a car; behind the others, goats.
- You pick a door, say No. 1, and the host, who knows what's behind the doors, opens another door, say No. 3, which has a goat.
- He then says to you, "Do you want to pick door No. 2?"
- **Is it to your advantage to switch your choice?**

Monte Hall problem

Door 1	Door 2	Door 3	Staying at Door 1	Switching to the door offered
Goat	Goat	**Car**	Wins goat	**Wins car**
Goat	**Car**	Goat	Wins goat	**Wins car**
Car	Goat	Goat	**Wins car**	Wins goat

Discrete random variables

- X and Y are **independent** if

$$f(x,y) = f_X(x)f_Y(y)$$

  for all x and y.
- Variables are dependent if they are not independent.
- Equivalently, X and Y are **independent** if

$$f(x|y) = f_X(x)$$

  for all x and y.
- These two definitions are equivalent.

$$f(x|y) = \frac{f(x,y)}{f_Y(y)} = \frac{f_X(x)f_Y(y)}{f_Y(y)} = f_X(x).$$

Discrete random variables

- Notation: $X_1, ..., X_n \overset{iid}{\sim} f(x)$ means that $X_1, ..., X_n$ are independent and identically distributed.

- This implies the joint PMF is

$$\text{Prob}(X_1 = x_1, ..., X_n = x_n) = \prod_{i=1}^{n} f(x_i).$$

- The same notation and definitions of independence apply to continuous random variables.

- In this class, assume independence unless otherwise noted.

Hurricane proportions by landfall (X) and category (Y)

	Category					
	1	2	3	4	5	Total
US (1)	0.0972	0.0903	0.0694	0.0069	0.0069	0.2708
Not US (0)	0.3194	0.1319	0.1389	0.1181	0.0208	0.7292
Total	0.4167	0.2222	0.2083	0.1250	0.0278	1.0000

Problem: Prove X and Y are dependent.

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Total	0.4167	0.2222	0.2083	0.1250	0.0278	1.0000

Problem: Prove X and Y are dependent.

- $\text{Prob}(X = 1) = 0.2708$, $\text{Prob}(Y = 1) = 0.4167$.

  So, $\text{Prob}(X = 1) \times \text{Prob}(Y = 1) = 0.2708 \times 0.4167 = 0.1128$.

  Thus, $\text{Prob}(X = 1) \times \text{Prob}(Y = 1) \neq \text{Prob}(X = 1, Y = 1)$.

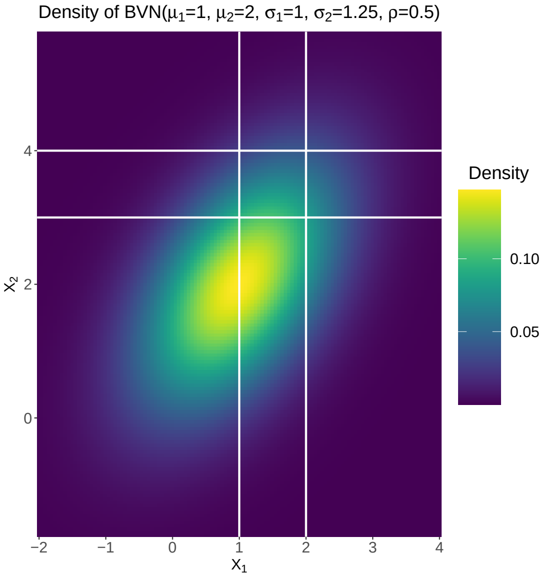
Continuous random variables

- Manipulating joint PDFs is similar to joint PMFs but sums are replaced by integrals.
- The joint PDF is denoted $f(x, y)$.
- Probabilities are computed as volume under the PDF:

$$\text{Prob}[(X, Y) \in A] = \int_A f(x, y)dxdy$$

where $A \subset \mathbb{R}^2$.

Continuous random variables



Continuous random variables

- Example: X=birthweight, Y=gestational age.
- Domain: $X \in (2, 10)$ lbs and $Y \in (20, 50)$ weeks.
- PDF: $f(x, y) = 0.26 \exp(-|x - 7| - |y - 40|)$.
- Find: $\text{Prob}(X > 7, Y > 40)$.

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- Find: $\text{Prob}(X > 7, Y > 40)$.
- $$\begin{aligned} & \text{Prob}(X > 7, Y > 40) \\ = \; & \int_7^{10} \int_{40}^{50} 0.26 \exp(-|x - 7| - |y - 40|) dx dy \\ = \; & 0.26 \left(\int_0^3 \exp(-x) dx \right) \times \left(\int_0^{10} \exp(-y) dy \right). \end{aligned}$$

Continuous random variables

- The **Marginal PDF** of X is

$$f_X(x) = \int f(x, y)dy.$$

- f_X is the univariate PDF for X as if we never considered Y.
- Find: $f_X(x)$ for the birthweight example.

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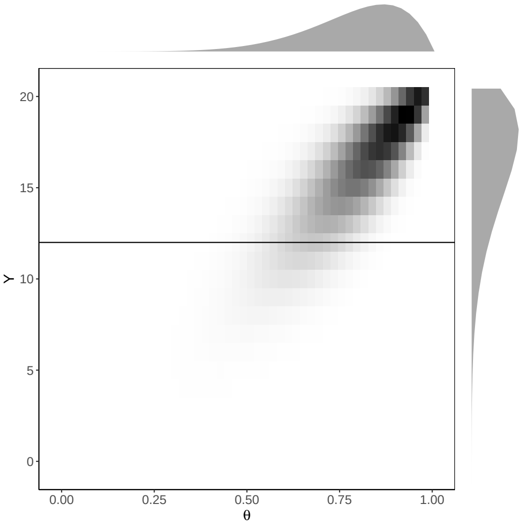
Joint and marginal distributions



Continuous random variables

- The **Conditional PDF** of Y given X is

$$f(y|x) = \frac{f(x,y)}{f_X(x)}.$$

- Proper: $\int f(y|x)dy = \int \frac{f(x,y)}{f_X(x)}dy = \frac{\int f(x,y)dy}{f_X(x)} = 1.$
- Find: $f(y|x)$ for the birthweight example.

Continuous random variables

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- Find: $f(y|x)$ for the birthweight example.
- $$\begin{aligned} f(y|x) &= \frac{f(x,y)}{f_X(x)} \\ &= 0.26\exp(-|x-7|-|y-40|)/\int_{20}^{50} 0.26\exp(-|x-7|-|y-40|)dy \\ &= \exp(-|y-40|)/\int_{20}^{50}\exp(-|y-40|)dy. \end{aligned}$$

Bivariate normal distribution

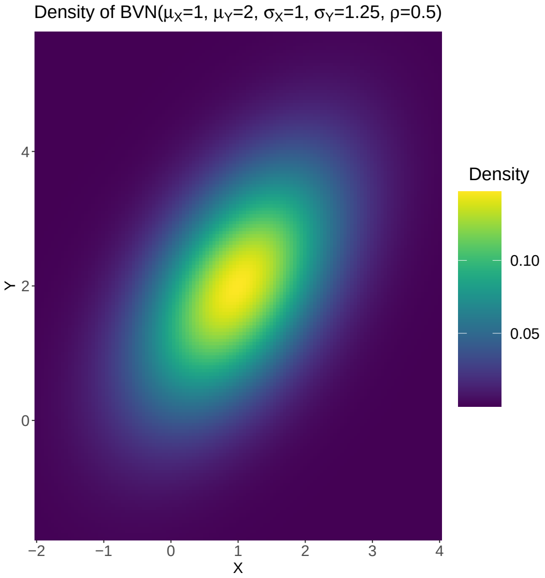
- The **bivariate normal distribution (BVN)** is the most common multivariate family.

- There are 5 parameters:
  - The marginal means of X and Y are μ_X and μ_Y.
  - The marginal variances of X and Y are $\sigma_X^2 > 0$ and $\sigma_Y^2 > 0$.
  - The correlation between X and Y is $\rho \in (-1, 1)$.

- The joint PDF is $f(x, y) =$

$$\frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left\{-\frac{\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)}{2(1-\rho^2)}\right\}$$

Bivariate normal distribution



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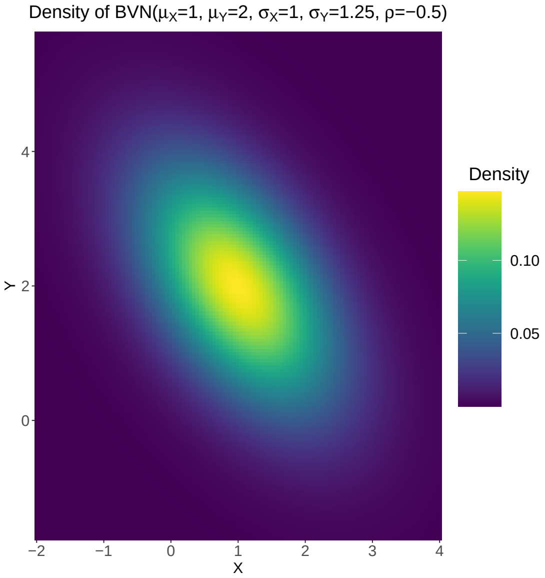

Bivariate normal distribution

- The marginal distributions are $X \sim \text{Normal}(\mu_X, \sigma_X^2)$ and $Y \sim \text{Normal}(\mu_Y, \sigma_Y^2)$
- The conditional distribution is

$$Y|X \sim \text{Normal}\left\{\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(X - \mu_X), \sigma_Y^2(1 - \rho^2)\right\}$$

- The standard BVN distribution has $\mu_X = \mu_Y = 0$ and $\sigma_X = \sigma_Y = 1$ and thus

$$Y|X \sim \text{Normal}\left(\rho X, 1 - \rho^2\right)$$

- Explain the role of ρ.
- See “BVN marginal” and “BVN conditional” in the derivations uploaded at Hello IIT-K.

Defining joint distributions conditionally

- Specifying joint distributions is hard.
- Every joint distribution can be written

$$f(x, y) = f(y|x)f(x).$$

- Therefore, any joint distribution can be defined by

  1. X's marginal distribution,
  2. The conditional distribution of $Y|X$.

- The joint problem reduces to two univariate problems.
- This idea forms the basis of hierarchical modeling.

Defining joint distributions conditionally

- Let X be the number of robins in the forest.
- Let Y be the number of robins we observe.
- Model $\text{Prob}(X = x) = 1/20$ for $x \in \{0, ..., 19\}$ and $Y|X \sim \text{Binomial}(X, 0.2)$.
- What is the support of (X, Y)?
- What is $\text{Prob}(X = 10, Y = 1)$?
- What is $\text{Prob}(Y = 0)$?

Defining joint distributions conditionally

- $f(x, y)$ is positive over $x, y \in \{0, 1, \ldots, 19\}$ such that $y \leq x$.
- $$\begin{aligned} & \text{Prob}(X = 10, Y = 1) \\ = \; & \text{Prob}(X = 10) \times \text{Prob}(Y = 1 | X = 10) \\ = \; & \frac{1}{20} \times \binom{10}{1} 0.2^1 0.8^9. \end{aligned}$$
- $$\text{Prob}(Y = 0) = \sum_{x=0}^{19} \text{Prob}(X = x, Y = 0)$$ $$= \sum_{x=0}^{19} \text{Prob}(X = x) \times \text{Prob}(Y = 0 | X = x)$$ $$= \sum_{x=0}^{19} \frac{1}{20} \times \binom{x}{0} 0.2^0 0.8^x = \frac{1}{20} \times \sum_{x=0}^{19} 0.8^x.$$

Lecture 3

Bayes' theorem and Bayesian learning

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Bayes' Theorem

- Suppose, you were having high fever.

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- Let there are only two possibilities: either you have seasonal flu or you have contracted Covid 19.

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- Let θ be the parameter of interest with

$$\theta = \begin{cases} 0 & \text{you have seasonal flu,} \\ 1 & \text{you have contracted Covid 19.} \end{cases}$$

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$$\theta = \begin{cases} 0 & \text{you have seasonal flu,} \\ 1 & \text{you have contracted Covid 19.} \end{cases}$$

- Then, you went to the health center and got tested for Covid 19.
- The data is Y, defined as

$$Y = \begin{cases} 0 & \text{test is negative,} \\ 1 & \text{test is positive.} \end{cases}$$

Bayes' Theorem

- Objective: Derive the probability that you have Covid 19 given the test results.

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- Since Y is binary, we use a Bernoulli PMF for the likelihood.
- We must specify the likelihood for both $\theta = 0$ and $\theta = 1$.
- $\text{Prob}(Y = 1|\theta = 0) = q_0$ is the false positive rate.
- $\text{Prob}(Y = 1|\theta = 1) = q_1$ is the true positive rate.
- How might we select q_0 and q_1?

Covid 19 example - Prior

- The prior represents our uncertainty about the parameters before we observe the data.

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Covid 19 example - Prior

- The prior represents our uncertainty about the parameters before we observe the data.
- Since θ is binary, we use a Bernoulli PMF for the prior.
- Prob$(\theta = 1) = p$ is the population prevalence of Covid 19.
- How might we select p?
- Several factors involved: current positivity rate at IIT-K, if anyone tested positive with whom you came in close contact, etc.

Covid 19 example - Posterior

- Derive the posterior probability that you have contracted Covid 19 given a positive test.
- That is Prob$(\theta = 1 | Y = 1)$.

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- $$\begin{aligned} & \text{Prob}(\theta = 1 | Y = 1) \\ = \; & \frac{\text{Prob}(\theta = 1, Y = 1)}{\text{Prob}(Y = 1)} \\ = \; & \frac{\text{Prob}(Y = 1 | \theta = 1)\text{Prob}(\theta = 1)}{\text{Prob}(Y = 1 | \theta = 1)\text{Prob}(\theta = 1) + \text{Prob}(Y = 1 | \theta = 0)\text{Prob}(\theta = 0)} \\ = \; & \frac{q_1 p}{q_1 p + q_0(1 - p)}. \end{aligned}$$

Covid 19 example - Posterior

- Derive the posterior probability that you have contracted Covid 19 given a positive test.
- That is Prob$(\theta = 1 | Y = 1)$.
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- This is Bayes' theorem: obtaining $p(\theta | y)$ from $f(y | \theta)$ and $\pi(\theta)$.

Bayes' Theorem

- In Bayesian statistics, we select the prior, $\pi(\theta)$, and the likelihood, $f(y|\theta)$.
- Based on these two pieces of information, our main goal is to compute the posterior $p(\theta|y)$.
- Bayes' theorem is the mathematical formula to convert the likelihood and prior to the posterior.
- Bayes theorem:

$$p(\theta|y) = \frac{f(y|\theta)\pi(\theta)}{m(y)}.$$

- This holds for discrete (PMF) and continuous (PDF) cases.

Bayes' theorem

- Bayes theorem in math:
$$p(\theta|y) = \frac{f(y|\theta)\pi(\theta)}{m(y)}.$$
- Bayes theorem in words:
$$p(\theta|y) = \frac{\text{Likelihood} * \text{Prior}}{\text{marginal distribution of Y}}.$$
- As in the formula for a conditional distribution, $m(y)$ is just the normalizing constant required so that $\int p(\theta|y)d\theta = 1$.
- Most of the time $m(y)$ can be ignored because it doesn't depend on θ and the objective is to study the posterior of θ.

Cricket example

- Team India plays half its games at home, wins 70% of its home games, and 40% of its away games. Given that the team wins a game, what's the probability it was a home game?

Cricket example

- Team India plays half its games at home, wins 70% of its home games, and 40% of its away games. Given that the team wins a game, what's the probability it was a home game?
- Let θ be the parameter of interest with

$$\theta = \begin{cases} 1 & \text{if it was a home match} \\ 0 & \text{if it was an away match} \end{cases}$$

- The data is Y, defined as

$$Y = \begin{cases} 1 & \text{if the team wins} \\ 0 & \text{if the team loses} \end{cases}$$

Cricket example

- Here, $\text{Prob}(\theta = 0) = \text{Prob}(\theta = 1) = 0.5$.
- Also, $\text{Prob}(Y = 1|\theta = 1) = 0.7$ and $\text{Prob}(Y = 1|\theta = 0) = 0.4$.
- $$\begin{aligned} & \text{Prob}(\theta = 1|Y = 1) \\ = \quad & \frac{\text{Prob}(\theta = 1, Y = 1)}{\text{Prob}(Y = 1)} \\ = \quad & \frac{\text{Prob}(Y = 1|\theta = 1)\text{Prob}(\theta = 1)}{\text{Prob}(Y = 1|\theta = 1)\text{Prob}(\theta = 1) + \text{Prob}(Y = 1|\theta = 0)\text{Prob}(\theta = 0)} \\ = \quad & \frac{0.7 \times 0.5}{0.7 \times 0.5 + 0.4 \times 0.5} \\ = \quad & \frac{7}{11}. \end{aligned}$$

Robins example

- Let X be the number of robins in the forest
- Let Y be the number of robins we observe
- Uniform prior: $\text{Prob}(X = x) = 1/20$ for $x \in \{0, ..., 19\}$
- Likelihood: $Y|\theta \sim \text{Binomial}(X, 0.2)$ (0.2 is the detection probability)
- Given that we do not observe any birds, what is the probability that no birds are in the forest?
- Intuitively, how would this change if the prior was $\text{Prob}(X = x) = 1/100$ for $x \in \{0, ..., 99\}$?
- Intuitively, how would this change if the detection probability increased from 0.2 to 0.9?

Robins example

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= \quad & \frac{\text{Prob}(Y = 0 | X = 0)\text{Prob}(X = 0)}{\sum_{x=0}^{19} \text{Prob}(Y = 0 | X = x)\text{Prob}(X = x)} \\
= \quad & \frac{1 \times 1/20}{\sum_{x=0}^{19} \binom{x}{0} 0.2^0 0.8^x \times 1/20} \\
= \quad & \frac{1}{\sum_{x=0}^{19} 0.8^x} = 0.2023.
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Robins example

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- So, the probability decreases. Why this is so?

Robins example

- Intuitively, how would this change if the prior was $\text{Prob}(X = x) = 1/100$ for $x \in \{0, ..., 99\}$?
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= \; & \frac{1}{\sum_{x=0}^{99} 0.8^x} = 0.2000.
\end{aligned}
$$

- So, the probability decreases. Why this is so?
- The prior probability of $\text{Prob}(X = 0)$ is smaller here.

Robins example

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- So, the probability increases. Why this is so?

Robins example

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= \; & \frac{1}{\sum_{x=0}^{19} 0.1^x} = 0.9000.
\end{aligned}
$$

- So, the probability increases. Why this is so?
- With better detection probability, we are more confident in our data.

Bayesian learning

- Bayesians quantify uncertainty about fixed but unknown parameters by treating them as random variables.

- This requires that we set a **prior distribution** $\pi(\theta)$ to summarize uncertainty before observing the data.

- The distribution of the observed data given the model parameters is the **likelihood function**, $f(Y|\theta)$.

- The likelihood function is the most important piece of a Bayesian analysis because it links the data and parameters.

Bayesian learning

- The **posterior distribution** $p(\theta|Y)$ summarizes uncertainty about the parameters given the prior and data.
- The reduction in uncertainty from prior to posterior represents **Bayesian learning**.
- **Bayes' Theorem** (Bayes' Rule) converts the likelihood and prior to the posterior.
- Bayes' Theorem:
$$p(\theta|Y) = \frac{f(Y|\theta)\pi(\theta)}{m(Y)}$$
where $m(Y) = \int f(Y|\theta)\pi(\theta)d\theta$ is the marginal distribution of the data and can usually be ignored.

How to select the prior?

- There is no “true” or “correct” prior.
- In some cases expert opinion or similar studies can be used to specify an informative prior.
- It would be a waste to discard this information.
- If prior information is unavailable, then the prior should be non-informative or weakly informative.
- The prior is best viewed as an initial value to a statistical procedure.

How to select the prior?

- As we will see later, as Bayesian learning continues, and further data are collected, the posterior concentrates around the true value for any reasonable prior.
- However, in a finite sample the prior can have some effect.
- We will study several systematic ways to select priors.
- However, there is inherent **subjectivity** to selecting the prior.
- That is, different analysts may pick different priors and thus have different results.

How to select the likelihood?

- The likelihood is the same as in a Freqentist analysis.
- For example, in a linear regression analysis, we might say

$$Y_i|\boldsymbol{\beta}, \sigma^2. \sim \text{Normal}\left(\sum_{j=1}^{p} X_{ij}\beta_j, \sigma^2\right).$$

- Is there "true" or "correct" likelihood?
- Is specification of the likelihood subjective like the prior?

How to select the likelihood?

Subjective decisions required to specify the likelihood:

- The errors are independent.
- $\log(Y_i)$ follows a normal distribution.
- The mean is a function of X_{i1}, X_{i3}, and X_{i3}^2.
- Y_{43} is an outlier and removed.
- Many, many more.

Opinions about subjectivity

- Perhaps we should aspire to objectivity, but in most real-life analyses we are forced to accept some subjectivity.
- A notable exception is a tightly controlled experiment, although even this is debatable.
- However, not all subjective decisions (assumptions) are equal.
- If readers disagree with your assumptions they will reject your findings.
- It is your job to justify your assumptions theoretically and empirically.
- Determining the sensitivity to key assumptions is an important step.

Thank you!

Lecture 4

Frequentist versus Bayesian

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

A motivating example

- Suppose, you are interested in balloon shooting in a village fair.
- Say, you shoot $n = 100$ times and hit $Y = 60$ balloons.
- A reasonable model for the number of hits would be $Y \sim \text{Binomial}(n, \theta)$.
- The parameter $\theta \in [0, 1]$ is the true probability of a successful shooting.
- What's your best guess about θ?
- What's the probability that θ is greater than a half?

Frequentist approach

- A **frequentist procedure** quantifies uncertainty in terms of *repeating the process that generated the data many times*.
- The parameters θ are fixed and unknown.
- The sample (data) Y is random.
- A frequentest would **never** say Prob($\theta > 0.5$) $= 0.60$ because θ is not a random variable.
- All probability statements should be made about randomness in the data.

Frequentist approach

- A **frequentist** procedure quantifies uncertainty in terms of *repeating the process that generated the data many times*.
- For an illustration see `https://www.rossmanchance.com/applets/2021/confsim/ConfSim.html`
- A **statistic** $\hat{\theta}$ is a summary of the sample.
- For example, the sample proportion $\hat{\theta} = Y/n$ is a statistic, and it is an **estimator** of the true proportion θ.
- The distribution of $\hat{\theta}$ that arises from repeating the process that generated the data many times is its **sampling distribution.**
- A frequentist would **never** say "the distribution of θ is Beta(6,4)".

Frequentist approach

- A **frequentist** procedure quantifies uncertainty in terms of *repeating the process that generated the data many times*.

- A **95% confidence interval** (l, u) is

  If we repeat the experiment 100,000 times, then, on an average, the true unknown θ falls within (l, u) for 95,000 of the times.

- A frequentist would **never** say "the probability that the true proportion is in the interval (0.4, 0.8) is 0.95".

Frequentist approach

- A **frequentist** procedure quantifies uncertainty in terms of *repeating the process that generated the data many times*.
- A common approach for testing a hypothesis is to reject the null if a test statistic exceeds a threshold.
- For example, we might reject $\mathcal{H}_0 : \theta \leq 0.5$ in favor of the alternative $\mathcal{H}_1 : \theta > 0.5$ if $\hat{\theta} = Y/n > T$.
- A **p-value** is

  Under the null hypothesis, the probability that the test statistic $\hat{\theta} = Y/n$ exceeds the observed value, i.e., 0.6.

- A frequentist would **never** say "the probability that the null hypothesis is true is 0.03".

Frequentist approach

There is currently an intense discussion of the merits of the p-value in the scientific community:

- https://www.nature.com/articles/506150a
- https://fivethirtyeight.com/features/not-even-scientists-can-easily-explain-p-values/
- https://fivethirtyeight.com/features/science-isnt-broken/
- https://www.tandfonline.com/doi/pdf/10.1080/01973533.2015.1012991

How about a frequentist answer these questions?

Before we start:

1. What's your best guess about θ?
2. What's the probability that θ is greater than a half?

After we have observed some n trials and sample proportion $\hat{\theta} = Y/n$:

1. What's your best guess about θ?
2. What's the probability that θ is greater than a half?

The Bayesian approach

- Bayesians also view θ as fixed and unknown.
- However, we express our uncertainty about θ using probability distributions.
- The distribution before observing the data is the **prior distribution**.
- Example: Prob$(\theta > 0.5) = 0.6$.
- Probability statements like this are more intuitive.
- This is subjective in that people may have different priors (we will also discuss objective Bayes).

The Bayesian approach

- Our uncertainty about θ is changed (hopefully reduced) after observing the data.
- The **Likelihood function** is the distribution of the observed data given the parameters.
- This is the same likelihood function used in a maximum likelihood analysis.
- Therefore, when the prior information is weak, Bayesian and maximum likelihood estimates are similar.
- Even in this case, the interpretations are different.

The Bayesian approach

- The uncertainty distribution of θ after observing the data is the **posterior distribution.**
- **Bayes theorem** provides the rule for updating the prior.

$$p(\theta|Y) = \frac{f(Y|\theta)\pi(\theta)}{m(Y)}.$$

- In words: Posterior $\propto$ Likelihood$\cdot$prior.
- A key difference between Bayesian and frequentist statistics is that all inference is conditional on the single data set we observed Y.

Back to the example

- Say we observed $Y = 60$ successes in $n = 100$ trials.
- The parameter $\theta \in [0, 1]$ is the true probability of success.
- In most cases we would select a prior that puts probability on all values between 0 and 1.
- If we have no relevant prior information we might use the prior

$$\theta \sim \text{Uniform}(0, 1),$$

  so that all values between 0 and 1 are equally likely.
- This is an example of an **non-informative prior**.

Posterior distribution

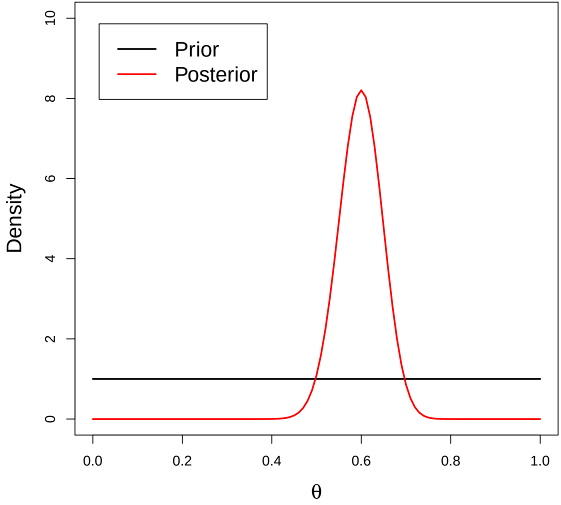
- The likelihood is $Y|\theta \sim \text{Binomial}(n, \theta)$.
- The uniform prior is $\theta \sim \text{Uniform}(0, 1)$.
- So, the posterior is

$$p(\theta|Y) \propto f(Y|\theta)\pi(\theta) \propto \theta^{Y}(1-\theta)^{n-Y} \times \mathbb{1}(\theta \in [0,1]).$$

- Then it turns out the posterior is

$$\theta|Y \sim \text{Beta}(Y+1, n-Y+1).$$

Bayesian learning: $Y = 60$ and $n = 100$



Beta prior

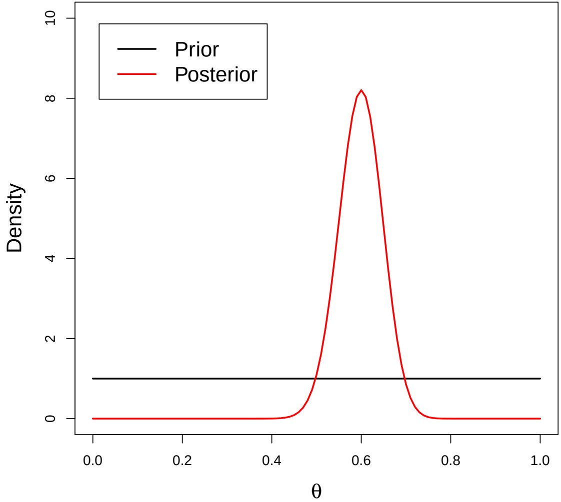
- The uniform prior represents prior ignorance.
- To encode prior information we need a more general prior.
- The beta distribution is a common prior for a parameter that is bounded between 0 and 1.
- If $\theta \sim \text{Beta}(a, b)$ then the posterior is

$$p(\theta|Y) \propto f(Y|\theta)\pi(\theta) \propto \theta^{Y}(1-\theta)^{n-Y} \times \theta^{a-1}(1-\theta)^{b-1}\mathbb{1}(\theta \in [0,1]).$$

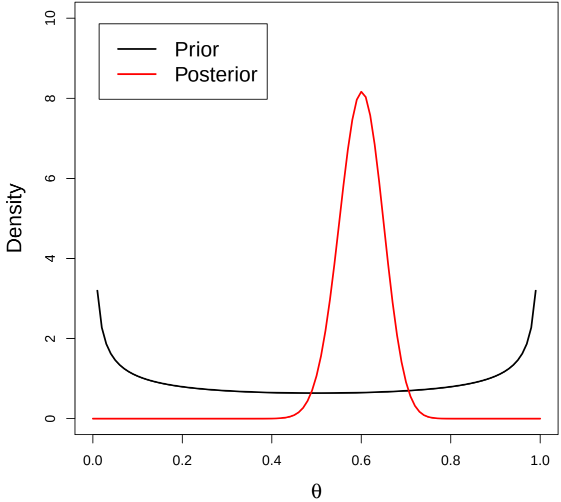
- Then it turns out the posterior is

$$\theta|Y \sim \text{Beta}(Y+a, n-Y+b).$$

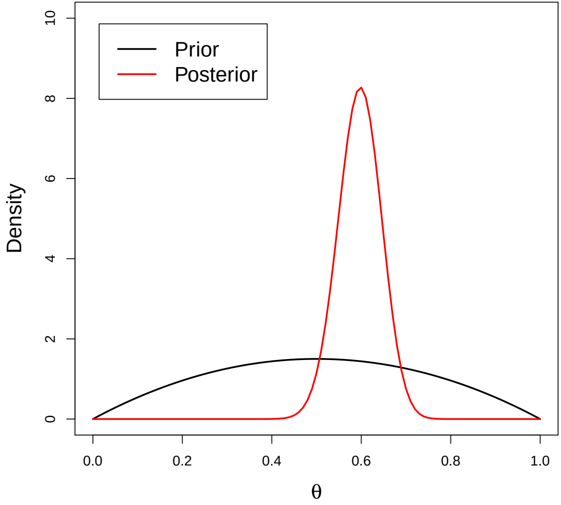
Prior 1: $\theta \sim \text{Beta}(1, 1)$



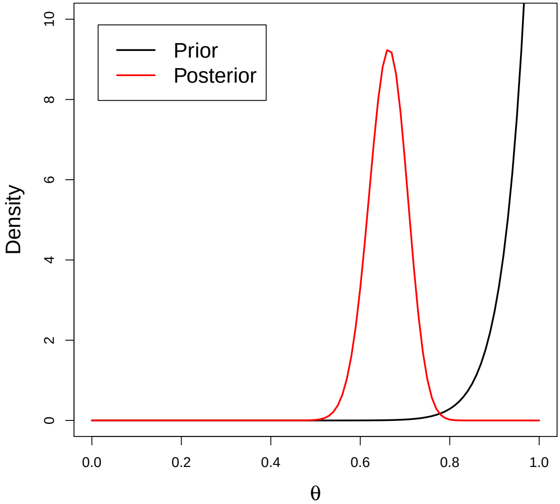
Prior 2: $\theta \sim \text{Beta}(0.5, 0.5)$



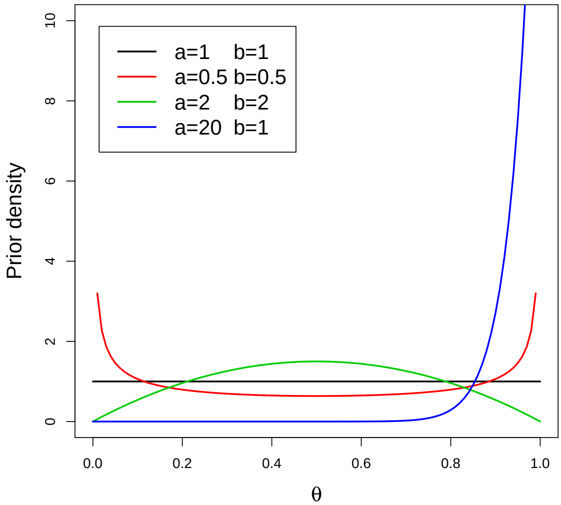
Prior 3: $\theta \sim \text{Beta}(2, 2)$



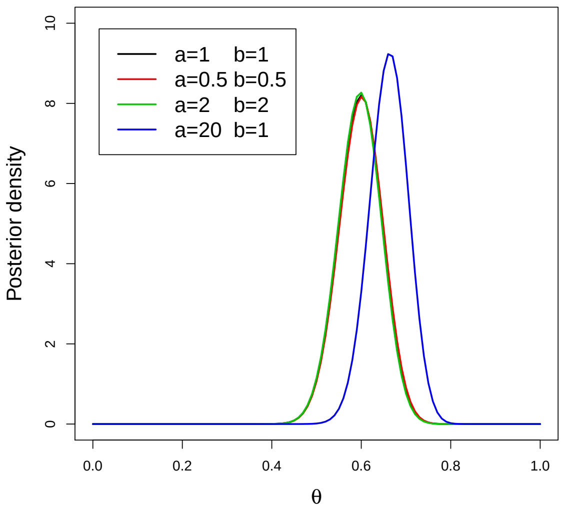
Prior 4: $\theta \sim \text{Beta}(20, 1)$



Plot of different beta priors



Plots of the corresponding posteriors



Senstivity to the prior

		Prior			Posterior		
a	b	Mean	SD	P$>$0.5	Mean	SD	P$>$0.5
1	1	0.50	0.29	0.50	0.60	0.05	0.98
0.5	0.5	0.50	0.50	0.50	0.60	0.05	0.98
2	2	0.50	0.22	0.50	0.60	0.05	0.98
20	1	0.95	0.05	1.00	0.66	0.04	1.00

Summary

- The first three priors give essentially the same results.
- Say the objective is to test $\mathcal{H}_o : \theta \leq 0.5$ versus $\mathcal{H}_A : \theta > 0.5$.
- In these three cases, we can say that, after observing the data, the probability of the null is only 0.02, and the alternative is 49 times more likely than the null.
- The final prior strongly favored large θ and gave different results.

Advantages of the Bayesian approach

- Bayesian concepts (posterior prob of the null) are arguably easier to interpret than frequentist ideas (p-value).
- We can incorporate scientific knowledge via the prior.
- Excellent at quantifying uncertainty in complex problems (e.g., missing data, correlation, etc.).
- In some cases (mainly for complex problems), the computing is easier.
- Provides a framework to incorporate data/information from multiple sources.

Disadvantages of Bayesian methods

- Less common/familiar.
- Picking a prior is subjective (we will study objective priors).
- Procedures with frequentist properties are desirable (we will study the frequentist properties of Bayesian methods).
- Computing can be slow or unstable for hard problems.
- Nonparametric methods are challenging.

Thank you!

Lecture 5

Univariate Posterior Summarization

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Bayes' Theorem

- In Bayesian statistics, we select the prior, $\pi(\theta)$, and the likelihood, $f(y|\theta)$.
- Based on these two pieces of information, our main goal is to compute the posterior $p(\theta|y)$.
- Bayes' theorem is the mathematical formula to convert the likelihood and prior to the posterior.
- Bayes theorem:
$$p(\theta|y) = \frac{f(y|\theta)\pi(\theta)}{m(y)}.$$
- This holds for discrete (PMF) and continuous (PDF) cases.
- As in the formula for a conditional distribution, $m(y)$ is just the normalizing constant required so that $\int p(\theta|y)d\theta = 1$.

Examples of a univariate posterior

- Here, we have only a single unknown parameter, θ.
- For example, we say the model is

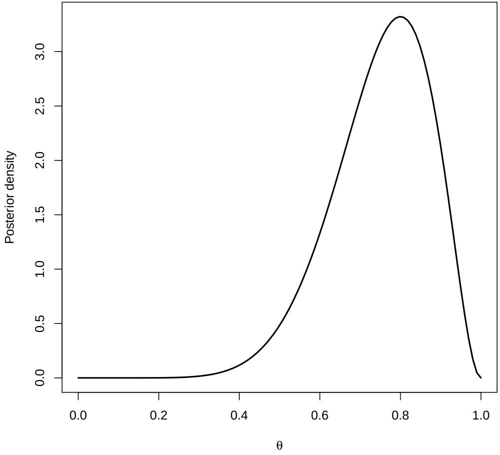
  Likelihood: $Y|\theta \sim \text{Binomial}(N, \theta)$,
  Prior: $\theta \sim \text{Beta}(a, b)$.

- We saw that the posterior is then

$$\theta|Y \sim \text{Beta}(Y + a, N - Y + b).$$

- The posterior is a *distribution* that can be plotted as on the next slide.

Summarizing a univariate posterior



In this beta/binomial example, $Y = 8$, $N = 10$ and $a = b = 1$.

Summarizing a univariate posterior

- A plot of the posterior tells the whole story.
- However, to be more concise we typically use a few numerical summaries of the distribution.
- This is particularly important when there are many parameters.
- The posterior can be summarized like any other distribution, by say the mean, variance, skewness, etc.

Point estimators

- A **point estimator** is a one number summary used to estimate the unknown parameter.
- For example, we might use the posterior mean (or median) as the "best guess" of θ.
- The posterior mean is

$$\hat{\theta} = \mathrm{E}(\theta|Y) = \int \theta p(\theta|Y) d\theta.$$

- For the Beta/Binomial example, we have $\hat{\theta} = \frac{Y+a}{n+a+b}$.
- This is an alternative to the sample proportion $\hat{\theta} = Y/n$.

MAP estimator

- The posterior mode is the called the maximum a posteriori (MAP) estimator.
- The MAP estimator is
$$\hat{\theta} = \arg\max_{\theta} p(\theta|Y) = \arg\max_{\theta} \{\log[f(Y|\theta)] + \log[\pi(\theta)]\}.$$
- If the prior is uniform (i.e., flat), the MAP is the MLE.
- The MAP is easier to compute than the posterior mean.

MAP estimator

- Let $Y|\theta \sim \text{Binomial}(n, \theta)$ and $\pi(\theta) = \mathbb{1}(\theta \in [0, 1])$. Find the MAP estimator of θ.

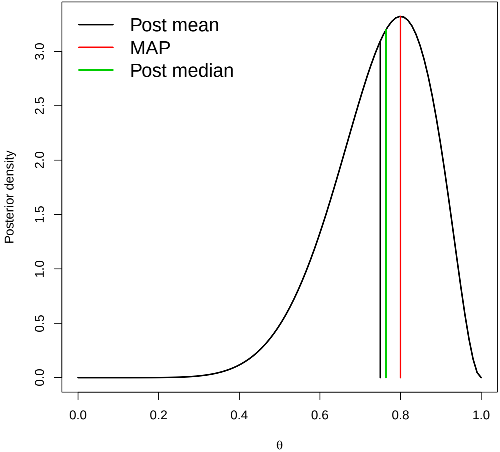
MAP estimator

- Let $Y|\theta \sim \text{Binomial}(n, \theta)$ and $\pi(\theta) = \mathbb{1}(\theta \in [0,1])$. Find the MAP estimator of θ.

- The MAP estimator is

$$
\begin{aligned}
\hat{\theta} = \arg\max_{\theta} p(\theta|Y) &= \arg\max_{\theta} \{\log[f(Y|\theta)] + \log[\pi(\theta)]\} \\
&= \arg\max_{\theta \in [0,1]} \{\log[f(Y|\theta)]\} \\
&= Y/n.
\end{aligned}
$$

MAP estimator

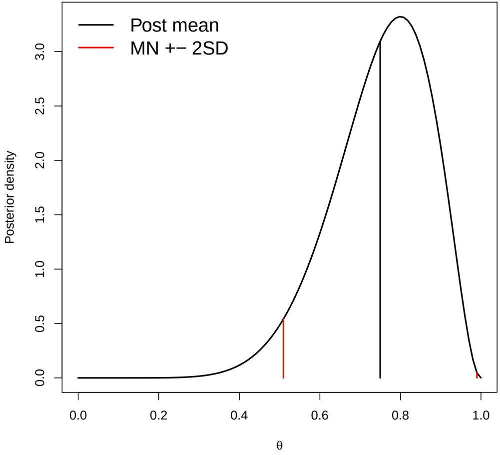


In this beta/binomial example $Y = 8$, $N = 10$ and $a = b = 1$.

Uncertainty measures

- Sometimes a point estimate is sufficient, but more often we need to quantify uncertainty.
- The **posterior standard deviation** is one measure of uncertainty.
- If the posterior is approximately normal, then the mean plus/minus two standard deviation units captures 95% of the posterior probability.
- The posterior standard deviation is analogous to but fundamentally different than the frequentist **standard error**.
- The standard error is the standard deviation of $\hat{\theta}$'s sampling distribution.

Uncertainty measures



In this beta/binomial example $Y = 8$, $N = 10$ and $a = b = 1$.

Credible intervals

- In addition to standard error, uncertainty can be quantified using a **credible interval**.
- The interval (l, u) is a $100(1-\alpha)\%$ posterior credible interval/set if
$$\text{Prob}(l < \theta < u | Y) = 1 - \alpha.$$
- Interpretation of a 95% credible interval: “given the data and prior, I am 95% certain that θ is between l and u”.
- This is analogous but different than a **confidence interval**.

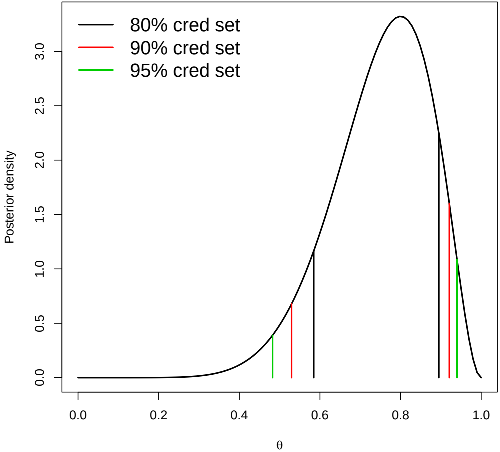
Credible intervals

- Credible sets are not unique.
- Let q_τ be the τ quantile of the posterior so that

$$\text{Prob}(\theta < q_\tau | Y) = \tau.$$

- Then $(q_{0.00}, q_{0.95})$, $(q_{0.01}, q_{0.96})$, etc. are all valid 95% credible sets.
- The **equal-tailed** interval is $(q_{\alpha/2}, q_{1-\alpha/2})$.
- The **highest posterior density** interval searches for the smallest interval that contains the proper probability.

Credible intervals



In this beta/binomial example $Y = 8$, $N = 10$ and $a = b = 1$.

Hypothesis testing

- **Hypothesis tests** are conducted by simply computing the posterior probability of each hypothesis.
- Say the null hypothesis is $\mathcal{H}_0 : \theta \leq 0.5$ and the alternative is $\mathcal{H}_1 : \theta > 0.5$.
- The posterior probability of the null hypothesis is

$$\text{Prob}(\theta < 0.5|Y) = \int_0^{0.5} p(\theta|Y)d\theta.$$

- We reject the null if its probability is small.
- In a Bayesian analysis we can say "Given the data and prior, the probability that the null hypothesis is true is 0.02".
- This is analogous to but different than the **p-value**.

Hypothesis testing

```
> # Data
> Y <- 8; n <- 10
> # The posterior is theta|Y~Beta(A,B)
> A <- Y+1; B <- n-Y+1
> # Posterior mean
> A/(A+B)
[1] 0.75
> # Posterior standard deviation
> sqrt(A*B/((A+B)^2*(A+B+1)))
[1] 0.1200961
> # Posterior 95% credible interval
> qbeta(c(0.025,0.975),A,B)
[1] 0.4822441 0.9397823
> # Posterior probability that theta<0.5
> pbeta(0.5,A,B)
[1] 0.03271484
```

Monte Carlo approximations

- **Monte Carlo (MC) sampling** is a useful tool for summarizing a posterior.
- For univariate cases, is it not particularly useful. But, in harder problems, it is the best approach available.
- In MC sampling, we draw S samples from the posterior:

$$\theta^{(1)}, ..., \theta^{(S)} \sim p(\theta|Y),$$

  and use these samples to approximate the posterior.
- For example, the posterior mean and variance are approximated by the sample mean and variance of the $\theta^{(s)}, s = 1, \ldots, S$.

Monte Carlo approximations

- MC sampling facilitates studying **transformations** of parameters.
- For example, the odds corresponding to θ are $\gamma = \theta/(1 - \theta)$.
- How to approximate the posterior mean and variance of γ?
- We simply transform each draw to the odds

$$\gamma^{(1)} = \frac{\theta^{(1)}}{1 - \theta^{(1)}}, \ldots, \gamma^{(S)} = \frac{\theta^{(S)}}{1 - \theta^{(S)}},$$

  and use these draws to approximate γ's posterior.

Monte Carlo approximations

```
> # Data
> Y <- 8; n <- 10
> # The posterior is theta|Y~Beta(A,B)
> A <- Y+1; B <- n-Y+1
> # MC sampling
> theta <- rbeta(100000,A,B)
> # Approximate the posterior mean and SD
> mean(theta);sd(theta)
[1] 0.749792
[1] 0.1201799
> # Transform to odds
> gamma <- theta/(1-theta)
> # Approximate the posterior mean and SD
> mean(gamma);sd(gamma)
[1] 4.483378
[1] 4.720541
```

Thank you!

Lecture 6

Multivariate Posterior Summarization

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Summarizing multivariate posteriors

- A univariate posterior can be fully understood by a simple plot.
- However, when there are many parameters, this is impossible.
- Say, $\boldsymbol{\theta} = (\theta_1, ..., \theta_p)$.
- Ideally, we reduce to the univariate marginal posteriors

$$p(\theta_1|Y) = \int ... \int p(\theta_1, ..., \theta_p|Y) d\theta_2, ..., d\theta_p.$$

- The same ideas we used for univariate models are now applicable.
- However, computing these integrals is often challenging.

Bayesian one-sample *t*-test

- In this section, we will study the one-sample *t*-test in depth.
- Likelihood: $Y_i|\mu,\sigma \overset{IID}{\sim} \mathrm{N}(\mu,\sigma^2)$, $i = 1, ..., n$.
- Priors: $\pi(\mu,\sigma^2) = \pi_1(\mu)\pi_2(\sigma^2)$, $\mu \sim \mathrm{N}(\mu_0,\sigma_0^2)$ and $\sigma^2 \sim$ Inverse-Gamma(a, b).
- The joint (bivariate PDF) of (μ,σ^2) is proportional to

$$\left\{\sigma^{-n}\exp\left[-\frac{\sum_{i=1}^{n}(Y_i-\mu)^2}{2\sigma^2}\right]\right\}\times\exp\left[-\frac{(\mu-\mu_0)^2}{2\sigma_0^2}\right]\times(\sigma^2)^{-a-1}\exp\left[-\frac{b}{\sigma^2}\right].$$

- How to summarize this complicated function?

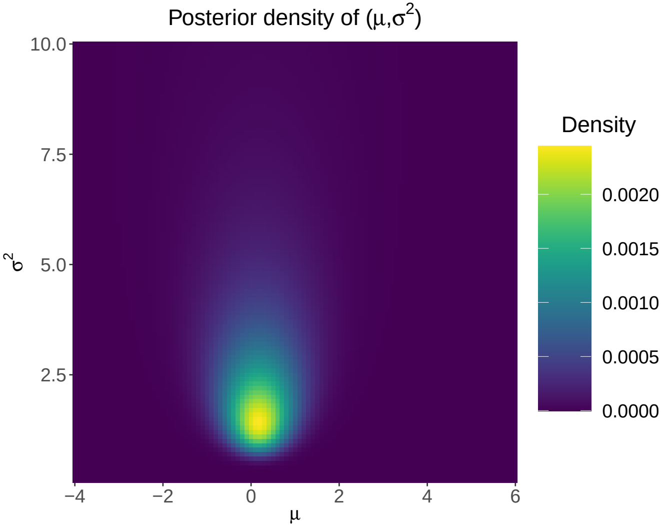
Plotting the posterior on a grid

- For models with only a few parameters, we could simply plot the posterior on a grid.
- That is, suppose, we compute $p(\mu, \sigma^2 | Y_1, ..., Y_n)$ for all combinations of m values of μ and m values of σ^2.
- The number of grid points is m^p where p is the number of parameters in the model.
- The posterior is plotted on the next slide for

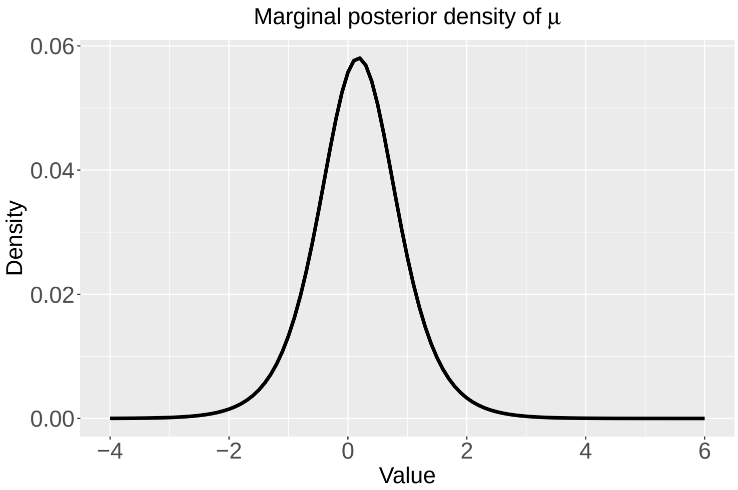
$$Y_1 = 2.68, Y_2 = 1.18, Y_3 = -0.97, Y_4 = -0.98, Y_5 = -1.03,$$

and priors $\mu|\sigma^2 \sim \text{N}(0, 100\sigma^2)$ and $\sigma^2 \sim \text{Inverse-Gamma}(0.01, 0.01)$.

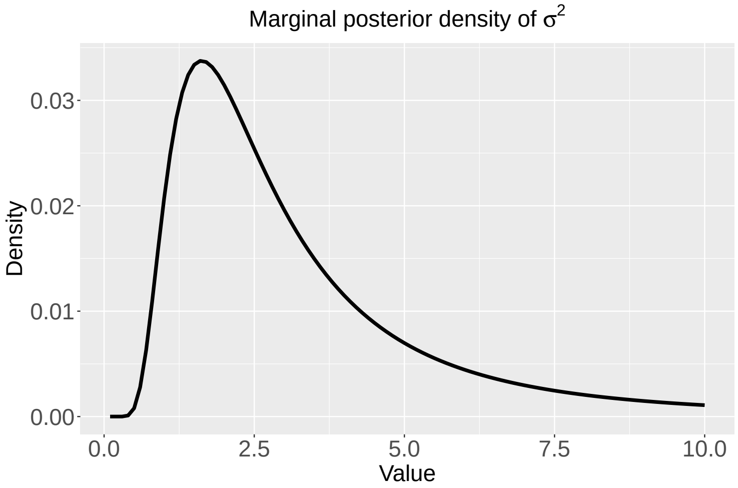
Bivariate posterior



Marginal posterior



Marginal posterior



Summarizing the results in a table

- Typically, we are interested in the marginal posterior

$$f(\mu|\mathbf{Y}) = \int_0^\infty p(\mu, \sigma^2|\mathbf{Y}) d\sigma^2,$$

  where $\mathbf{Y} = (Y_1, ..., Y_n)$.

- This accounts for our uncertainty about σ^2.
- We could also report the marginal posterior of σ^2.
- Results are usually given in a table with marginal mean, SD, and 95% interval for all parameters of interest.
- The marginal posteriors can be computed using numerical integration.

Summarizing the results in a table

	Posterior mean	Posterior SD	95% credible set
μ	0.17	1.31	(-2.49, 2.83)
σ	2.57	1.37	(1.10, 6.54)

Frequentist analysis of a normal mean

- In frequentist statistics, the estimate of the mean is $\bar{Y}$.
- If σ is known, the 95% interval is

$$\bar{Y} \pm z_{0.975} \frac{\sigma}{\sqrt{n}},$$

where z is the quantile of a normal distribution.

- If σ is unknown, the 95% interval is

$$\bar{Y} \pm t_{0.975,n-1} \frac{s}{\sqrt{n}},$$

where t is the quantile of a Student's t-distribution.

Bayesian analysis of a normal mean

- The Bayesian estimate of μ is its marginal posterior mean.
- The interval estimate is the 95% posterior credible interval.
- If σ is known, the posterior of $\mu|\mathbf{Y}$ is Gaussian (**check**) and the 95% credible interval is
$$\mathrm{E}(\mu|\mathbf{Y}) \pm z_{0.975}\mathrm{SD}(\mu|\mathbf{Y}).$$
- If σ is unknown, the marginal (over σ^2) posterior of μ is Student's t with $\nu = n + 2a$ degrees of freedom (**check**).
- Therefore, the 95% credible interval is
$$\mathrm{E}(\mu|\mathbf{Y}) \pm t_{0.975,\nu}\mathrm{SD}(\mu|\mathbf{Y}).$$

Case 1: σ is known

The posterior is

$$
\begin{aligned}
p(\mu|\mathbf{Y}) &\propto \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(Y_i-\mu)^2\right]\times\exp\left[-\frac{1}{2\sigma_0^2}(\mu-\mu_0)^2\right] \\
&\propto \exp\left[-\frac{n\mu^2}{2\sigma^2}\right]\times\exp\left[\frac{\mu\sum_{i=1}^{n}Y_i}{\sigma^2}\right]\times\exp\left[-\frac{\mu^2}{2\sigma_0^2}\right]\times\exp\left[\frac{\mu\mu_0}{\sigma_0^2}\right] \\
&= \exp\left[-\frac{1}{2}\mu^2\left(\frac{n}{\sigma^2}+\frac{1}{\sigma_0^2}\right)\right]\times\exp\left[\mu\left(\frac{\sum_{i=1}^{n}Y_i}{\sigma^2}+\frac{\mu_0}{\sigma_0^2}\right)\right].
\end{aligned}
$$

By matching the terms with those in the density of a normal distribution, we have

$$
\mu|\mathbf{Y}\sim \text{Normal}\left(\left[\frac{\sum_{i=1}^{n}Y_i}{\sigma^2}+\frac{\mu_0}{\sigma_0^2}\right]\Big/\left[\frac{n}{\sigma^2}+\frac{1}{\sigma_0^2}\right], 1\Big/\left[\frac{n}{\sigma^2}+\frac{1}{\sigma_0^2}\right]\right).
$$

Case 2: σ is unknown

The priors are: $\mu|\sigma^2 \sim \text{Normal}(\mu_0, c\sigma^2)$ and $\sigma^2 \sim \text{Inverse-Gamma}(a, b)$.

The posterior is

$$\begin{aligned} p(\mu, \sigma^2|\mathbf{Y}) &\propto (\sigma^2)^{-n/2} \exp\left[-\frac{\sum_{i=1}^n (Y_i - \mu)^2}{2\sigma^2}\right] \exp\left[-\frac{(\mu - \mu_0)^2}{2c\sigma^2}\right] (\sigma^2)^{-a-1} \exp\left[-\frac{b}{\sigma^2}\right] \\ &= (\sigma^2)^{-(a+n/2)-1} \exp\left[-\frac{1}{\sigma^2}\left\{\frac{\sum_{i=1}^n (Y_i - \mu)^2}{2} + \frac{(\mu - \mu_0)^2}{2c} + b\right\}\right]. \end{aligned}$$

The density has the same kernel as an Inverse-Gamma distribution with shape parameter $a + n/2$ and rate parameter $\frac{\sum_{i=1}^n (Y_i - \mu)^2}{2} + \frac{(\mu-\mu_0)^2}{2c} + b$. Thus, after integrating out with respect to σ^2, we have

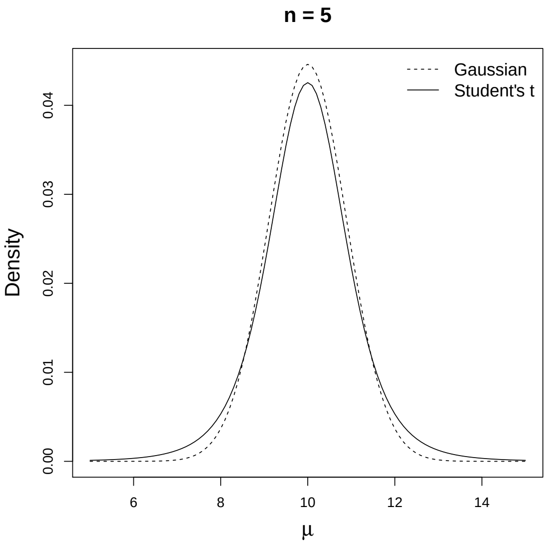
$$p(\mu|\mathbf{Y}) = \int p(\mu, \sigma^2|\mathbf{Y}) d\sigma^2 \propto \left\{\frac{\sum_{i=1}^n (Y_i - \mu)^2}{2} + \frac{(\mu - \mu_0)^2}{2c} + b\right\}^{-\frac{n+2a}{2}},$$

which is similar to the kernel of a Student's t distribution with df $n + 2a$.

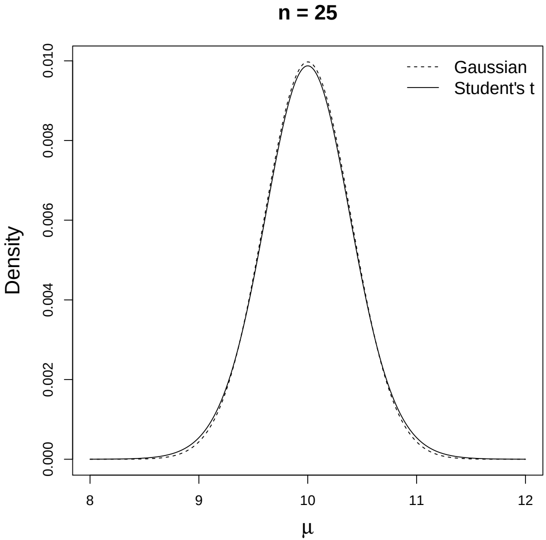
Bayesian analysis of a normal mean

- The following two slides give the posterior of μ for a data set with sample mean 10 and sample variance 4.
- The Gaussian analysis assumes $\sigma^2 = 4$ is known.
- The Student's *t* analysis integrates over uncertainty in σ^2.
- As expected, the latter interval is a bit wider.

Bayesian analysis of a normal mean



Bayesian analysis of a normal mean



Example with two parameters

- Assume $N = 10$ patients are given treatment and $N = 10$ are given control.
- Let θ_1 and θ_2 be the survival probabilities for the control and treatment groups, respectively.
- We pick priors $\theta_1, \theta_2 \stackrel{IID}{\sim} \text{Uniform}(0, 1)$.
- We observe $Y_1 = 5$ survivals and $Y_2 = 8$ survivals for control and treatment groups, respectively.
- Our goal is to determine if treatment improves survival, i.e., $\theta_2 > \theta_1$.
- The requires summarizing the bivariate posterior $p(\theta_1, \theta_2 | Y_1, Y_2)$.

Methods for dealing with multiple parameters

- We want to compute $\text{Prob}(\theta_2 > \theta_1 | Y_1, Y_2)$.
- We could do an integral or grid approximation.
- Both often fail when there are many parameters.
- We need new tools!
- Monte Carlo sampling will be a key tool.
- We'll spend a month on this in the computing section.

Comparing proportions

- The model is
  - $Y_1|\theta_1 \sim \text{Binomial}(N, \theta_1)$
  - $Y_2|\theta_2 \sim \text{Binomial}(N, \theta_2)$
  - $\theta_1, \theta_2 \overset{IID}{\sim} \text{Beta}(1, 1)$
- The full posterior is

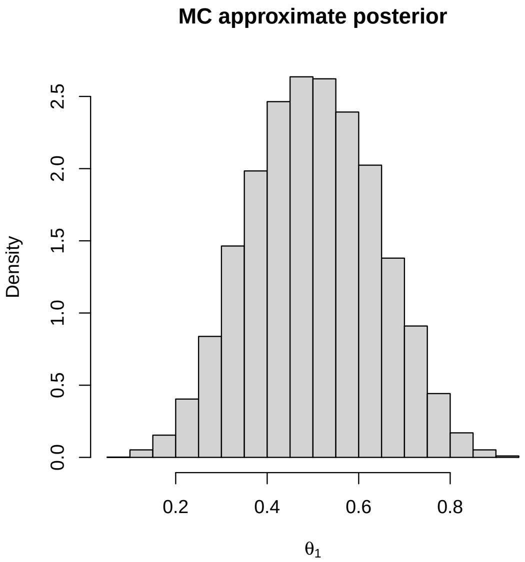
$$
\begin{aligned}
p(\theta_1, \theta_2 | Y_1, Y_2) &\propto f(Y_1, Y_2|\theta_1, \theta_2)\pi(\theta_1, \theta_2) \\
&= f(Y_1|\theta_1)f(Y_2|\theta_2)\pi(\theta_1)\pi(\theta_2) \\
&= \{f(Y_1|\theta_1)\pi(\theta_1)\}\{f(Y_2|\theta_2)\pi(\theta_2)\}. \\
&\propto p(\theta_1|Y_1)p(\theta_2|Y_2).
\end{aligned}
$$

- The posterior is $\theta_1|Y_1, Y_2 \sim Beta(Y_1 + 1, N - Y_1 + 1)$ independent of $\theta_2|Y_1, Y_2 \sim Beta(Y_2 + 1, N - Y_2 + 1)$.
- The next slide approximates $\text{Prob}(\theta_2 > \theta_1|Y_1, Y_2)$ using MC sampling.

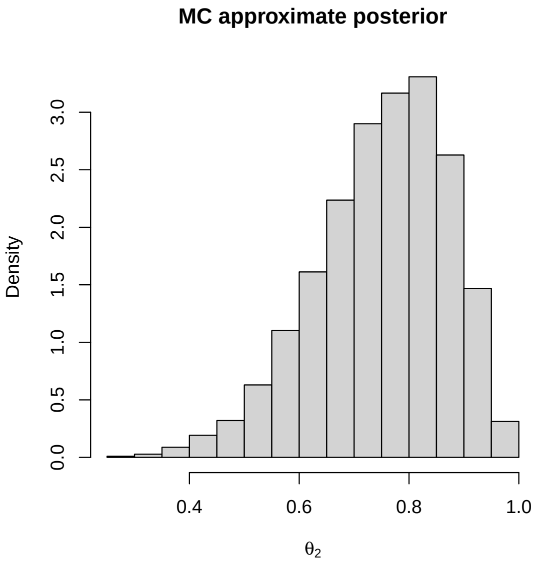
Summarizing a posterior using MC sampling

```
> N <- 10; Y1 <- 5; Y2 <- 8 # Data
> set.seed(100) # Keeps the results reproducible
> S <- 10000 # Number of MC samples
> theta1 <- rbeta(S,Y1+1,N-Y1+1); theta2 <- rbeta(S,Y2+1,N-Y2+1)
>
> hist(theta1,xlab=expression(theta[1]),
+      main="MC approximate posterior", probability = TRUE)
> (Y1+1)/(N+2)# True post mean
[1] 0.5
> mean(theta1) # MC estimate
[1] 0.5010727
> plot(theta1,theta2,xlim=c(0,1),ylim=c(0,1),
+      xlab=expression(theta[1]), ylab=expression(theta[2]))
> abline(a=0, b = 1) # adds diagonal line
> mean(theta2>theta1)
[1] 0.9103
```

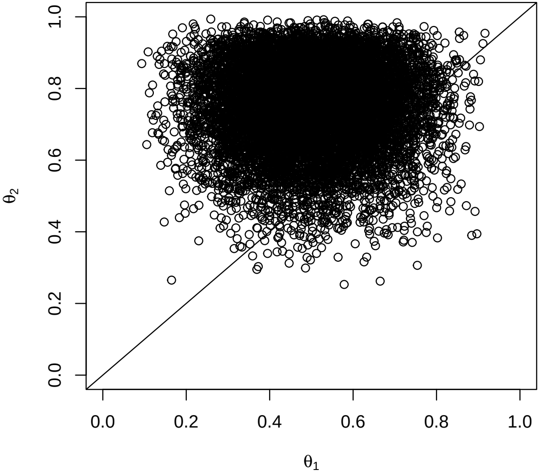
Marginal posterior, $p(\theta_1 \mid Y_1, Y_2)$



Marginal posterior, $p(\theta_2 | Y_1, Y_2)$



Joint posterior, $p(\theta_1, \theta_2 | Y_1, Y_2)$

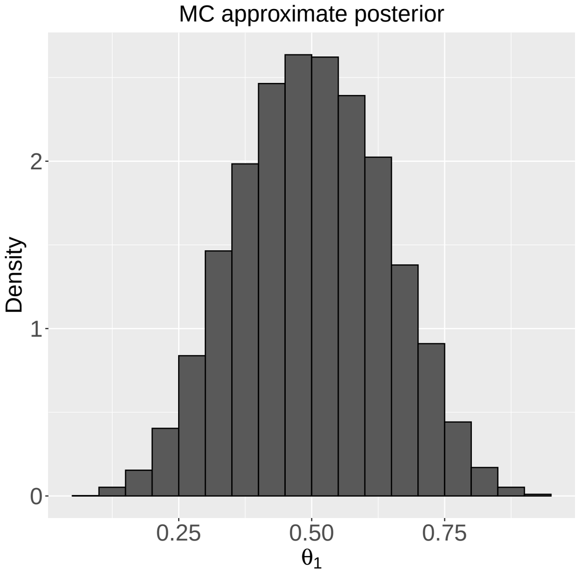


Supplementary slides for `ggplot`

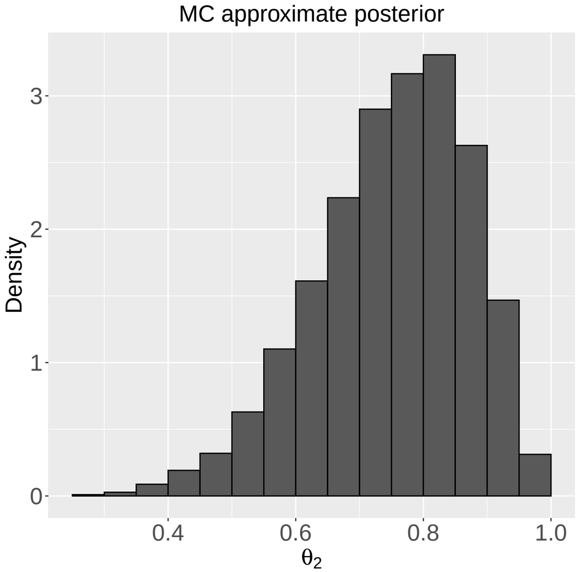
ggplot codes

```
p <- ggplot() +
     geom_histogram(aes(x= theta1, y=..density..),
                   breaks = hist(theta1, plot = F)$breaks,
                   colour="black") +
  ggtitle("MC approximate posterior") +
  theme(plot.title = element_text(hjust = 0.5)) +
  xlab(expression(theta[1])) + ylab("Density") +
  theme(axis.text=element_text(size=20),
        axis.title=element_text(size=20),
        plot.title = element_text(size=20))
ggsave("ggplot_pos_dist_theta1.pdf", p, height = 7, width = 7)
```

Marginal posterior, $p(\theta_1 \,|\, Y_1, Y_2)$



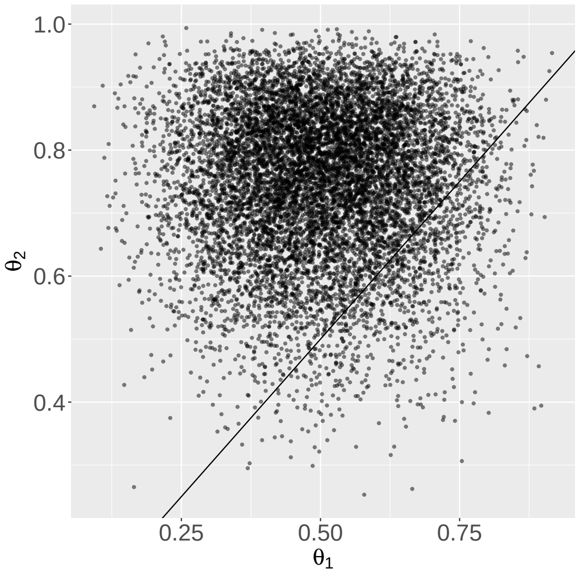
Marginal posterior, $p(\theta_2 | Y_1, Y_2)$



ggplot codes

```
p <- ggplot() + geom_point(aes(x = theta1, y = theta2),
                            size = 1, alpha = 0.5) +
  xlab(expression(theta[1])) + ylab(expression(theta[2])) +
  theme(axis.text=element_text(size=20),
        axis.title=element_text(size=20)) +
  geom_abline(slope = 1)
ggsave(p, filename = "ggplot_pos_dist_theta1_theta2.pdf",
       width = 7, height = 7)
```

Joint posterior, $p(\theta_1, \theta_2 | Y_1, Y_2)$



Thank you!

Lecture 7

Posterior Predictive Distribution

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Bayesian prediction

- Often the objective is to predict a future event.
- Example: Last spring we planted $n = 10$ seedlings and $Y = 2$ survived the winter. If we plant $n = 10$ plants again this year, what is the probability at least one will survive the winter?
- Let Y^* be the predicted value and θ be the true survival probability.
- If the parameters were known, then we would predict

$$Y^*|\theta \sim \text{Binomial}(10, \theta),$$

  and thus, $\text{Prob}(Y > 0) = 1 - (1 - \theta)^{10}$.
- Of course, if we knew the parameters, we would be doing probability and not statistics.

Two types of uncertainty

- Sampling: Even if we knew θ, we could not predict Y exactly because there is inherent randomness in which plants survive. This is quantified using the likelihood distribution, $f(Y|\theta)$.

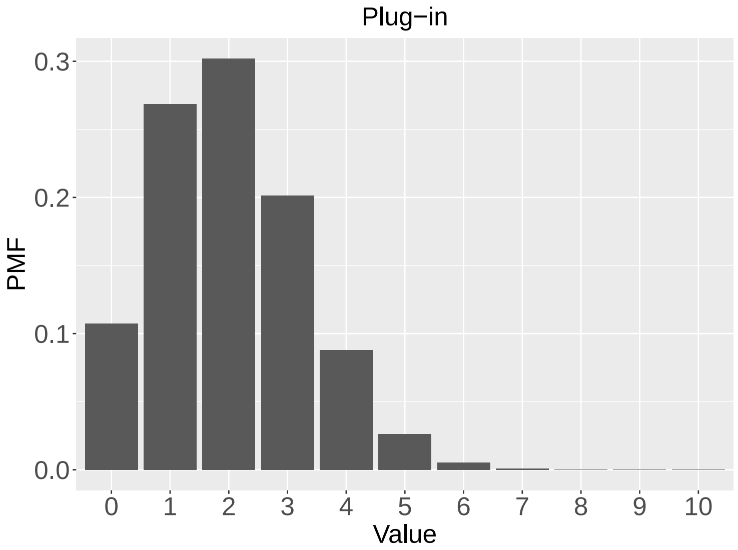
- Parametric: We can never know θ exactly. This uncertainty is quantified by its prior and posterior distributions.

- How to resolve each type of uncertainty?

Plug-in approach

- One approach to prediction is the "plug-in" approach.
- That is, if $\hat{\theta}$ is an estimate, then use prediction distribution
$$Y^* \sim f(Y|\hat{\theta}).$$
- Example: Here $\hat{\theta} = 2/10$ and then we calculate
$$\text{Prob}(Y > 0) = 1 - (1 - 0.2)^{10}.$$
- If $\hat{\theta}$ has small uncertainty, then this is fine.
- Otherwise, this underestimates uncertainty in Y^*.

Plug-in approach



The PPD

- For the sake of prediction, the parameters are not of interest.
- They are vehicles by which the data inform about the predictive model.
- The **Posterior Predictive Distribution** (PPD) averages over their posterior uncertainty
$$f(Y^*|Y) = \int f(Y^*|\theta)p(\theta|Y)d\theta.$$
- This properly accounts for parametric uncertainty.
- The input is data, and the output is a prediction distribution.

An example of PPD

- Suppose, our model is

  Likelihood: $Y|\theta \sim \text{Binomial}(N, \theta)$,
  Prior: $\theta \sim \text{Beta}(a, b)$.

- We saw that the posterior is then

$$\theta|Y \sim \text{Beta}(Y + a, N - Y + b).$$

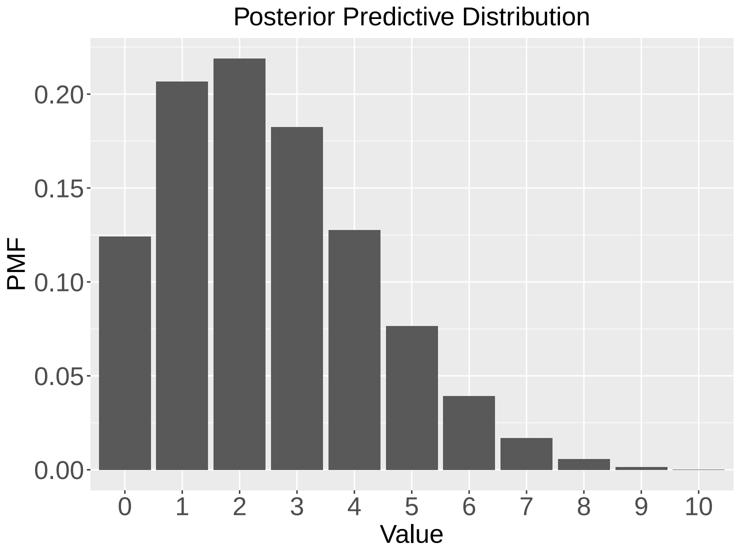
- The PPD is

$$f(Y^*|Y) = \int f(Y^*|\theta)p(\theta|Y)d\theta$$

$$= \int \binom{N}{Y^*}\theta^{Y^*}(1-\theta)^{N-Y^*}\frac{\Gamma(N+a+b)}{\Gamma(Y+a)\Gamma(N-Y+b)}\theta^{Y+a-1}(1-\theta)^{N-Y+b-1}d\theta$$

$$= \binom{N}{Y^*}\frac{\Gamma(N+a+b)}{\Gamma(Y+a)\Gamma(N-Y+b)}\frac{\Gamma(Y+Y^*+a)\Gamma(2N-Y-Y^*+b)}{\Gamma(2N+a+b)}.$$

PPD


Posterior Predictive Distribution
PMF
0.20
0.15
0.10
0.05
0.00
0
1
2
3
4
5
6
7
8
9
10
Value

Computing the PPD

- Monte Carlo sampling approximates the PPD.
- Say $\theta^{(1)}, ..., \theta^{(S)}$ are samples from the posterior.
- If we make a sample for Y^* for each $\theta^{(s)}$,

$$Y^{*(s)} \sim f(Y|\theta^{(s)}),$$

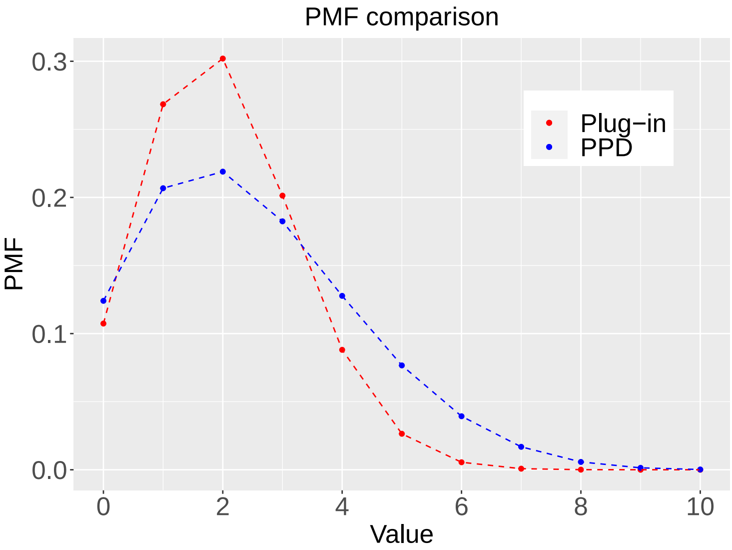
  then the $Y^{*(s)}$ are samples from the PPD.
- The posterior predictive mean is approximated by the sample mean of the $Y^{*(s)}$.
- The probability that $Y^* > 0$ is approximated by the sample proportion of the $Y^{*(s)}$ that are non-zero.

Bayesian prediction

```
> # Data
> Y <- 2; n <- 10

> # The posterior is theta|Y~Beta(A,B)
> A <- Y+1; B <- n-Y+1
>
> # Plug in estimate of P(Ystar>0)
> 1-dbinom(0,10,.2)
[1] 0.8926258
>
> # Approximate the PPD using MC sampling
> theta <- rbeta(100000,A,B)
> Ystar <- rbinom(100000,10,theta)
> mean(Ystar>0)
[1] 0.87454
```

Plug-in versus PPD

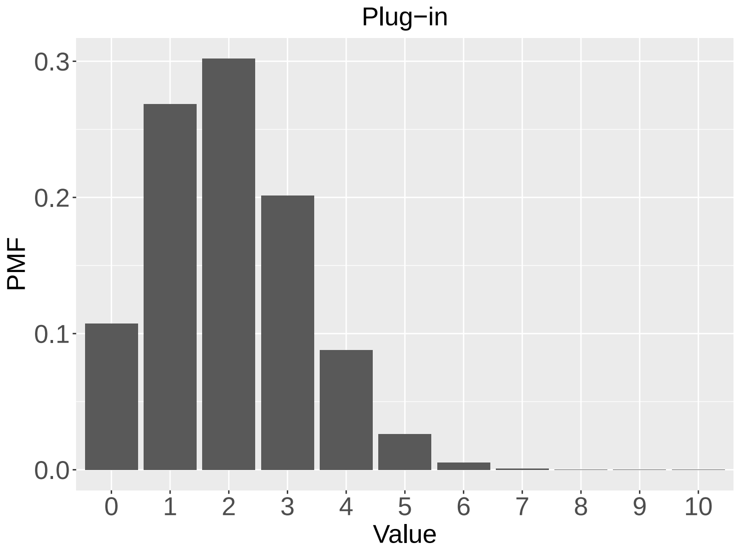


Another example

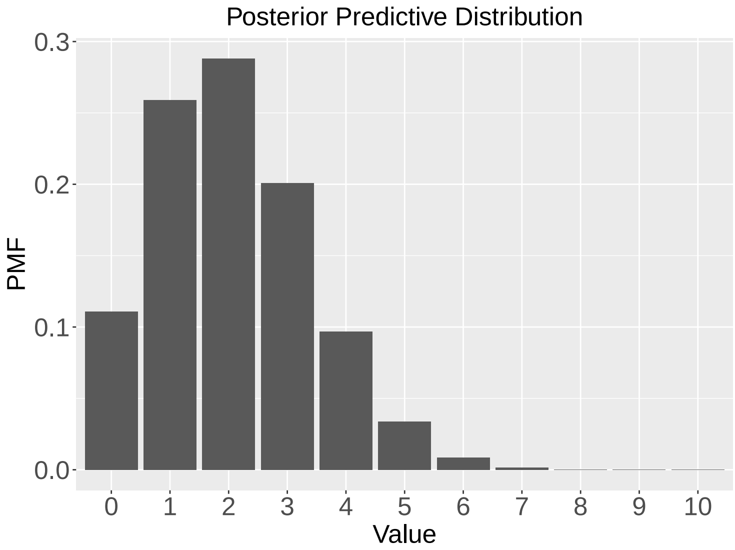
- Example: Last spring we planted $n = 100$ seedlings and $Y = 20$ survived the winter. If we plant $n = 10$ plants again this year, what is the probability at least one will survive the winter?

- We again compare the plug-in and the PPD approaches.

Plug-in approach

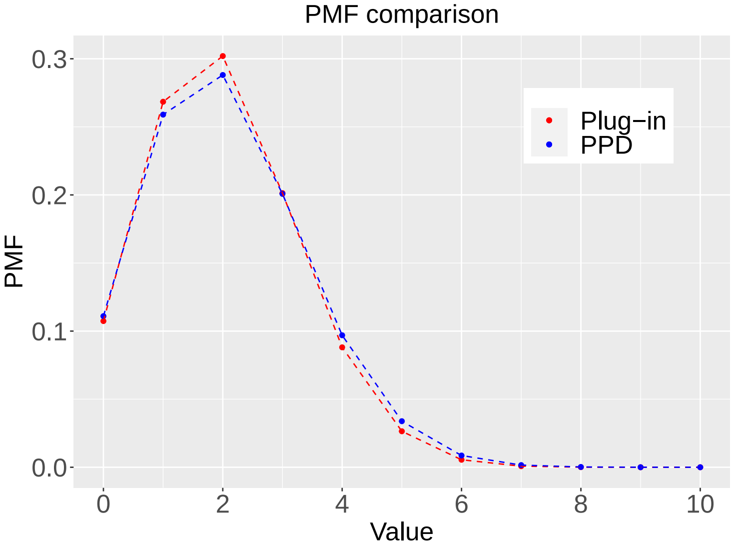


PPD



Posterior Predictive Distribution
0.3
0.2
0.1
0.0
PMF
0
1
2
3
4
5
6
7
8
9
10
Value

Plug-in versus PPD



Thank you!

1. If X has support $X \in S = [1, \infty)$, find the constant c (as a function of θ) that makes $f(x) = ce^{-x/\theta}$ a valid PDF.

A: $\int_1^\infty f(x)\,dx = 1 \Rightarrow \int_1^\infty c e^{-x/\theta}\,dx = 1 \Rightarrow c.\left[\frac{e^{-x/\theta}}{-1/\theta}\right]_1^\infty = 1$

$\Rightarrow c.\left[\frac{e^{-\infty/\theta}}{-1/\theta} - \frac{e^{-1/\theta}}{-1/\theta}\right] = 1 \Rightarrow c.\frac{e^{-1/\theta}}{1/\theta} = 1$

$\Rightarrow c.\theta e^{-1/\theta} = 1 \Rightarrow c = \theta^{-1} e^{1/\theta}$.

2. Assume that $X \sim U(a,b)$ so the support is $S = [a,b]$ and the PDF is $f(x) = 1/(b-a)$ for any $x \in S$.

(a) prove that this is a valid PDF.

A: $f(x) = \frac{1}{b-a}, x \in [a,b]$ Here $b > a$ (otherwise X and S are invalid)

So, $f(x) > 0 \ \forall x \in S$ and $f(x) = 0 \ \forall x \notin S$

overall, $f(x) \geq 0$.

Now, $\int_S f(x)\,dx = \int_S \frac{1}{b-a}\,dx = \int_a^b \frac{1}{b-a}\,dx = \frac{b-a}{b-a} = 1$.

So, $f(x)$ is a valid PDF.

(b) Derive the mean and variance of X.

A: $E(X) = \int_a^b x \frac{1}{b-a}\,dx = \frac{1}{b-a}\int_a^b x\,dx = \frac{1}{b-a}\left[\frac{x^2}{2}\right]_a^b$

$= \frac{1}{b-a}\frac{b^2-a^2}{2} = \frac{a+b}{2}$.

$V(X) = E(X^2) - E(X)^2$

$E(X^2) = \int_a^b x^2 f(x)\,dx = \int_a^b x^2 \frac{1}{b-a}\,dx = \frac{1}{b-a}\int_a^b x^2\,dx$

$= \frac{1}{b-a}\left.\frac{x^3}{3}\right|_a^b = \frac{1}{b-a}\cdot\frac{b^3-a^3}{3} = \frac{\cancel{(b-a)}(b^2+ab+a^2)}{3\cancel{(b-a)}}$

$V(X) = \frac{b^2+ab+a^2}{3} - \left(\frac{a+b}{2}\right)^2 = \frac{b^2+ab+a^2}{3} - \frac{a^2+2ab+b^2}{4}$

$= \frac{1}{12}\left[4b^2+4ab+4a^2-3a^2-6ab-3b^2\right] = \frac{1}{12}\left[a^2-2ab+b^2\right] = \frac{1}{12}(a-b)^2$.

3. Expert Knowledge dictates that a parameter must be positive and that is prior distribution should have the mean 5 and Variance 3. Find a prior distribution that Satisfies these Constraints.

A: This problem does not have a Unique Solution, rather just an example needs to be provided.

Because, it is positive, one good choice would be a gamma distribution.

Now, $\pi(\theta) \equiv \text{Gamma}(a, b)$

$E(\theta) = \frac{a}{b}$, $V(\theta) = \frac{a}{b^2}$ [Check Lecture 1]

So, $\frac{a}{b} = 5$, $\frac{a}{b^2} = 3 \Rightarrow \dfrac{\frac{a}{b}}{\frac{a}{b^2}} = \frac{5}{3} \Rightarrow b = \frac{5}{3}$.

So, $a = 5b = 5 \times \frac{5}{3} = \frac{25}{3}$. $\boxed{a = \frac{25}{3}, b = \frac{5}{3}}$

4. Joint PMF of X_1 and X_2 is

x_1	x_2	$P(X_1 = x_1, X_2 = x_2)$
0	0	0.15
1	0	0.15
2	0	0.15
0	1	0.15
1	1	0.20
2	1	0.20

(a) Compute the marginal distribution of X_1.

A: X_1 can take 3 values: 0, 1, 2., X_2 takes 2 values: 0, 1.

$$Pr(X_1 = 0) = \sum_{x_2 \in \{0,1\}} Pr(X_1 = 0, X_2 = x_2) = P(X_1=0, X_2=0) + P(X_1=0, X_2=1) = 0.15 + 0.15 = 0.30$$

$$Pr(X_1 = 1) = \sum_{x_2 \in \{0,1\}} Pr(X_1 = 1, X_2 = x_2) = P(X_1=1, X_2=0) + P(X_1=1, X_2=1) = 0.15 + 0.20 = 0.35$$

$$Pr(X_1 = 2) = \sum_{x_2 \in \{0,1\}} Pr(X_1 = 2, X_2 = x_2) = Pr(X_1 = 2, X_2 = 0) + Pr(X_1 = 2, X_2 = 1) = 0.15 + 0.20 = 0.35$$

(b) Compute the marginal distribution of X_2.

A: X_2 takes 2 values : 0, 1.

$$P(X_2=0) = \sum_{x_1 \in \{0,1,2\}} \Pr(X_1 = x_1, X_2 = 0)$$

$$= \Pr(X_1=0, X_2=0) + \Pr(X_1=1, X_2=0) + \Pr(X_1=2, X_2=0)$$

$$= 0.15 + 0.15 + 0.15 = 0.45.$$

$$P(X_2=1) = \sum_{x_1 \in \{0,1,2\}} \Pr(X_1 = x_1, X_2 = 1)$$

$$= \Pr(X_1=0, X_2=1) + \Pr(X_1=1, X_2=1) + \Pr(X_1=2, X_2=1)$$

$$= 0.15 + 0.20 + 0.20 = 0.55.$$

(c) Compute the Conditional distribution of $X_1 | X_2$.

A: X_1 takes values $\{0,1,2\}$, X_2 takes values $\{0,1\}$.

So, when we talk about $\Pr(X_1 = x_1 | X_2 = x_2)$, we have to Consider 6 cases.

$$\Pr(X_1=0 | X_2=0) = \frac{\Pr(X_1=0, X_2=0)}{\Pr(X_2=0)} = \frac{0.15}{0.45} = \frac{1}{3}.$$

$$\Pr(X_1=1 | X_2=0) = \frac{\Pr(X_1=1, X_2=0)}{\Pr(X_2=0)} = \frac{0.15}{0.45} = \frac{1}{3}.$$

$$\Pr(X_1=2 | X_2=0) = \frac{\Pr(X_1=2, X_2=0)}{\Pr(X_2=0)} = \frac{0.15}{0.45} = \frac{1}{3}.$$

$$\Pr(X_1=0 | X_2=1) = \frac{\Pr(X_1=0, X_2=1)}{\Pr(X_2=1)} = \frac{0.15}{0.55} = \frac{3}{11}.$$

$$\Pr(X_1=1 | X_2=1) = \frac{\Pr(X_1=1, X_2=1)}{\Pr(X_2=1)} = \frac{0.20}{0.55} = \frac{4}{11}.$$

$$\Pr(X_1=2 | X_2=1) = \frac{\Pr(X_1=2, X_2=1)}{\Pr(X_2=1)} = \frac{0.20}{0.55} = \frac{4}{11}.$$

(d) Compute the Conditional distribution of $X_2 | X_1$.

$$Pr(X_2=0|X_1=0) = \frac{Pr(X_1=0, X_2=0)}{Pr(X_1=0)} = \frac{0.15}{0.3} = \frac{1}{2}.$$

$$Pr(X_2=1|X_1=0) = \frac{Pr(X_1=0, X_2=1)}{Pr(X_1=0)} = \frac{0.15}{0.3} = \frac{1}{2}$$

$$Pr(X_2=0|X_1=1) = \frac{Pr(X_1=1, X_2=0)}{Pr(X_1=1)} = \frac{0.15}{0.35} = \frac{3}{7}$$

$$Pr(X_2=1|X_1=1) = \frac{Pr(X_1=1, X_2=1)}{Pr(X_1=1)} = \frac{0.2}{0.35} = \frac{4}{7}.$$

$$Pr(X_2=0|X_1=2) = \frac{Pr(X_1=2, X_2=0)}{Pr(X_1=2)} = \frac{0.15}{0.35} = \frac{3}{7}.$$

$$Pr(X_2=1|X_1=2) = \frac{Pr(X_1=2, X_2=1)}{Pr(X_1=2)} = \frac{0.2}{0.35} = \frac{4}{7}$$

(e) Are X_1 and X_2 independent? Justify your answer.

A: $Pr(X_1=0, X_2=0) = 0.15$

$Pr(X_1=0) = 0.3$, $Pr(X_2=0) = 0.45$, $0.3 \times 0.45 \neq 0.15$

So, $Pr(X_1=0, X_2=0) \neq Pr(X_1=0) \times Pr(X_2=0)$

So, X_1 and X_2 are not independent.

5/ If (X_1, X_2) is bivariate normal with $E(X_1) = E(X_2) = 0$, $Var(X_1) = Var(X_2) = 1$, and $Cor(X_1, X_2) = \rho$.

(a) Derive the marginal distribution of X_1.

$$f_X(x_1, x_2) = \frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}(x_1^2+x_2^2-2\rho x_1 x_2)} \quad ; x_1, x_2 \in \mathbb{R}$$

So, the marginal distribution of X_1 (density function) is

$$f_{X_1}(x_1) = \int f_X(x_1, x_2)\, dx_2$$

$$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int e^{-\frac{1}{2(1-\rho^2)}(x_1^2+x_2^2-2\rho x_1 x_2)}\, dx_2$$

$$f_{X_1}(x_1) = \frac{1}{2\pi\sqrt{(1-\rho^2)}} \int e^{-\frac{1}{2(1-\rho^2)}\left[(x_2-\rho x_1)^2 + (1-\rho^2)x_1^2\right]} dx_2$$

$$= \frac{1}{\sqrt{2\pi}\cancel{\sqrt{1-\rho^2}}} \cdot e^{-\frac{1}{2\cancel{(1-\rho^2)}}\cancel{(1-\rho^2)}x_1^2} \times \underbrace{\int \frac{1}{\sqrt{2\pi(1-\rho^2)}} e^{-\frac{1}{2(1-\rho^2)}(x_2-\rho x_1)^2} dx_2}_{\text{density of } N(\rho x_1,\, 1-\rho^2) \text{ distribution}} \times \cancel{\sqrt{1-\rho^2}}$$

$$= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x_1^2}.$$

So, Marginal distribution of X_1 is $X_1 \sim N(0,1)$. Similarly, $X_2 \sim N(0,1)$.

(b) Derive the Conditional distribution of $X_1|X_2$.

$$f_{X_1|X_2=x_2}(x_1) = \frac{f_X(x_1,x_2)}{f_{X_2}(x_2)}$$

$$= \frac{\frac{1}{2\pi\sqrt{1-\rho^2}} e^{-\frac{1}{2(1-\rho^2)}\cdot(x_1^2+x_2^2-2\rho x_1 x_2)}}{\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}\cdot x_2^2}}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \cdot e^{-\frac{1}{2(1-\rho^2)}\cdot\left[x_1^2+x_2^2-2\rho x_1 x_2 - (1-\rho^2)x_2^2\right]}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \cdot e^{-\frac{1}{2(1-\rho^2)}\left[x_1^2 - 2\rho x_1\cdot(\rho x_2) + (\rho x_2)^2\right]}$$

$$= \frac{1}{\sqrt{2\pi}\sqrt{1-\rho^2}} \cdot e^{-\frac{1}{2(1-\rho^2)}(x_1-\rho x_2)^2},$$

which is the density of a Normal$(\rho x_2,\, 1-\rho^2)$ distribution.

So, $X_1|X_2 \sim N(\rho X_2, \boxed{1-\rho^2})$. variance, not SD.

6. Assume (X_1, X_2) have bivariate PDF

$$f(x_1,x_2) = \frac{1}{2\pi}\left(1+x_1^2+x_2^2\right)^{-3/2}$$

(a) plot the conditional distribution of $X_1|X_2=x_2$ for $x_2 \in \{-3,-2,-1,0,1,2,3\}$.

A: For plotting, we need a software. But, let's just derive the expression of $f_{X_1|X_2=x_2}(x_1)$.

$$f(x_1,x_2) = \frac{1}{2\pi}\left(1+x_1^2+x_2^2\right)^{-3/2}$$

$$f_{X_2}(x_2) = \int \frac{1}{2\pi}\left(1+x_1^2+x_2^2\right)^{-3/2} dx_1$$

$$= \int \frac{1}{2\pi}\cdot\left[(1+x_2^2)\left(1+\frac{x_1^2}{1+x_2^2}\right)\right]^{-3/2} dx_1$$

$$= \frac{1}{2\pi}\left(1+x_2^2\right)^{-3/2}\int_{-\infty}^{+\infty}\left(1+\frac{x_1^2}{1+x_2^2}\right)^{-3/2} dx_1$$

the PDF of a Student's t-distribution (with df ν) is

$$f(x) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\sqrt{\nu\pi}\,\Gamma\left(\frac{\nu}{2}\right)}\cdot\left(1+\frac{x^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad \text{Also, } \int_{-\infty}^{+\infty} f(x)\,dx = 1.$$

We next try to match the forms:

$$f_{X_2}(x_2) = \frac{1}{2\pi}\left(1+x_2^2\right)^{-3/2}\int_{-\infty}^{+\infty}\left[1+\frac{\left(\frac{x_1}{\sqrt{(1+x_2^2)/2}}\right)^2}{2}\right]^{-\frac{2+1}{2}} d\left(\frac{x_1}{\sqrt{(1+x_2^2)/2}}\right) \times \sqrt{(1+x_2^2)/2}$$

$\rightarrow$ $\Gamma(1.5) = 0.5\,\Gamma(0.5) = 0.5\sqrt{\pi}$

$$= \frac{1}{2\pi}\left(1+x_2^2\right)^{-3/2} \times \sqrt{(1+x_2^2)/2} \times \frac{\sqrt{2\pi}\times\Gamma\left(\frac{2}{2}\right)}{\Gamma\left(\frac{2+1}{2}\right)}.$$

$$= \frac{1}{2\pi}\left(1+x_2^2\right)^{-3/2+1/2}\cdot\frac{1}{\sqrt{2}}\times\frac{\sqrt{2\pi}\times 1}{0.5\sqrt{\pi}}$$

$$= \frac{1}{2\pi}\times\frac{\cancel{\sqrt{2\pi}}}{0.5\cancel{\sqrt{2\pi}}}\left(1+x_2^2\right)^{-1}$$

$$= \frac{1}{\pi}\left(1+x_2^2\right)^{-1},$$

So, X_2 follows a Standard Cauchy distribution.

$$\text{So, } f_{X_1|X_2=x_2}(x_1) = \frac{\frac{1}{2\pi}\left(1+x_1^2+x_2^2\right)^{-3/2}}{\frac{1}{\pi}\left(1+x_2^2\right)^{-1}} = \frac{1}{2}\,\frac{1+x_2^2}{\left(1+x_1^2+x_2^2\right)^{3/2}}.$$

$$= \frac{1}{2}\,\frac{1}{\sqrt{1+x_2^2}}\cdot\frac{\left(1+x_2^2\right)^{3/2}}{\left(1+x_1^2+x_2^2\right)^{3/2}} = \frac{1}{2}\cdot\frac{1}{\sqrt{1+x_2^2}}\cdot\left(1+\frac{x_1^2}{1+x_2^2}\right)^{-3/2}$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{1+x_2^2}} \left(1 + \left[\frac{x_1}{\sqrt{(1+x_2^2)/2}}\right]^2 \Big/ 2\right)^{-\frac{2+1}{2}}$$

$$\frac{\Gamma\left(\frac{2+1}{2}\right)}{\sqrt{2\pi}\,\Gamma\left(\frac{2}{2}\right)} = \frac{\Gamma\left(\frac{3}{2}\right)}{\sqrt{2\pi} \times 1} = \frac{0.5\sqrt{\pi}}{\sqrt{2\pi} \times 1} = \frac{0.5}{\sqrt{2}} = \frac{1}{2\sqrt{2}}$$

$$= \frac{1}{2} \cdot \frac{1}{\sqrt{(1+x_2^2)/2}} \times \frac{1}{\sqrt{2}} \left(1 + \frac{\left(x_1 / \sqrt{(1+x_2^2)/2}\right)^2}{2}\right)^{-\frac{2+1}{2}}$$

$$= \frac{1}{2\sqrt{2}} \cdot \frac{1}{\sqrt{(1+x_2^2)/2}} \left(1 + \frac{\left(x_1 / \sqrt{(1+x_2^2)/2}\right)^2}{2}\right)^{-\frac{2+1}{2}}$$

$$= \frac{\Gamma\left(\frac{2+1}{2}\right)}{\sqrt{2\pi}\,\Gamma\left(\frac{2}{2}\right)} \frac{1}{\sqrt{(1+x_2^2)/2}} \left(1 + \frac{\left(x_1 / \sqrt{(1+x_2^2)/2}\right)^2}{2}\right)^{-\frac{2+1}{2}}$$

$$X_1 \mid X_2 = x_2 \sim \sqrt{\frac{1+x_2^2}{2}} \cdot t_2$$

So, $\pi(X_1 \mid X_2 = x_2)$ depends on X_2. So, X_1 and X_2 are not independent.

X_1 and X_2 are identically distributed as standard Cauchy distributions. So, $E(X_1)$ and $E(X_2)$ does not exist.

But, the appearance still possibly makes sense, as is it not possible to judge, by just looking at the plot, the existence of correlation.

$E(X_1 \mid X_2 = x_2) \propto \sqrt{1+x_2^2}$. Because it is symmetric around 0 the correlation would appear zero.

Besides, $f(x_1, x_2) = \frac{1}{2\pi}\left(1 + \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}' \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}\right)^{-3/2}$

Correlation matrix

Contour plots are circular. as density is a function of (x^2+y^2).

7/ The 2017 auto theft rate was 135 per 10,000 ~~rate~~ residents in Raleigh. Compared to 214 per 10,000 residents in Durham. Assuming Raleigh's population is twice as large as Durham and a car has been stolen somewhere in the triangle (one of these areas: Raleigh and Durham), what is the probability it was stolen in Raleigh?

A: $Pr(\text{Theft} \mid \text{Raleigh}) = \frac{135}{10{,}000}$

$Pr(\text{Theft} \mid \text{Durham}) = \frac{214}{10{,}000}$

$Pr(\text{Raleigh}) = \frac{2}{3}$, $Pr(\text{Durham}) = \frac{1}{3}$

$$Pr(\text{Raleigh} \mid \text{Theft}) = \frac{Pr(\text{Theft} \mid \text{Raleigh})\, Pr(\text{Raleigh})}{Pr(\text{Theft} \mid \text{Raleigh})\, Pr(\text{Raleigh}) + Pr(\text{Theft} \mid \text{Durham})\, Pr(\text{Durham})}$$

$$= \frac{\frac{135}{10{,}000} \times \frac{2}{3}}{\frac{135}{10{,}000} \times \frac{2}{3} + \frac{214}{10{,}000} \times \frac{1}{3}} = \frac{2 \times 135}{2 \times 135 + 214}$$

$$= \frac{270}{270 + 214} = \frac{270}{484} = 0.5578.$$

8/ Your daily commute is distributed uniformly between 15 and 20 minutes if there is no convention in the Downtown. However, are scheduled for roughly 1 in 4 days, and your commute time is distributed uniformly from 15 and 30 minutes if there is a convention. Let Y = commute time this morning.

(a) What is the probability that there was a convention in the Downtown given $Y = 18$?

A:- So, support of $Y = [15, 30]$.

$Pr(\text{Convention}) = \frac{1}{4}$, $Pr(\text{No Convention}) = \frac{3}{4}$.

$Y \mid \text{No Convention} \sim U(15, 20)$, $f_0(y) = \frac{1}{5}\, \mathbb{I}(y \in (15, 20))$

$Y \mid \text{Convention} \sim U(15, 30)$, $f_1(y) = \frac{1}{15}\, \mathbb{I}(y \in (15, 30))$

$$f(y) = Pr(\text{No Convention})\, f_0(y) + Pr(\text{Convention})\, f_1(y)$$

$$= \frac{3}{4} \times \frac{1}{5}\, \mathbb{I}(y \in (15, 20)) + \frac{1}{4} \times \frac{1}{15}\, \mathbb{I}(y \in (15, 30))$$

$$= \frac{3}{20}\,\mathbb{I}(y\in(15,20)) + \frac{1}{60}\,\mathbb{I}(y\in(15,30))$$

indicator function

$$\int_{15}^{30} f(y)\,dy = \int_{15}^{20} f(y)\,dy + \int_{20}^{30} f(y)\,dy$$

$$= \int_{15}^{20}\left(\frac{3}{20}+\frac{1}{60}\right)dy + \int_{20}^{30}\frac{1}{60}\,dy$$

$$= \frac{10}{60}\times(20-15) + \frac{1}{60}\times(30-20)$$

$$= \frac{10\times 5}{60} + \frac{1\times 10}{60} = 1.$$

$$f(y) = \frac{3}{20}\,\mathbb{I}(y\in(15,20)) + \frac{1}{60}\,\mathbb{I}(y\in(15,30))$$

is a valid PDF.

So, $$\pi(\text{Convention}\mid Y=18) = \frac{f(18\mid\text{Convention})\times\text{Prob}(\text{Convention})}{f(18\mid\text{Convention})\times\text{Prob}(\text{Conv.})+f(18\mid\text{No Conv})\times\Pr(\text{No Conv})}$$

$$= \frac{\frac{1}{15}\times\frac{1}{4}}{\frac{1}{15}\times\frac{1}{4}+\frac{1}{5}\times\frac{3}{4}} = \frac{1/60}{\frac{1}{60}+\frac{3}{20}} = \frac{1}{10} = 0.1.$$

What is the prob. that there was a Convention in the Downtown given $Y=28$?

$$\pi(\text{Convention}\mid Y=28) = \frac{f(28\mid\text{Convention})\times\Pr(\text{Conv.})}{f(28\mid\text{Conv.})\Pr(\text{Conv.})+f(28\mid\text{No Conv.})\Pr(\text{No Conv.})}$$

$$= \frac{\frac{1}{15}\times\frac{1}{4}}{\frac{1}{15}\times\frac{1}{4}+0\times\frac{3}{4}} = 1.$$

The answer is natural, as if there was no Convention, the commute would be less than 20. So, because commute > 20, we know the convention is happening certainly.

9. For this problem, pretend that we are dealing with a language with a six-word dictionary.

$$\{fun, sun, sit, sat, fan, for\}$$

An extensive study of literature written in this language reveals that all words are equally likely except that "for" is α times as likely as the other words. Further study reveals that:

i. Each keystroke is an error with probability θ.
ii. All letters are equally likely to produce errors.
iii. Given that a letter is typed incorrectly, it is equally likely to be any other letter.
iv. Errors are independent across letters.

prob. of correctly typing a word is $(1-\theta)^3$.

prob. of typing "bun" or "fon" when intending to type "fun" is $\theta(1-\theta)^2$.

prob. of typing "foo" or "nnn" when intending to type "fun" is $\theta^2(1-\theta)$.

Use Bayes' rule to develop a simple spell checker for this language. For each typed words "sun", "the", "foo", give the probability that each word in the dictionary was the intended word. Perform this for the parameters:

(a) $\alpha = 2$ and $\theta = 0.1$.
(b) $\alpha = 50$ and $\theta = 0.1$.
(c) $\alpha = 2$ and $\theta = 0.95$.

A: All words are of 3 letters. So, # keystroke error $\sim Bin(3, \theta)$.

Word	fun	sun	sit	sat	fan	for
"sun" →	1	0	2	2	2	3
"the" →	3	3	3	3	3	3
"foo" →	2	3	3	3	2	1

a) $\alpha = 2 \rightarrow$ "fun" "sun" "sit" "sat" "fan" "for"

	"fun"	"sun"	"sit"	"sat"	"fan"	"for"	
weights $\rightarrow$	1	1	1	1	1	2	$\rightarrow$ total $= 7$
prob $\rightarrow$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{2}{7}$	

we will just do one case:

"sun" $\rightarrow$	1	0	2	2	2	3
Likelihood $\rightarrow$	$\binom{3}{1}\theta(1-\theta)^2$	$\binom{3}{0}\theta^0(1-\theta)^3$	$\binom{3}{2}\theta^2(1-\theta)$	"	"	$\binom{3}{3}\theta^3(1-\theta)^0$
	$= 3\theta(1-\theta)^2$	$= (1-\theta)^3$	$= 3\theta^2(1-\theta)$	"	"	$= \theta^3$

$$Pr(\text{"fun"} \mid \text{"sun"}) = \frac{Pr(\text{"sun"} \mid \text{"fun"})\, Pr(\text{"fun"})}{Pr(\text{"sun"} \mid \text{"fun"})\, Pr(\text{"fun"}) + Pr(\text{"sun"} \mid \text{"sun"})\, Pr(\text{"sun"}) + \dots + Pr(\text{"sun"} \mid \text{"for"})\, Pr(\text{"for"})}$$

$$= \frac{3\theta(1-\theta)^2 \times \frac{1}{7}}{\left[3\theta(1-\theta)^2 \times \frac{1}{7} + (1-\theta)^3 \times \frac{1}{7} + 3\theta^2(1-\theta) \times \frac{1}{7} + 3\theta^2(1-\theta)\times\frac{1}{7} + 3\theta^2(1-\theta)\times\frac{1}{7} + \theta^3 \times \frac{2}{7}\right]}$$

[For specific θ, this expression can be calculated.

Similarly, for all words ... until "for" as follows.

$$Pr(\text{"for"} \mid \text{"sun"}) = \frac{Pr(\text{"sun"} \mid \text{"for"}) \times Pr(\text{"for"})}{Pr(\text{"sun"} \mid \text{"fun"})\, Pr(\text{"fun"}) + \dots + Pr(\text{"sun"} \mid \text{"for"})\, Pr(\text{"for"})}$$

$$= \frac{\theta^3 \times \frac{2}{7}}{3\theta(1-\theta)^2 \times \frac{1}{7} + \dots + \theta^3 \times \frac{2}{7}}$$

Good exercise to do using R.

when α increases $\rightarrow$ prior for "for" increases and so posterior should also increase.

when θ increases $\rightarrow$ errors are more likely and so words with many errors have higher probability.

10. $X_1 \sim \text{Bernoulli}(\theta)$ → indicator ~~of~~ for a tree species occupying a forest. θ is the prior occupancy probability. A reseacher collects a sample of n trees from the forest and X_2 belong to the species of interest. The model is $X_2 | X_1 \sim \text{Binomial}(n, \lambda X_1)$.

A: First think of the validity of the model:

If $X_1 = 0 \Rightarrow X_2 \sim \text{Binomial}(n, 0) \Rightarrow X_2 = 0$ w.p. 1.

that is reasonable, because if the species is not occupying the forest, there is no chance that the sample would contain any sample of that species.

otherwise, if $X_1 = 1 \Rightarrow X_2 \sim \text{Binomial}(n, \lambda)$

⇒ Binomial is a very reasonable model on the number of samples out on n samples that belong to that species.

So, perfect model!

$$X_1 = \begin{cases} 1 & \text{w.p. } \theta \\ 0 & \text{w.p } 1-\theta \end{cases}$$

(a) $\Pr(X_1 = X_2 = 0)$

$= \Pr(X_1 = 0) \Pr(X_2 = 0 | X_1 = 0)$

$= (1-\theta) \times 1$

$= (1-\theta)$.

$X_2 | X_1 = 0 \sim \text{Bin}(n, 0)$

$= 0$ w.p. 1

So, $\Pr(X_2 = 0 | X_1 = 0) = 1$

(b) $\Pr(X_1 = 0) = 1 - \theta$.

$X_2 | X_1 = 1 \sim \text{Bin}(n, \lambda)$

(c) $\Pr(X_2 = 0)$

$= \Pr(X_2 = 0 | X_1 = 0) \Pr(X_1 = 0) + \Pr(X_2 = 0 | X_1 = 1) \Pr(X_1 = 1)$

$= 1 \times (1-\theta) + \binom{n}{0} \lambda^0 (1-\lambda)^{n-0} \times \theta$

$= 1 - \theta + \theta (1-\lambda)^n$.

(d) $\Pr(X_1 = 0 | X_2 = 0) = \dfrac{\Pr(X_2 = 0 | X_1 = 0) \Pr(X_1 = 0)}{\Pr(X_2 = 0)}$

$= \dfrac{1 \times (1-\theta)}{1 - \theta + \theta(1-\lambda)^n} = \dfrac{1-\theta}{1-\theta+\theta(1-\lambda)^n}$.

(e) $\Pr(X_2 = 0 \mid X_1 = 0) = 1$

(f) $\Pr(X_1 = 0 \mid X_2 = 1) = \dfrac{\Pr(X_2 = 1 \mid X_1 = 0)\Pr(X_1 = 0)}{\Pr(X_2 = 1)}$

Now, $X_2 \mid X_1 = 0 \sim Bin(n, 0) \Rightarrow \Pr(X_2 = 1 \mid X_1 = 0) = 0$.

So, $\Pr(X_1 = 0 \mid X_2 = 1) = 0$.

(g) $\Pr(X_2 = 0 \mid X_1 = 1) \overset{X_2 \mid X_1 = 1 \sim Bin(n,\lambda)}{=} \binom{n}{0}\lambda^0(1-\lambda)^{n-0} = (1-\lambda)^n$

(h) $\boxed{d}$: $\Pr(X_1 = 0 \mid X_2 = 0)$:

n samples were drawn, but no sample of that species was observed: So, one possibility is that ~~there is no tree of that species~~ the species does not occupy the forest. Another possibility is that the species actually occupy the forest but was not observed in the sample.

So, the bigger θ we have, the bigger is the chance of the species occupying the forest. That is, it is natural that the chance of not occupying the forest is smaller.

So, keeping λ and n fixed, the term should decrease with θ. and smaller prior $\equiv$ smaller posterior.

If n is bigger (keeping θ and λ fixed):

within a bigger sample, no observation belong to that species. $\Rightarrow$ the chance of that species occupying the forest is even smaller. $\Rightarrow$ the chance of $X_1 = 0$ is more.

Here, also $n \uparrow$, $\dfrac{1-\theta}{1-\theta+\theta(1-\lambda)^n} \uparrow$.

So, logical in both ways.

Now, if λ is bigger $\Rightarrow$ if $X_1 = 1$, there is less chance of X_2 being zero. Thus, given $X_2 = 0$, (which is a less likely event when λ is large), it is more likely that the species actually does not occupy. Mathematically, $\lambda \uparrow \Rightarrow$ denominator $\downarrow$

$\Rightarrow \Pr(X_1 = 0 \mid X_2 = 0) \uparrow$.

So, logical in both ways!

(i) So, $\dfrac{1-\theta}{1-\theta+\theta(1-\lambda)^n} \geq 0.95$

$\Rightarrow \dfrac{1-0.5}{1-0.5+0.5(1-0.1)^n} \geq 0.95$

$\Rightarrow \dfrac{0.5}{0.95} \geq 0.5(1+0.9^n)$

$\Rightarrow \dfrac{1}{0.95} \geq 1+0.9^n$

$\Rightarrow 0.9^n \leq \dfrac{1}{0.95} - 1 = \dfrac{0.05}{0.95} = \dfrac{1}{19}$

~~$\Rightarrow n \log(0.9) \; \log(19)$~~

~~$\Rightarrow n = \log$~~

$\Rightarrow n \log(0.9) \geq \log(1/19)$

$\Rightarrow n \geq \dfrac{-\log(19)}{\log(0.9)}$. Note the sign change as $\log(0.9) < 0$.

So, $n = \left\lceil \dfrac{-\log(19)}{\log(0.9)} \right\rceil$ ← Ceiling operator.

11. In a study that uses Bayes methods to forecast the number of species that will be discovered in future years, some reseraechers report that the number of marine bivalve species discovered in each year between 2010-2015 was 64, 13, 33, 18, 30, and 20.

Y_t = # species discovered in year t

$Y_t | \lambda \overset{iid}{\sim} Poi(\lambda)$ and $\lambda \sim Unif(0, 100)$. Plot the posterior.

A: Here we just calculate the posterior. $\underset{\sim}{Y} = (64, 13, 33, 18, 30, 20)$

$$p(\lambda | \underset{\sim}{Y}) \propto f(\underset{\sim}{Y} | \lambda)\, \pi(\lambda)$$

$$= \prod_{i=1}^{6} e^{-\lambda} \cdot \frac{\lambda^{y_i}}{y_i!} \times \frac{1}{100} \mathbb{I}(\lambda \in (0,100))$$

$$\propto e^{-6\lambda} \cdot \frac{\lambda^{\Sigma y_i}}{y_i!} \; \mathbb{I}(\lambda \in (0,100))$$

Note that the Kernel of $p(\lambda|Y)$ is similar to a

Gamma distribution with Shape parameter $\sum y_i + 1$ and rate parameter 6, and truncated between 0 and 100.

So, the final density would be

$$p(\lambda|\underset{\sim}{Y}) = \left[\frac{6^{\sum y_i+1}}{\Gamma(\sum y_i+1)} \lambda^{\sum y_i} \cdot e^{-6\lambda}\right] \Big/ \int_0^{100} \left[\frac{6^{\sum y_i+1}}{\Gamma(\sum y_i+1)} \lambda^{\sum y_i} e^{-6\lambda} d\lambda\right]$$

(coeff's cancel out) $= \boxed{\dfrac{\lambda^{\sum y_i} \cdot e^{-6\lambda}}{\int_0^{100} \lambda^{\sum y_i} e^{-6\lambda} d\lambda}}$ ← Plot this one.

Gamma density: $X \sim \text{Gamma}(a, b)$

$$\Rightarrow f_X(x) = \frac{b^a}{\Gamma(a)} \cdot x^{a-1} \cdot e^{-bx}.$$

12. $\underset{\sim}{Z} = (X, Y) \sim \text{BVN}(0, 0, 1, 1, \rho)$ $\qquad \underset{\sim}{Z}_i = (x_i, y_i)$

marginal mean of X; marginal mean of Y; marginal variance of X; marginal variance of Y; Correlation.

Observed data: $\underbrace{(-3.3, -2.6)}_{=\underset{\sim}{Z}_1}$, $\underbrace{(0.1, -0.2)}_{=\underset{\sim}{Z}_2}$, $\underbrace{(-1.1, -1.5)}_{=\underset{\sim}{Z}_3}$, $\underbrace{(2.7, 1.5)}_{\underset{\sim}{Z}_4}$, $\underbrace{(2.0, 1.9)}_{\underset{\sim}{Z}_5}$, $\underbrace{(-0.4, -0.3)}_{\underset{\sim}{Z}_6}$.

$$p(\rho|\underset{\sim}{Z}_i, i=1,\dots,6) \propto \underbrace{\left(\frac{1}{2\pi\sqrt{1-\rho^2}}\right)^6 \cdot e^{-\frac{1}{2(1-\rho^2)}\sum_{i=1}^{6}(x_i^2+y_i^2-2\rho x_i y_i)}}_{\text{Likelihood}} \times \underbrace{\frac{1}{2}\, \mathbb{I}(\rho \in (-1,1))}_{\text{prior}}$$

this is not in any known distributional form.

$$\text{So, } p(\rho|\underset{\sim}{Z}_i) = \frac{(1-\rho^2)^{-3} e^{-\frac{1}{2(1-\rho^2)}\sum_{i=1}^{6}(x_i^2+y_i^2-2\rho x_i y_i)}}{\int_{-1}^{+1}(1-\rho^2)^{-3} \cdot e^{-\frac{1}{2(1-\rho^2)}\sum_{i=1}^{6}(x_i^2+y_i^2-2\rho x_i y_i)}\, d\rho}$$

13. NDVI (normalized difference vegetation index) is a classifier of land cover based on Satellite data. Claim: NDVI ~ Beta(25, 10) for pixels in a rain forest, NDVI ~ Beta(10, 15) for pixels that are deforested. Assumption: 10% of the rain forest is deforested.

(a) Plot the PDF of NDVI for forested and deforested pixels.

A: Let $X = \text{NDVI}$, $X \sim \text{Beta}(25, 10)$ for forested pixels.

So, $f_{X|\text{forested}}(x) = \dfrac{\Gamma(35)}{\Gamma(25)\Gamma(10)} x^{25-1}(1-x)^{10-1}$ for forested pixels.

$X \sim \text{Beta}(10, 15)$

$f_{X|\text{deforested}}(x) = \dfrac{\Gamma(25)}{\Gamma(10)\Gamma(15)} x^{10-1}(1-x)^{15-1}$ for deforested pixels

overall, Prob(forested pixel) = 0·9

Prob(deforested pixel) = 0·1

So, overall,

$$f_X(x) = f_{X|\text{forested}}(x)\ \text{Prob}(\text{forested}) + f_{X|\text{deforested}}(x)\ \text{Prob}(\text{deforested})$$

$$= 0{\cdot}9\ \frac{\Gamma(35)}{\Gamma(25)\Gamma(10)} x^{25-1}(1-x)^{10-1} + 0{\cdot}1\ \frac{\Gamma(25)}{\Gamma(10)\Gamma(15)} x^{10-1}(1-x)^{15-1}.$$

(b) Give an expression of the probability that a pixel is deforested given its NDVI.

$$\text{Prob}(\text{deforested} \mid X=x) = \frac{f_{X|\text{deforested}}(x) \times \Pr(\text{deforested})}{f_{X|\text{deforested}}(x) \times \Pr(\text{deforested}) + f_{X|\text{forested}}(x) \times (1-\Pr(\text{deforested}))}$$

$$= \frac{0{\cdot}1 \times \frac{\Gamma(25)}{\Gamma(10)\Gamma(15)} x^{10-1}(1-x)^{15-1}}{0{\cdot}1 \times \frac{\Gamma(25)}{\Gamma(10)\Gamma(15)} x^{10-1}(1-x)^{15-1} + 0{\cdot}9 \times \frac{\Gamma(35)}{\Gamma(25)\Gamma(10)} x^{25-1}(1-x)^{10-1}}.$$

(c) Suppose that value is c. and range would be $(0, c)$

(Small $X \Rightarrow$ deforestation)

Prob(deforested $| X=x$) is a function of x.

and a decreasing function of x.

So, Solve Prob(deforested $| X=c$) $\geq 0{\cdot}9$

$$\Rightarrow \frac{0{\cdot}1\ \text{dbeta}(c, 10, 15)}{0{\cdot}1\ \text{dbeta}(c, 10, 15) + 0{\cdot}9\ \text{dbeta}(c, 25, 10)} \geq 0{\cdot}9.$$

$\Rightarrow 0.1 \text{ dbeta}(c,10,15) \geq 0.09 \text{ dbeta}(c,10,15) + 0.81 \text{ dbeta}(c,25,10)$

$\Rightarrow 0.01 \text{ dbeta}(c,10,15) \geq 0.81 \text{ dbeta}(c,25,10)$

Now, need to solve numerically.

14. n = unknown # customers.

Y = # customers that make a purchase

$Y|n \sim Bin(n,\theta)$, θ = prob. of purchasing given that the customer visited the store

Prior: $n \sim Poi(5)$.

Assume θ is known and n is unknown, plot the posterior distribution of n.

$$Pr(n=n^* | Y=y) = \frac{Pr(Y=y|n=n^*)\,Pr(n=n^*)}{\sum_{n'=0}^{\infty} Pr(Y=y|n=n')\,Pr(n=n')},$$

$$= \frac{\binom{n^*}{y}\theta^y(1-\theta)^{n^*-y} \times e^{-5}.\,5^{n^*}/n^*!}{\sum_{n'=0}^{\infty}\binom{n'}{y}\theta^y(1-\theta)^{n'-y} \times e^{-5}.\,\frac{5^{n'}}{n'!}}.$$

Let's think the effect of θ logically, numerically not possible.

Suppose, $Y=y$ is fixed (given).

If $\theta\uparrow$, prob. of purchasing increases. Now, # customers making purchase remain the same $\Rightarrow$ n decreases stochastically.

So, $\theta\uparrow \Rightarrow Pr(n=n^*|Y=y)$ weights shifted to smaller values of n^*.

15. Last spring your lab planted 10 seedlings and two survived the winter. Let θ be the prob. that a seedling survives the winter.

(a) Assuming a uniform prior distribution for θ, compute its posterior mean and sd.

A: Y = # seedlings survived the winter. $Y \sim Bin(10,\theta)$

We observed $Y=2$. Here, prior $\pi(\theta) = Uniform(0,1) \equiv Beta(1,1)$

$p(\theta|Y) \propto f(Y|\theta)\pi(\theta) \propto \theta^2(1-\theta)^8 \times \theta^{1-1}(1-\theta)^{1-1}$

$= \theta^{2+1-1}(1-\theta)^{8+1-1} \propto$ density of $Beta(3,9)$

So, $\theta | Y = 2 \sim \text{Beta}(3, 9)$

So, posterior mean $E(\theta|Y=2) = \frac{3}{3+9} = \frac{3}{12} = 0.25$.

posterior sd $SD(\theta|Y=2) = \sqrt{\frac{3 \times 9}{(3+9)^2(3+9+1)}} = \sqrt{\frac{27}{12^2 \times 13}}$.

(b) Assuming the same prior for θ, compute and compare the equal-tailed and highest density 95% posterior credible intervals (CI)

So, equal tailed 95% posterior CI is (in terms of R code)

$\big(\text{qbeta}(0.025, 3, 9),\ \text{qbeta}(0.975, 3, 9)\big)$

2.5th percentile of Beta(3,9) distribution — 97.5th percentile of Beta(3,9) distribution

Next, highest density 95% posterior CI:

Let, it starts at → $\text{qbeta}(c, 3, 9)$. Then it should end at $\text{qbeta}(c+0.95, 3, 9)$

So, length $= \text{qbeta}(c+0.95, 3, 9) - \text{qbeta}(c, 3, 9)$

We have choose c so that length is minimum.

Suppose c^* (for exam) $= \underset{c}{\arg\min}\left[\text{qbeta}(c+0.95, 3, 9) - \text{qbeta}(c, 3, 9)\right]$.

In R, we might run this minimization over a grid $c(0.0001, 0.0002, \ldots, 0.0499)$, or use "optimize" function in R (for project). Final interval: $(\text{qbeta}(c^*, 3, 9), \text{qbeta}(c^*+0.95, 3, 9))$.

(c) If you plant another 10 seedlings next year, what is the posterior predictive probability that at least one will survive the winter?

$Y^* | \theta \sim \text{Bin}(10, \theta)$

$$f(Y^*|Y) = \int f(Y^*|\theta)\, p(\theta|Y)\, d\theta$$

$$= \int_0^1 \binom{10}{Y^*} \theta^{Y^*} (1-\theta)^{10-Y^*} \times \frac{\Gamma(12)}{\Gamma(3)\Gamma(9)} \theta^{3-1} (1-\theta)^{9-1}\, d\theta$$

$$= \binom{10}{Y^*} \times \frac{\Gamma(12)}{\Gamma(3)\Gamma(9)} \int_0^1 \theta^{Y^*+3-1} (1-\theta)^{10-Y^*+9-1}\, d\theta$$

$$= \binom{10}{Y^*} \frac{\Gamma(12)}{\Gamma(3)\Gamma(9)} \times \frac{\Gamma(Y^*+3)\Gamma(19-Y^*)}{\Gamma(22)}, \quad Y^* = 0, 1, \ldots, 10.$$

$$\Pr(Y^* > 0 | Y) = 1 - \Pr(Y^* = 0 | Y) = 1 - \binom{10}{0} \frac{\Gamma(12)}{\cancel{\Gamma(3)}\Gamma(9)} \times \frac{\cancel{\Gamma(3)}\Gamma(19)}{\Gamma(22)} = 1 - \frac{\Gamma(12)\Gamma(19)}{\Gamma(9)\Gamma(22)}$$

16. X_1 and X_2 are binary indicators of failure for two parts of a machine. Independent tests have shown that $X_1 \sim \text{Bernoulli}(1/2)$ and $X_2 \sim \text{Ber}(1/3)$. Y_1 and Y_2 are binary ~~system~~ indicators of two System failures.

So, ~~B~~ using AND/OR gates, $Y_1 = \begin{cases} 1 & \text{if both } X_1 = 1, X_2 = 1 \\ 0 & \text{otherwise} \end{cases}$

$Y_2 = \begin{cases} 0 & \text{if both } X_1 = 0, X_2 = 0 \\ 1 & \text{otherwise.} \end{cases}$

Compute the following probabilities:

(a) The probability that $X_1 = 1$ and $X_2 = 1$ given $Y_1 = 1$.

$Y_1 = 1$ if $X_1 = 1$, and $X_2 = 1$ and $Y_1 \neq 1$ otherwise. So, given $Y_1 = 1$

$P(X_1=1, X_2=1 \mid Y=1) = 1$.

(b) The probability that $X_1 = 1$ and $X_2 = 1$ given $Y_2 = 1$.

$Y_2 = 1 \Rightarrow \begin{cases} X_1 = 1, X_2 = 1 \\ X_1 = 1, X_2 = 0 \\ X_1 = 0, X_2 = 1 \end{cases}$ One of these cases.

So, $Pr(Y_2 = 0) = Pr(X_1 = 0, X_2 = 0)$. No reason to assume X_1 and X_2 are dependent

$= Pr(X_1 = 0)\, Pr(X_2 = 0)$

$= \frac{1}{2} \times \frac{2}{3} = \frac{1}{3}$.

So, $Pr(Y_2 = 1) = \frac{2}{3}$.

Now, $Pr(X_1=1, X_2=1 \mid Y_2 = 1)$ $X_1=1, X_2=1 \Rightarrow Y_2 = 1$

$= \frac{Pr(X_1=1, X_2=1, Y_2=1)}{P(Y_2=1)}$

$= \frac{Pr(X_1=1, X_2=1)}{Pr(Y_2=1)} = \frac{Pr(X_1=1)\, Pr(X_2=1)}{2/3} = \frac{\frac{1}{2} \times \frac{1}{3}}{2/3}$

$= \frac{1}{2} \times \frac{1}{3} \times \frac{3}{2} = \frac{1}{4}$.

$\{X_1=1, Y_1=1\} = \{X_1=1, Y_1=1\}$

(c) Probability that $X_1 = 1$ given $Y_1 = 1$.

$Pr(X_1 = 1 \mid Y_1 = 1) = \frac{Pr(X_1=1, Y_1=1)}{Pr(Y_1=1)} = \frac{Pr(X_1=1, X_2=1)}{Pr(Y_1=1)}$

$= \frac{Pr(X_1=1, X_2=1)}{Pr(X_1=1, X_2=1)} = 1$. Also, logical: $Y_1 = 1 \Rightarrow X_1=1, X_2=1$

(d) The probability that $X_1 = 1$ given $Y_2 = 1$.

$$Pr(X_1 = 1 | Y_2 = 1) = \frac{Pr(X_1 = 1, Y_2 = 1)}{Pr(Y_2 = 1)}$$

$$Pr(X_1 = 1, Y_2 = 1) = Pr(X_1 = 1, X_2 = 0) + Pr(X_1 = 1, X_2 = 1)$$

$$= Pr(X_1 = 1)\, Pr(X_2 = 0) + Pr(X_1 = 1)\, Pr(X_2 = 1)$$

$$= Pr(X_1 = 1) \times [Pr(X_2 = 0) + Pr(X_2 = 1)]$$

$$= Pr(X_1 = 1)$$

$$= \frac{1}{2}.$$

So, $Pr(Y_2 = 1) = \frac{2}{3}$.

So, $Pr(X_1 = 1 | Y_2 = 1) = \frac{1/2}{2/3} = \frac{3}{4}$.

17/

Player	Overall prop. (θ_i)	Clutch makes (Y_i)	Clutch attempts (N_i)
P1	0.845	64	75
P2	0.847	72	95
P3	0.880	55	63
P4	0.674	27	39
P5	0.909	75	83
P6	0.898	24	26
P7	0.770	28	41
P8	0.801	66	82
P9	0.802	40	54
P10	0.875	13	16

Clutch: free throws taken in pressure situations.
The overall proportion is computed using a large sample size, assume it is fixed and analyze the clutch data separately for each player using Bayesian methods. Assume uniform priors.

(a) Describe your model for studying the clutch success probability including the likelihood and prior.

A:- Likelihood : $Y_i | N_i, \theta_i \sim Bin(N_i, \theta_i)$

Prior : $\theta_i \sim Uniform(0,1) \equiv Beta(1,1)$.

(b) Plot the posteriors of the clutch success probabilities.

Plot using some software.

$\theta_i | Y_i \sim Beta(Y_i+1, N_i - Y_i + 1)$.

(c) Summarize the posteriors in a table.

Report posterior mean, posterior SD, 95% equal-tailed posterior credible region.

$\theta_i | Y_i \sim Beta(Y_i+1, N_i - Y_i + 1)$

So, $E(\theta_i | Y_i) = \dfrac{Y_i+1}{N_i+2}$, $V(\theta_i | Y_i) = \dfrac{(Y_i+1)(N_i - Y_i + 1)}{(N_i+2)^2 (N_i+3)}$

$CI = (qbeta(0.025, Y_i+1, N_i - Y_i + 1), qbeta(0.975, Y_i+1, N_i - Y_i + 1))$.

(d) Do you find evidence that any of the players have a different clutch percentage than overall percentage?

A: Check if the overall proportion falls between CI in part (c). If not, report that.

(e) Are the results sensitive to prior?

Try two weakly-informative priors: $Beta(0.5, 0.5)$, $Beta(2,2)$ only for players with small information (less number of tries, less number of successes), it would affect more. This is natural.

$\theta_i | Y_i \sim Beta(Y_i + 0.5, N_i - Y_i + 0.5)$ when $\theta_i \sim Beta(0.5, 0.5)$

$\theta_i | Y_i \sim Beta(Y_i+2, N_i - Y_i + 2)$, when $\theta_i \sim Beta(2,2)$

18. In the early twentieth century, it was generally agreed that Hamilton and Madison wrote 51 and 14 Federalist Papers, resply. There was dispute over how to attribute 12 other papers between these two authors. In the 51 papers attributed to Hamilton, the word "upon" was used 3.24 times per 1,000 words, compared to 0.23 times per 1,000 words in the 14 papers attributed to Madison.

(a) If the word "upon" is used three times in a disputed text of length 1000 words and we assume the prior prob. 0.5, what is the posterior prob. that the paper was written by Hamilton?

A:- $Pr(\text{Hamilton}) = \frac{1}{2}$, $Pr(\text{Madison}) = \frac{1}{2}$.

$Y = \#$ "upon" used in the text of length 1000 words

$$Y \mid \text{Hamilton} \sim \text{Bin}\left(1000, \frac{3.24}{1000}\right)$$

$$Y \mid \text{Madison} \sim \text{Bin}\left(1000, \frac{0.23}{1000}\right)$$

$$Pr(Y=3 \mid \text{Hamilton}) = \binom{1000}{3} \left(\frac{3.24}{1000}\right)^3 \left(1 - \frac{3.24}{1000}\right)^{997}$$

$$Pr(Y=3 \mid \text{Madison}) = \binom{1000}{3} \left(\frac{0.23}{1000}\right)^3 \left(1 - \frac{0.23}{1000}\right)^{997}$$

$$Pr(\text{Hamilton} \mid Y=3) = \frac{Pr(Y=3 \mid \text{Hamilton})\, Pr(\text{Hamilton})}{Pr(Y=3 \mid \text{Hamilton})\, Pr(\text{Hamilton}) + Pr(Y=3 \mid \text{Madison})\, Pr(\text{Madison})}$$

$$= \frac{\cancel{\binom{1000}{3}} \left(\frac{3.24}{1000}\right)^3 \left(1 - \frac{3.24}{1000}\right)^{997} \times \cancel{\frac{1}{2}}}{\cancel{\binom{1000}{3}} \left(\frac{3.24}{1000}\right)^3 \left(1 - \frac{3.24}{1000}\right)^{997} \times \cancel{\frac{1}{2}} + \cancel{\binom{1000}{3}} \left(\frac{0.23}{1000}\right)^3 \left(1 - \frac{0.23}{1000}\right)^{997} \times \cancel{\frac{1}{2}}}$$

$$= \frac{3.24^3 (1000 - 3.24)^{997}}{3.24^3 (1000 - 3.24)^{997} + 0.23^3 (1000 - 0.23)^{997}}.$$

(b) Give one assumption you are making in (a) that is likely unreasonable. Justify your answer.

A:- We assume $Y \mid \text{Hamilton} \sim \text{Bin}(1000, p)$

$\Rightarrow$ writing Binomial as a sum of Bernoulli's,

$$Y \mid \text{Hamilton} = \sum_{i=1}^{1000} X_i, \quad X_i \overset{iid}{\sim} \text{Bernoulli}(p)$$

Now, $X_1 = \begin{cases} 1 & \text{if 1st word is "upon"} \\ 0 & \text{if 1st word 1 is not "upon"} \end{cases}$

$X_2 = \begin{cases} 1 & \text{if 2nd word is "upon"} \\ 0 & \text{if 2nd word is not "upon"} \end{cases}$

Now, given that the first word is "upon", it is unlikely that the second word would also be "upon". $\Rightarrow$ X_i are unlikely to be indep.

$$\Rightarrow \sum_{i=1}^{1000} X_i \not\sim \text{Binomial}(1000, p).$$

(e) In (a), if the # instances of "upon" to one, do you expect the posterior prob. to increase, decrease or remain the same? why?

$$\Pr(\text{Hamilton} \mid Y=1) = \frac{3.24 \times (1000 - 3.24)}{3.24 \times (1000 - 3.24) + 0.23 \times (1000 - 0.23)}$$

$$\Pr(Y=3 \mid \text{Hamilton}) = \binom{1000}{3} \cdot \left(\frac{3.24}{1000}\right)^3 \left(1 - \frac{3.24}{1000}\right)^{997}$$

$$\Pr(Y=1 \mid \text{Hamilton}) = \binom{1000}{1} \cdot \left(\frac{3.24}{1000}\right) \times \left(1 - \frac{3.24}{1000}\right)$$

It is natural that $\underbrace{\Pr(Y=1 \mid \text{Hamilton})}_{1 \text{ is far from } E(Y)=3.24} < \underbrace{\Pr(Y=3 \mid \text{Hamilton})}_{3 \text{ is close to } E(Y)=3.24}$

Besides, $\Pr(Y=1 \mid \text{Madison}) = \binom{1000}{1} \left(\frac{0.23}{1000}\right)\left(1 - \frac{0.23}{1000}\right)$

$$\Pr(Y=3 \mid \text{Madison}) = \binom{1000}{3} \left(\frac{0.23}{10^3}\right)^3 \left(1 - \frac{0.23}{10^3}\right)^{997}$$

It is natural that $\underbrace{\Pr(Y=1 \mid \text{Madison})}_{1 \text{ is close to } E(Y)=0.23} > \underbrace{\Pr(Y=3 \mid \text{Madison})}_{3 \text{ is far from } E(Y)=0.23}$

$$\Pr(\text{Hamilon} \mid Y=1) = \frac{\Pr(Y=1 \mid \text{Hamiton})}{\Pr(Y=1 \mid \text{Hamilton}) + \Pr(Y=1 \mid \text{Madison})}$$

$$= \frac{1}{1 + \frac{\Pr(Y=1 \mid \text{Madison})}{\Pr(Y=1 \mid \text{Hamilton})}}$$

We have, $\Pr(Y=1 \mid \text{Hamilton}) < \Pr(Y=3 \mid \text{Hamilton})$

$\Pr(Y=1 \mid \text{Madison}) > \Pr(Y=3 \mid \text{Madison})$

$$\Rightarrow \frac{\Pr(Y=1 \mid \text{Madison})}{\Pr(Y=1 \mid \text{Hamilton})} > \frac{\Pr(Y=3 \mid \text{Madison})}{\Pr(Y=3 \mid \text{Hamilton})}$$

$$\Rightarrow \Pr(\text{Hamilton} \mid Y=1) < \Pr(\text{Hamilton} \mid Y=3)$$

$\Rightarrow$ the posterior prob. decrease.

(d) In (a), if we changed the text length to 10,000 words and number of instances of "upon" to 30, do you expect the posterior to increase, decrease or stay the same? Why?

A:- $$\Pr(\text{Hamilton} \mid Y=30) = \frac{3.24^{30}(1000-3.24)^{970}}{3.24^{30}(1000-3.24)^{970} + 0.23^{30}(1000-0.23)^{970}}$$

$$= \frac{1}{1+\left(\frac{0.23}{3.24}\right)^{30}\left(\frac{1000-0.23}{1000-3.24}\right)^{970}}$$

$$\Pr(\text{Hamilton} \mid Y=3) = \frac{1}{1+\left(\frac{0.23}{3.24}\right)^{3}\left(\frac{1000-0.23}{1000-3.24}\right)^{997}}$$

Mathematically, we can always compare.

Logically, $Y=30$ is highly unlikely for for both Hamilton and Madison. However, it is way more unlikely for Madison.

So, $$\frac{\Pr(Y=30 \mid \text{Madison})}{\Pr(Y=30 \mid \text{Hamilton})} < \frac{\Pr(Y=3 \mid \text{Madison})}{\Pr(Y=3 \mid \text{Hamilton})}$$

$$\Rightarrow \Pr(\text{Hamilton} \mid Y=30) = \frac{1}{1+\frac{\Pr(Y=30 \mid \text{Madison})}{\Pr(Y=30 \mid \text{Hamilton})}}$$

$$> \frac{1}{1+\frac{\Pr(Y=3 \mid \text{Madison})}{\Pr(Y=3 \mid \text{Hamilton})}} = \Pr(\text{Hamilton} \mid Y=3)$$

(e) Let Y be the number of the observed # instances of "upon" in 1000 words. Compute the posterior prob. the paper was written by Hamilton for each $Y \in \{0, 1, \ldots, 20\}$

$$\Pr(\text{Hamilton} \mid Y=y) := \frac{3.24^{y}(1000-3.24)^{1000-y}}{3.24^{y}(1000-3.24)^{1000-y} + 0.23^{y}(1000-0.23)^{1000-y}}$$

$$\Pr(\text{Madison} \mid Y=y) = \frac{0.23^{y}(1000-0.23)^{1000-y}}{3.24^{y}(1000-3.24)^{1000-y} + 0.23^{y}(1000-0.23)^{1000-y}}$$

a: plot $\Pr(\text{Hamilton} \mid Y=y)$. $\Rightarrow$ use some software, R, for example.

Q: Give a rule for the number of instances of "upon" needed before the paper should be attributed to Hamilton.

A:- $Pr(\text{Hamilton} \mid Y=y) > Pr(\text{Madison} \mid Y=y)$

$$\Rightarrow \frac{Pr(\text{Hamilton} \mid Y=y)}{Pr(\text{Madison} \mid Y=y)} > 1$$

$$\Rightarrow \frac{3.24^y (1000-3.24)^{1000-y}}{0.23^y (1000-0.23)^{1000-y}} > 1$$

$$\Rightarrow \left(\frac{3.24}{0.23}\right)^y \left(\frac{1000-3.24}{1000-0.23}\right)^{1000-y} > 1$$

$$\Rightarrow \left[\frac{3.24 \times (1000-0.23)}{0.23 \times (1000-3.24)}\right]^y > \left(\frac{1000-0.23}{1000-3.24}\right)^{1000}$$

$$\Rightarrow y > \frac{1000 \log\left(\frac{1000-0.23}{1000-3.24}\right)}{\log\left[\frac{3.24 \times (1000-0.23)}{0.23 \times (1000-3.24)}\right]} = c \text{ (say)}$$

So, if $y > c \Rightarrow$ paper should be attributed to Hamilton.

Lecture 9

Conjugate priors: Part 1

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(Edited from the lecture notes by Brian J. Reich, NC State)

Conjugate priors

- A prior is called **conjugate** if the posterior belongs to the same parametric family as that of the prior.
- We have seen that in case of a binomial likelihood and a beta prior, the posterior is also a beta.
- This requires a mathematical pairing of the likelihood and prior.
- There is a long list of conjugate priors
  `https://en.wikipedia.org/wiki/Conjugate_prior`
- The advantage of a conjugate prior is that the posterior has a closed form expression.

Why conjugate priors?

- Let us see an example of a non-conjugate prior.
- Suppose, $Y \sim \mathit{Poisson}(\lambda)$ with $\lambda \in (0, 1)$, and $\lambda \sim \text{Beta}(a, b)$.
- The posterior is

$$f(\lambda | Y) \propto \left\{ \exp(-\lambda)\lambda^{Y} \right\} \left\{ \lambda^{a-1}(1-\lambda)^{b-1} \right\} \mathbb{1}(\lambda \in [0, 1])$$

- This is not a beta PDF, so the prior is not conjugate.
- In fact, this is not a member of any known family of distributions.
- For some model parameters, there is no known conjugate prior.

Why conjugate priors?

- The final form of the posterior is

$$f(\lambda|Y) = \frac{\left\{\exp(-\lambda)\lambda^{Y}\right\}\left\{\lambda^{a-1}(1-\lambda)^{b-1}\right\}}{\int_0^1 \left\{\exp(-\lambda_0)\lambda_0^{Y}\right\}\left\{\lambda_0^{a-1}(1-\lambda)^{b-1}\right\} d\lambda_0}\mathbb{1}(\lambda_0 \in [0,1])$$

- To calculate the posterior mean, you need to evaluate

$$E(\lambda|Y) = \frac{\int_0^1 \lambda_0 \left\{\exp(-\lambda)\lambda^{Y}\right\}\left\{\lambda^{a-1}(1-\lambda)^{b-1}\right\} d\lambda_0}{\int_0^1 \left\{\exp(-\lambda_0)\lambda_0^{Y}\right\}\left\{\lambda_0^{a-1}(1-\lambda)^{b-1}\right\} d\lambda_0}.$$

- Impossible to evaluate without a software.

Example of a conjugate prior

Estimating a proportion using the beta/binomial model

- A fundamental task in statistics is to estimate a proportion using a series of trials:
  - What is the success probability of a new cancer treatment?
  - What proportion of voters support my candidate?
  - What proportion of the population has a rare gene?
- Let $\theta \in [0, 1]$ be the proportion we are trying to estimate (e.g., the success probability).
- We conduct n independent trials, each with success probability θ, and observe $Y \in \{0, ..., n\}$ successes.
- We would like obtain the posterior of θ, a 95% interval, and a test that θ equals some predetermined value θ_0.

Frequentist analysis

- The maximum likelihood estimate is the sample proportion

$$\hat{\theta} = Y/n.$$

- For large Y and $n - Y$, the sampling distribution of $\hat{\theta}$ is approximately

$$\hat{\theta} \sim \text{Normal}\left(\theta, \frac{\theta(1-\theta)}{n}\right).$$

- The standard error (standard deviation of the sampling distribution) is approximated as

$$\text{SE}(\hat{\theta}) \approx \sqrt{\frac{\hat{\theta}(1-\hat{\theta})}{n}}.$$

- A 95% CI is then

$$\hat{\theta} \pm 2\text{SE}(\hat{\theta}).$$

Bayesian analysis - Likelihood

- Since Y is the number of successes in n independent trials, each with success probability θ, its distribution is

$$Y|\theta \sim \text{Binomial}(n, \theta).$$

- PMF: $P(Y = y|\theta) = \binom{n}{y}\theta^y(1-\theta)^{n-y}$.

- Mean: $\text{E}(Y|\theta) = n\theta$.

- Variance: $\text{V}(Y|\theta) = n\theta(1-\theta)$.

Bayesian analysis - Prior

- The parameter θ can take all possible values between 0 and 1. Therefore, a natural prior is
$$\theta \sim \text{Beta}(a, b).$$
- PDF: $f(\theta) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} \theta^{a-1}(1-\theta)^{b-1}$.
- Mean: $\text{E}(\theta) = \frac{a}{a+b}$.
- Variance: $\text{V}(\theta) = \frac{ab}{(a+b)^2(a+b+1)}$.

Derivation of the posterior

- The posterior is

$$\begin{aligned}
p(\theta|Y) &\propto f(Y|\theta)\pi(\theta) \\
&\propto \theta^{Y}(1-\theta)^{n-Y} \times \theta^{a-1}(1-\theta)^{b-1}\mathbb{1}(\theta \in [0,1]) \\
&\propto \theta^{Y+a-1}(1-\theta)^{n-Y+b-1}\mathbb{1}(\theta \in [0,1]).
\end{aligned}$$

- By matching the form of the kernel with that of a beta distribution, the posterior is

$$\theta|Y \sim \text{Beta}(a+Y, b+n-Y).$$

- Our prior choice for θ was beta and the posterior we obtain is again beta.
- Thus, Beta(a, b) is a conjugate prior for θ.

Shrinkage

- The posterior mean is

$$\hat{\theta}_B = \mathrm{E}(\theta|Y) = \frac{y+a}{n+a+b}.$$

- The posterior mean is between the sample proportion Y/n and the prior mean $a/(a+b)$:

$$\hat{\theta}_B = w\frac{Y}{n} + (1-w)\frac{a}{a+b},$$

  where the weight on the sample proportion is $w = \frac{n}{n+a+b}$.

- $\hat{\theta}_B$ close to Y/n when a and b are negligible in comparison of n.

- $\hat{\theta}_B$ shrinks towards the prior mean $a/(a+b)$ when n is negligible in comparison of $a+b$.

Selecting the prior

- The posterior is $\theta | Y \sim \text{Beta}(a + Y, b + n - Y)$.
- Therefore, a and b can be interpreted as the “prior number of success and failures”.
- If we have no information about θ before collecting data, we should select a and b to be tiny in comparison of n.
- If historical data/expert opinion indicates that θ is likely between 0.6 and 0.8, we should select a and b in such a way that $\text{Prob}(0.6 < \theta < 0.8)$ is high.

Related problem

- Suppose, you are playing a computer game where you have n (known) lives.
- In each turn, either you win a point or you lose a life (computer wins).
- You keep on playing until you lose all n lives and suppose your score is Y.
- The success probability of independent stages of the game is θ.
- Y is the number of successes before we observe n failures.
- Then $Y|\theta \sim$ Negative-Binomial(n, θ) and

$$\text{Prob}(Y = y|\theta) = \binom{y+n-1}{y}\theta^y(1-\theta)^n.$$

- Assume the prior $\theta \sim$ Beta(a, b) and find the posterior.
- R code: `https://stat.ethz.ch/R-manual/R-devel/library/stats/html/NegBinomial.html`

Derivation of the posterior

- The posterior is

$$
\begin{aligned}
p(\theta|Y) &\propto f(Y|\theta)\pi(\theta) \\
&\propto \theta^Y(1-\theta)^n \times \theta^{a-1}(1-\theta)^{b-1}\mathbb{1}(\theta \in [0,1]) \\
&\propto \theta^{Y+a-1}(1-\theta)^{n+b-1}\mathbb{1}(\theta \in [0,1]).
\end{aligned}
$$

- By matching the form of the kernel with that of a beta distribution, the posterior is

$$\theta|Y \sim \mathrm{Beta}(a+Y, b+n).$$

- Our prior choice for θ was beta and the posterior we obtain is again beta.
- Thus, $\mathrm{Beta}(a,b)$ is a conjugate prior for θ.

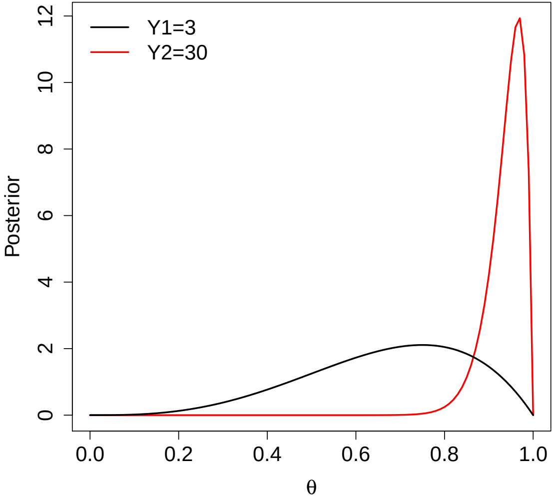
Smoking example

- Two smokers have just quit.
- Say subject i has probability θ_i of abstaining each day.
- The number of days until relapse for two patients is 3 and 30 days.
- Can we conclude the patients have different probabilities of relapse?
- What is probability that their next attempts will exceed 30 days?

Smoking example

- The likelihood is $Y_i \sim \text{Negative-Binomial}(1, \theta_i)$.
- Assume uniform priors $\theta_i \sim \text{Beta}(1, 1)$.
- The posteriors are $\theta_i | Y_i \sim \text{Beta}(Y_i + 1, 2)$.
- The posterior are plotted on the next slide.
- The following slide uses Monte Carlo sampling to address the two motivating questions.

Smoking example



Smoking example

```
> S  <- 1000000
> theta1 <- rbeta(S, 3+1, 2); theta2 <- rbeta(S, 30+1, 2)
> mean(theta2 > theta1)
[1] 0.957222
>
> samp1 <- rnbinom(S,1,prob=1-theta1)
> samp2 <- rnbinom(S,1,prob=1-theta2)
> quantile(samp1,c(0.05,0.5,0.95))
5% 50% 95%
0   1  15
> quantile(samp2,c(0.05,0.5,0.95))
5% 50% 95%
0  13 109
> mean(samp1>30); mean(samp2>30)
[1] 0.015781
[1] 0.254129
```

Related problem

- Virat Kohli captained Team India for 68 tests. The team won 40 matches, lost 17 matches and 11 matches were draw.
- Now, we have more than two categories.
- The winning, losing, and draw probabilities of independent matches are θ_1, θ_2, and θ_3, respectively. We must have $\theta_1 + \theta_2 + \theta_3 = 1$. Let $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)'$.
- $\boldsymbol{Y} = (Y_1, Y_2, Y_3)'$ is the vector of counts of different categories out of n outcomes.
- Then $\boldsymbol{Y}|\boldsymbol{\theta} \sim \text{Multinomial}(n, \boldsymbol{\theta})$ and for $\boldsymbol{y} = (y_1, y_2, y_3)'$,

$$\text{Prob}(Y = y|\theta) = \frac{n!}{y_1!y_2!y_3!}\theta_1^{y_1}\theta_2^{y_2}\theta_3^{y_3}.$$

- R code: `https://stat.ethz.ch/R-manual/R-devel/library/stats/html/Multinom.html`

Bayesian analysis - Prior

- The parameters θ_1, θ_2, and θ_3 can take all possible values between 0 and 1 along with the constraint $\theta_1 + \theta_2 + \theta_3 = 1$. Therefore, a natural prior is

$$\boldsymbol{\theta} \sim \text{Dirichlet}(a_1, a_2, a_3).$$

- The density function is

$$\pi(\theta) = \frac{\Gamma(a_1 + a_2 + a_3)}{\Gamma(a_1)\Gamma(a_2)\Gamma(a_3)} \theta_1^{a_1 - 1} \theta_2^{a_2 - 1} \theta_3^{a_3 - 1}.$$

- R code: `https://www.rdocumentation.org/packages/DirichletReg/versions/0.6-3/topics/Dirichlet`
- Mean: $\mathrm{E}(\theta_i) = \frac{a_i}{\sum_i a_i}$.
- Variance: $\mathrm{V}(\theta_i) = \frac{a_i}{\sum_i a_i} \times \left(1 - \frac{a_i}{\sum_i a_i}\right) \times \frac{1}{(1+\sum_i a_i)}$.

Derivation of the posterior

- The posterior is

$$
\begin{aligned}
p(\boldsymbol{\theta}|\boldsymbol{Y}) &\propto f(\boldsymbol{Y}|\boldsymbol{\theta})\pi(\boldsymbol{\theta}) \\
&\propto \theta_1^{Y_1}\theta_2^{Y_2}\theta_3^{Y_3} \times \theta_1^{a_1-1}\theta_2^{a_2-1}\theta_3^{a_3-1} \\
&\propto \theta_1^{Y_1+a_1-1}\theta_2^{Y_2+a_2-1}\theta_3^{Y_3+a_3-1}.
\end{aligned}
$$

- By matching the form of the kernel with that of a Dirichlet distribution, the posterior is
$$\boldsymbol{\theta}|\boldsymbol{Y} \sim \text{Dirichlet}(Y_1 + a_1, Y_2 + a_2, Y_3 + a_3).$$
- Our prior choice for $\boldsymbol{\theta}$ was Dirichlet and the posterior we obtain is again Dirichlet.
- Thus, Dirichlet(a_1, a_2, a_3) is a conjugate prior for $\boldsymbol{\theta}$.

Thank you!

Lecture 10

Conjugate priors: Part 2

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Conjugate priors: Recap

- A prior is called **conjugate** if the posterior belongs to the same parametric family as that of the prior.
- The advantage of a conjugate prior is that the posterior has a closed form expression.
- We have seen that in case of a binomial or negative binomial likelihood and a beta prior, the posterior is also a beta distribution.
- In case of a multinomial likelihood and a Dirichlet prior, the posterior distribution is also Dirichlet.

Estimating rate/intensity using the Poisson/gamma model

- Estimating a rate has many applications:
  - Number of IED explosions per day in Afghanistan
  - Number of Covid 19 cases per day at IIT-K campus
  - Number of diseased trees per square mile in a forest
- Let $\lambda > 0$ be the rate we are trying to estimate.
- We make observations over a period (or region) of length (or area) N and observe $Y \in \{0, 1, 2, ...\}$ events.
- The expected number of events is $N\lambda$ so that λ is the expected number of events per unit time.
- We would like obtain the posterior of λ, a 95% interval, and a test that λ equals some predetermined value λ_0.

Frequentist/Bayesian analysis - Likelihood

- Since Y is a count with mean $N\lambda$, a natural model is
$$Y|\lambda \sim \text{Poisson}(N\lambda).$$
- PMF: $P(Y = y|\lambda) = \frac{\exp(-N\lambda)(N\lambda)^y}{y!}$.
- Mean: $\text{E}(Y|\lambda) = N\lambda$.
- Variance: $\text{V}(Y|\lambda) = N\lambda$.

Frequentist analysis

- The maximum likelihood estimate is the sample rate
$$\hat{\lambda} = Y/N.$$
- For large N and Y, the sampling distribution of $\hat{\lambda}$ is approximately
$$\hat{\lambda} \sim \text{Normal}\left(\lambda, \frac{\lambda}{N}\right).$$
- The standard error (standard deviation of the sampling distribution) is approximated as
$$\text{SE}(\hat{\lambda}) \approx \sqrt{\frac{\hat{\lambda}}{N}}.$$
- A 95% CI (for large sample case) is then
$$\hat{\lambda} \pm 2\text{SE}(\hat{\lambda}).$$

Bayesian analysis - Prior

- The parameter λ is continuous and positive, therefore a natural prior is
$$\lambda \sim \text{Gamma}(a, b).$$
- PDF: $f(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda)$.
- Mean: $\text{E}(\lambda) = \frac{a}{b}$.
- Variance: $\text{V}(\lambda) = \frac{a}{b^2}$.

Derivation of the posterior

- The posterior is

$$
\begin{aligned}
p(\lambda|Y) &\propto f(Y|\lambda)\pi(\lambda) \\
&\propto \exp(-N\lambda)\lambda^{Y}\lambda^{a-1}\exp(-b\lambda) \\
&\propto \exp(-[N+b]\lambda)\lambda^{Y+a-1}.
\end{aligned}
$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is
$$\lambda|Y \sim \text{Gamma}(a+Y, b+N).$$

- Our prior choice for λ was gamma and the posterior we obtain is again gamma.

- Thus, Gamma(a, b) is a conjugate prior for λ.

Shrinkage

- The posterior mean is
$$\hat{\lambda}_B = \mathrm{E}(\lambda|Y) = \frac{Y+a}{N+b}.$$
- The posterior mean is between the sample rate Y/N and the prior mean a/b:
$$\hat{\lambda}_B = w\frac{Y}{N} + (1-w)\frac{a}{b},$$
where the weight on the sample rate is $w = \frac{N}{N+b}$.
- $\hat{\lambda}_B$ close to Y/N, when b is negligible in comparison of N.
- $\hat{\lambda}_B$ shrinks towards the prior mean a/b, when N/b is negligible.

Selecting the prior

- The posterior is $\lambda | Y \sim \text{Gamma}(a + Y, b + N)$.
- Therefore, a and b can be interpreted as the "prior number of events and observation time".
- This is useful for specifying the prior.
- If we have no information about λ before collecting data, we should select a and b to be tiny; for example, $a = 0.1, b = 0.1$.
- If historical data/expert opinion indicates that λ is likely between 0.6 and 0.8, we should select a and b in such a way that $\text{Prob}(0.6 < \lambda < 0.8)$ is high.

Posterior with two observations

- Derive the posterior if $Y_1 \sim \text{Poisson}(N_1\lambda)$; $Y_2 \sim \text{Poisson}(N_2\lambda)$; and $\lambda \sim \text{Gamma}(a, b)$. If not mentioned, assume Y_1 and Y_2 to be independent.
- The posterior is

$$\begin{aligned} p(\lambda|Y_1, Y_2) &\propto f(Y_1, Y_2|\lambda)\pi(\lambda) \\ &\propto f(Y_1|\lambda)f(Y_2|\lambda)\pi(\lambda) \\ &\propto \exp(-N_1\lambda)\lambda^{Y_1}\exp(-N_2\lambda)\lambda^{Y_2}\lambda^{a-1}\exp(-b\lambda) \\ &\propto \exp(-[N_1 + N_2 + b]\lambda)\lambda^{Y_1+Y_2+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\lambda|Y_1, Y_2 \sim \text{Gamma}(a + Y_1 + Y_2, b + N_1 + N_2).$$

Posterior with two observations

- Derive the posterior if $Y_1 \sim \text{Poisson}(N_1\lambda)$; $Y_2 \sim \text{Poisson}(N_2\lambda)$; and $\lambda \sim \text{Gamma}(a, b)$. If not mentioned, assume Y_1 and Y_2 to be independent.

- The posterior is

$$\begin{aligned} p(\lambda|Y_1, Y_2) &\propto f(Y_1, Y_2|\lambda)\pi(\lambda) \\ &\propto f(Y_1|\lambda)f(Y_2|\lambda)\pi(\lambda) \\ &\propto \exp(-N_1\lambda)\lambda^{Y_1}\exp(-N_2\lambda)\lambda^{Y_2}\lambda^{a-1}\exp(-b\lambda) \\ &\propto \exp(-[N_1 + N_2 + b]\lambda)\lambda^{Y_1+Y_2+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\lambda|Y_1, Y_2 \sim \text{Gamma}(a + Y_1 + Y_2, b + N_1 + N_2).$$

Posterior with m observations

- Derive the posterior if $Y_1, ..., Y_m \sim \text{Poisson}(N\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.
- The posterior is

$$
\begin{aligned}
p(\lambda|Y_1,\ldots,Y_m) &\propto f(Y_1,\ldots,Y_m|\lambda)\pi(\lambda) \\
&\propto f(Y_1|\lambda)\times\ldots\times f(Y_m|\lambda)\pi(\lambda) \\
&\propto \exp(-N\lambda)\lambda^{Y_1}\times\ldots\times\exp(-N\lambda)\lambda^{Y_m}\lambda^{a-1}\exp(-b\lambda) \\
&\propto \exp(-[mN+b]\lambda)\lambda^{\sum_{i=1}^m Y_i+a-1}.
\end{aligned}
$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\lambda|Y_1,\ldots,Y_m \sim \text{Gamma}\left(a+\sum_{i=1}^m Y_i, b+mN\right).$$

Posterior with m observations

- Derive the posterior if $Y_1, ..., Y_m \sim \text{Poisson}(N\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.
- The posterior is

$$
\begin{aligned}
p(\lambda|Y_1, \ldots, Y_m) &\propto f(Y_1, \ldots, Y_m|\lambda)\pi(\lambda) \\
&\propto f(Y_1|\lambda) \times \ldots \times f(Y_m|\lambda)\pi(\lambda) \\
&\propto \exp(-N\lambda)\lambda^{Y_1} \times \ldots \times \exp(-N\lambda)\lambda^{Y_m}\lambda^{a-1}\exp(-b\lambda) \\
&\propto \exp(-[mN + b]\lambda)\lambda^{\sum_{i=1}^m Y_i + a - 1}.
\end{aligned}
$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$
\lambda|Y_1, \ldots, Y_m \sim \text{Gamma}\left(a + \sum_{i=1}^m Y_i, b + mN\right).
$$

AB testing example

- A tech company runs their regular user interface for $N_1 = 8$ hours and gets $Y_1 = 4721$ clicks.
- The next day they launch a new user interface for $N_2 = 8$ hours and get $Y_2 = 5209$ clicks.
- Assuming non-informative conjugate priors, determine if the new user interface has a higher click rate.

AB testing example

- Period 1: the likelihood is $Y_1|\lambda_1 \sim \text{Poisson}(N_1\lambda_1)$.
- The conjugate prior is $\lambda_1 \sim \text{Gamma}(a, b)$.
- The posterior is $\lambda_1|Y_1 \sim \text{Gamma}(Y_1 + a, N_1 + b)$.
- Period 2: Similarly, $\lambda_2|Y_2 \sim \text{Gamma}(Y_2 + a, N_2 + b)$.

Monte Carlo approximation

```
> S <- 100000
> a <- b <- 0.1
> N1 <- N2 <- 8
> Y1 <- 4721
> Y2 <- 5209
>
> # MC samples
> lambda1 <- rgamma(S,Y1+a,N1+b)
> lambda2 <- rgamma(S,Y2+a,N2+b)
>
> # Prob(new interface is better|data)
> mean(lambda2>lambda1)
[1] 1
> # The new interface almost surely works!
```

Exponential models

- The first continuous distribution we will discuss is the exponential distribution, and we denote it by $Y \sim \text{Exponential}(\lambda)$ where $\lambda > 0$.
- Domain: $Y \in (0, \infty)$.
- PDF: $f(y) = \lambda \exp(-\lambda y)$.
- Mean: $\text{E}(Y|\lambda) = \frac{1}{\lambda}$.
- Variance: $\text{V}(Y|\lambda) = \frac{1}{\lambda^2}$.

Bayesian analysis - Prior

- The parameter λ is continuous and positive, therefore a natural prior is
$$\lambda \sim \text{Gamma}(a, b).$$
- PDF: $f(\lambda) = \frac{b^a}{\Gamma(a)} \lambda^{a-1} \exp(-b\lambda)$.
- Mean: $\text{E}(\lambda) = \frac{a}{b}$.
- Variance: $\text{V}(\lambda) = \frac{a}{b^2}$.

Derivation of the posterior

- The posterior is

$$\begin{aligned} p(\lambda|Y) &\propto f(Y|\lambda)\pi(\lambda) \\ &\propto \lambda \exp\left(-\lambda Y\right) \lambda^{a-1} \exp(-b\lambda) \\ &\propto \exp(-[Y+b]\lambda)\lambda^{1+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is
$$\lambda|Y \sim \text{Gamma}(a+1, b+Y).$$
- Our prior choice for λ was gamma and the posterior we obtain is again gamma.
- Thus, Gamma(a, b) is a conjugate prior for λ.

Posterior with n observations

- Derive the posterior if $Y_1, ..., Y_n \sim \text{Exponential}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.
- The posterior is

$$\begin{aligned} p(\lambda | Y_1, \ldots, Y_n) &\propto f(Y_1, \ldots, Y_n | \lambda)\pi(\lambda) \\ &\propto f(Y_1|\lambda) \times \ldots \times f(Y_n|\lambda)\pi(\lambda) \\ &\propto \lambda \exp(-\lambda Y_1) \times \ldots \times \lambda \exp(-\lambda Y_n)\lambda^{a-1}\exp(-b\lambda) \\ &\propto \exp\left(-\left[\sum_{i=1}^{n} Y_i + b\right]\lambda\right)\lambda^{n+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\lambda | Y_1, \ldots, Y_n \sim \text{Gamma}\left(a + n, b + \sum_{i=1}^{n} Y_i\right).$$

Posterior with n observations

- Derive the posterior if $Y_1, ..., Y_n \sim \text{Exponential}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.
- The posterior is

$$\begin{aligned} p(\lambda|Y_1,\ldots,Y_n) &\propto f(Y_1,\ldots,Y_n|\lambda)\pi(\lambda) \\ &\propto f(Y_1|\lambda)\times\ldots\times f(Y_n|\lambda)\pi(\lambda) \\ &\propto \lambda\exp(-\lambda Y_1)\times\ldots\times\lambda\exp(-\lambda Y_n)\lambda^{a-1}\exp(-b\lambda) \\ &\propto \exp\left(-\left[\sum_{i=1}^n Y_i + b\right]\lambda\right)\lambda^{n+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\lambda|Y_1,\ldots,Y_n \sim \text{Gamma}\left(a+n, b+\sum_{i=1}^n Y_i\right).$$

Gamma models with known shape parameter

- The next continuous distribution we will discuss is the gamma distribution, and we denote it by $Y \sim \text{Gamma}(\alpha, \beta)$ where $\alpha > 0$ is known and $\beta > 0$.
- Domain: $Y \in (0, \infty)$.
- PDF: $f(y) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} \exp(-\beta y)$.
- Mean: $E(Y|\beta) = \frac{\alpha}{\beta}$.
- Variance: $V(Y|\beta) = \frac{\alpha}{\beta^2}$.

Bayesian analysis - Prior

- The parameter β is continuous and positive, therefore a natural prior is

$$\beta \sim \mathrm{Gamma}(a, b).$$

- PDF: $f(\beta) = \frac{b^a}{\Gamma(a)} \beta^{a-1} \exp(-b\beta)$.

- Mean: $\mathrm{E}(\beta) = \frac{a}{b}$.

- Variance: $\mathrm{V}(\beta) = \frac{a}{b^2}$.

Derivation of the posterior

- The posterior is

$$\begin{aligned} p(\beta|Y) &\propto f(Y|\beta)\pi(\beta) \\ &\propto \beta^{\alpha}\exp(-\beta Y)\beta^{a-1}\exp(-b\beta) \\ &\propto \exp(-[Y+b]\beta)\beta^{\alpha+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\beta|Y \sim \text{Gamma}(a+\alpha, b+Y).$$

- Our prior choice for β was gamma and the posterior we obtain is again gamma.

- Thus, Gamma(a, b) is a conjugate prior for β.

Posterior with n observations

- Derive the posterior if $Y_1, ..., Y_n \sim \text{Gamma}(\alpha, \beta)$ and $\beta \sim \text{Gamma}(a, b)$.
- The posterior is

$$\begin{aligned} p(\beta|Y_1,\ldots,Y_n) &\propto f(Y_1,\ldots,Y_n|\beta)\pi(\beta) \\ &\propto f(Y_1|\beta)\times\ldots\times f(Y_n|\beta)\pi(\beta) \\ &\propto \beta^{\alpha}\exp(-\beta Y_1)\times\ldots\times\beta^{\alpha}\exp(-\beta Y_n)\beta^{a-1}\exp(-b\beta) \\ &\propto \exp\left(-\left[\sum_{i=1}^{n} Y_i + b\right]\beta\right)\beta^{n\alpha+a-1}. \end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\beta|Y_1,\ldots,Y_n \sim \text{Gamma}\left(a+n\alpha, b+\sum_{i=1}^{n} Y_i\right).$$

Posterior with n observations

- Derive the posterior if $Y_1, ..., Y_n \sim \text{Gamma}(\alpha, \beta)$ and $\beta \sim \text{Gamma}(a, b)$.
- The posterior is

$$\begin{aligned}
p(\beta | Y_1, \ldots, Y_n) &\propto f(Y_1, \ldots, Y_n | \beta)\pi(\beta) \\
&\propto f(Y_1|\beta) \times \ldots \times f(Y_n|\beta)\pi(\beta) \\
&\propto \beta^{\alpha} \exp(-\beta Y_1) \times \ldots \times \beta^{\alpha} \exp(-\beta Y_n)\beta^{a-1} \exp(-b\beta) \\
&\propto \exp\left(-\left[\sum_{i=1}^{n} Y_i + b\right]\beta\right)\beta^{n\alpha+a-1}.
\end{aligned}$$

- By matching the form of the kernel with that of a gamma distribution, the posterior is

$$\beta | Y_1, \ldots, Y_n \sim \text{Gamma}\left(a + n\alpha, b + \sum_{i=1}^{n} Y_i\right).$$

Thank you!

Lecture 11

Conjugate priors: Part 3

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Conjugate priors: Recap

- A prior is called **conjugate** if the posterior belongs to the same parametric family as that of the prior.
- We have seen that in case of a binomial or negative binomial likelihood and a beta prior, the posterior is also a beta distribution.
- In case of a multinomial likelihood and a Dirichlet prior, the posterior distribution is also Dirichlet.
- In case of a Poisson or exponential or gamma (with known shape parameter) likelihood and a gamma prior, the posterior distribution is also gamma.

Gaussian models (univariate)

- The final (and the most important) distribution we will discuss is the Gaussian (normal) distribution, and we denote it by $Y \sim \text{Normal}(\mu, \sigma^2)$.
  - Domain: $Y \in (-\infty, \infty)$.
  - PDF: $f(y) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2\right]$.
  - Mean: $E(Y|\mu, \sigma^2) = \mu$.
  - Variance: $V(Y|\mu, \sigma^2) = \sigma^2$.
- Today we will discuss:
  - Estimating the mean assuming the variance is known.
  - Estimating the variance assuming the mean is known.
- For estimating mean and variance both, we will use MCMC!

Estimating a normal mean with known variance - Likelihood

- We assume the data consist of n independent and identically distributed observations $Y_1, \dots, Y_n$. Let $\mathbf{Y} = (Y_1, \dots, Y_n)'$.
- Each Y_i is Gaussian (given μ),

$$Y_i|\mu \sim \text{Normal}(\mu, \sigma^2),$$

  where σ is known.
- The likelihood is then

$$\prod_{i=1}^{n} f(y_i|\mu) = \left(\frac{1}{\sqrt{2\pi\sigma}}\right)^n \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2\right].$$

Bayesian analysis - Prior

- The parameter μ is continuous over the entire real line, therefore a natural prior is
$$\mu \sim \text{Normal}(\theta, \eta^2).$$
- The prior mean θ is the best guess before we observe data.
- The math is slightly more interpretable if we set $\eta^2 = \frac{\sigma^2}{m}$.
- As we will see, the prior variance controls the strength of the prior via $m > 0$.

Derivation of the posterior

- The posterior is

$$
\begin{aligned}
p(\mu|\mathbf{Y}) &\propto \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(Y_i-\mu)^2\right] \times \exp\left[-\frac{m}{2\sigma^2}(\mu-\theta)^2\right] \\
&\propto \exp\left[-\frac{n\mu^2}{2\sigma^2}\right] \times \exp\left[\frac{\mu\sum_{i=1}^{n}Y_i}{\sigma^2}\right] \times \exp\left[-\frac{m\mu^2}{2\sigma^2}\right] \times \exp\left[\frac{m\mu\theta}{\sigma^2}\right] \\
&= \exp\left[-\frac{1}{2}\mu^2\frac{n+m}{\sigma^2}\right] \times \exp\left[\mu\frac{\left(\sum_{i=1}^{n}Y_i+m\theta\right)}{\sigma^2}\right].
\end{aligned}
$$

By matching the terms with those in the density of a normal distribution, we have

$$
\mu|\mathbf{Y} \sim \text{Normal}\left(\frac{\sum_{i=1}^{n}Y_i+m\theta}{n+m}, \frac{\sigma^2}{n+m}\right).
$$

Shrinkage

- The posterior mean is

$$\hat{\mu}_B = \mathrm{E}(\mu|\mathbf{Y}) = \frac{\sum_{i=1}^{n} Y_i + m\theta}{n+m} = w\bar{Y} + (1-w)\theta,$$

where $w = n/(n+m)$.

- Therefore, if m is small then $\hat{\mu}_B \approx \bar{Y}$, and if m is large $\hat{\mu}_B \approx \theta$.

- If no prior information is available, take m to be small and thus the prior is non-informative.

- Small m gives large prior variance (relative to σ).

Shrinkage

- The posterior variance is
$$\mathrm{V}(\mu|\mathbf{Y}) = \frac{\sigma^2}{n+m}.$$
- The sampling variance of $\bar{Y}$ is $\frac{\sigma^2}{n}$.
- Therefore, we can loosely interpret m as the "prior number of observations".
- If m is small then $\mathrm{V}(\mu|\mathbf{Y}) \approx \mathrm{V}(\bar{Y})$, and if m is large $\mathrm{V}(\mu|\mathbf{Y}) \approx \mathrm{V}(\mu)$.

Estimating a normal variance with known mean - Likelihood

- We assume the data consist of n independent and identically distributed observations $Y_1, ..., Y_n$. Let $\mathbf{Y} = (Y_1, \ldots, Y_n)'$.
- Each Y_i is Gaussian (given σ^2),
$$Y_i|\sigma^2 \sim \text{Normal}(\mu, \sigma^2),$$
where μ is known.
- The likelihood is then
$$\prod_{i=1}^{n} f(y_i|\sigma^2) = \left(\frac{1}{\sqrt{2\pi}\sigma}\right)^n \exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mu)^2\right].$$

Bayesian analysis - Prior

- The parameter σ^2 is continuous over $(0, \infty)$, therefore a natural prior is $\sigma^2 \sim \text{Gamma}(a, b)$.
- However, the math is easier if we pick a gamma prior for the inverse variance (precision) $1/\sigma^2$.
- If $1/\sigma^2 \sim \text{Gamma}(a, b)$, then $\sigma^2 \sim \text{Inverse-Gamma}(a, b)$.
- This is the definition of the inverse-gamma distribution (Lecture 1).
- The inverse-gamma prior for σ^2 has PDF

$$f(\sigma^2) = \frac{b^a}{\Gamma(a)}(\sigma^2)^{-a-1}\exp(-b/\sigma^2).$$

Derivation of the posterior

- The posterior is

$$\begin{aligned}p(\sigma^2|\mathbf{Y}) &\propto (\sigma^2)^{-n/2}\exp\left[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(Y_i-\mu)^2\right]\times(\sigma^2)^{-a-1}\exp(-b/\sigma^2)\\ &= (\sigma^2)^{-(n/2+a)-1}\exp\left[-\frac{1}{\sigma^2}\left(\frac{1}{2}\sum_{i=1}^{n}(Y_i-\mu)^2+b\right)\right].\end{aligned}$$

- The posterior is

$$\sigma^2|\mathbf{Y}\sim\text{Inverse-Gamma}\left(n/2+a, SSE/2+b\right),$$

where $SSE=\sum_{i=1}^{n}(Y_i-\mu)^2$.

Shrinkage

- The mean of an Inverse-Gamma(a, b) distribution only exists if $a > 1$.
- The prior mean (if it exists) is $b/(a-1)$.
- The posterior mean is
$$\frac{SSE + 2b}{n + 2a - 2}$$
- It is common to take a and b to be small to give a non-informative prior.
- For large n and comparatively negligible a and b, the posterior mean is approximately same as the sample variance SSE/n.

Conjugate prior for a normal precision

- The precision is the inverse variance, $\tau = 1/\sigma^2$.
- If Y_i have mean μ and precision τ, the likelihood is proportional to

$$\prod_{i=1}^{n} f(y_i|\tau) \propto \tau^{n/2} \exp\left[-\frac{\tau}{2}\sum_{i=1}^{n}(y_i-\mu)^2\right]$$

- If $\tau \sim \text{Gamma}(a, b)$, then

$$\tau|Y \sim \text{Gamma}(n/2 + a, SSE/2 + b).$$

- This is the exact same analysis as the inverse-gamma prior for the variance.

Gaussian models (multivariate)

- The final multivariate distribution we will discuss is the multivariate normal distribution, and we denote the P-variate case by $\mathbf{Y} \sim \text{Normal}_P(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.
  - Domain: $\mathbf{Y} \in \mathbb{R}^P = (-\infty, \infty)^P$.
  - PDF: $f(\mathbf{y}) = |2\pi\boldsymbol{\Sigma}|^{-1/2} \exp\left[-\frac{1}{2}(\mathbf{y}-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{y}-\boldsymbol{\mu})\right]$.
  - Mean vector: $\text{E}(\mathbf{Y}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \boldsymbol{\mu}$.
  - Covariance matrix: $\text{Cov}(\mathbf{Y}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \boldsymbol{\Sigma}$. Here, $\boldsymbol{\Sigma}$ is a positive definite (PD) matrix.
- Today we will discuss:
  - Estimating the mean vector assuming the covariance matrix is known.
  - Estimating the covariance matrix assuming the mean vector is known.
- For estimating mean vector and covariance matrix both, we will use MCMC!

Estimating a normal mean vector with known covariance - Likelihood

- We assume the data consist of n independent and identically distributed observation vectors $\mathbf{Y}_1, \ldots, \mathbf{Y}_n$. Let $\mathcal{Y} = \{\mathbf{Y}_1, \ldots, \mathbf{Y}_n\}$.
- Each $\mathbf{Y}_i$ is Gaussian (given $\boldsymbol{\mu}$),

$$\mathbf{Y}_i | \boldsymbol{\mu} \sim \text{Normal}_P(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

  where $\boldsymbol{\Sigma}$ is known.
- The likelihood is then

$$\prod_{i=1}^{n} f(\mathbf{y}_i|\boldsymbol{\mu}) = |2\pi\boldsymbol{\Sigma}|^{-n/2} \exp\left[-\frac{1}{2}\sum_{i=1}^{n}(\mathbf{y}_i - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{y}_i - \boldsymbol{\mu})\right].$$

Bayesian analysis - Prior

- The parameter vector $\boldsymbol{\mu}$ is continuous over $\mathbb{R}^P$, therefore a natural prior is

$$\boldsymbol{\mu} \sim \text{Normal}_P(\boldsymbol{\mu}_0, \boldsymbol{\Sigma}_0).$$

- The prior mean $\boldsymbol{\mu}_0$ is the best guess before we observe data.

- The math is slightly more interpretable if we set $\boldsymbol{\Sigma}_0 = \frac{1}{m}\boldsymbol{\Sigma}$.

- As we will see, the prior variance controls the strength of the prior via $m > 0$.

Derivation of the posterior

- The posterior is

$$
\begin{aligned}
p(\boldsymbol{\mu}|\mathcal{Y}) &\propto \exp\left[-\frac{1}{2}\sum_{i=1}^{n}(\mathbf{Y}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{Y}_i-\boldsymbol{\mu})\right]\times\exp\left[-\frac{m}{2}(\boldsymbol{\mu}-\boldsymbol{\mu}_0)'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu}-\boldsymbol{\mu}_0)\right] \\
&\propto \exp\left[-\frac{n+m}{2}\boldsymbol{\mu}'\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}\right]\times\exp\left[\boldsymbol{\mu}'\boldsymbol{\Sigma}^{-1}\left(\sum_{i=1}^{n}\mathbf{Y}_i+m\boldsymbol{\mu}_0\right)\right].
\end{aligned}
$$

By matching the terms with those in the density of a multivariate normal distribution, we have

$$
\boldsymbol{\mu}|\mathcal{Y}\sim \text{Normal}_P\left(\frac{\sum_{i=1}^{n}\mathbf{Y}_i+m\boldsymbol{\mu}_0}{n+m},\frac{1}{n+m}\boldsymbol{\Sigma}\right).
$$

Shrinkage

- The posterior mean is

$$\hat{\mu}_B = \mathrm{E}(\mu|\mathcal{Y}) = \frac{\sum_{i=1}^{n} \mathbf{Y}_i + m\mu_0}{n+m} = w\bar{\mathbf{Y}} + (1-w)\mu_0,$$

  where $w = n/(n+m)$ and $\bar{\mathbf{Y}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{Y}_i$.

- Therefore, if m is small then $\hat{\mu}_B \approx \bar{\mathbf{Y}}$, and if m is large $\hat{\mu}_B \approx \mu_0$.

- If no prior information is available, take m to be small and thus the prior is non-informative.

- Small m gives large prior variances and covariances (relative to the components of $\boldsymbol{\Sigma}$).

Shrinkage

- The posterior covariance matrix is

$$\text{Cov}(\boldsymbol{\mu}|\mathcal{Y}) = \frac{1}{n+m}\boldsymbol{\Sigma}.$$

- The sampling variance of $\bar{\mathbf{Y}}$ is $\frac{1}{n}\boldsymbol{\Sigma}$.

- Therefore, we can loosely interpret m as the “prior number of observations”.

- If m is small then $\text{Cov}(\boldsymbol{\mu}|\mathcal{Y}) \approx \text{Cov}(\bar{\mathbf{Y}})$, and if m is large $\text{Cov}(\boldsymbol{\mu}|\mathcal{Y}) \approx \text{Cov}(\boldsymbol{\mu})$.

Estimating a normal covariance matrix with known mean - Likelihood

- We assume the data consist of n independent and identically distributed observation vectors $\mathbf{Y}_1, \ldots, \mathbf{Y}_n$. Let $\mathcal{Y} = \{\mathbf{Y}_1, \ldots, \mathbf{Y}_n\}$.
- Each $\mathbf{Y}_i$ is Gaussian (given $\boldsymbol{\Sigma}$),

$$\mathbf{Y}_i | \boldsymbol{\Sigma} \sim \text{Normal}_P(\boldsymbol{\mu}, \boldsymbol{\Sigma}),$$

  where $\boldsymbol{\mu}$ is known.
- The likelihood is then

$$\prod_{i=1}^{n} f(\mathbf{y}_i | \boldsymbol{\Sigma}) = |2\pi\boldsymbol{\Sigma}|^{-n/2} \exp\left[-\frac{1}{2} \sum_{i=1}^{n} (\mathbf{y}_i - \boldsymbol{\mu})' \boldsymbol{\Sigma}^{-1} (\mathbf{y}_i - \boldsymbol{\mu}) \right].$$

Bayesian analysis - Prior

- The parameter is neither scalar nor vector and we need to assign a prior for a random matrix $\boldsymbol{\Sigma}$ that is PD.
- Here, a natural prior is $\text{Wishart}_P(\boldsymbol{W}, \nu)$ where $\nu > P - 1$ and $\boldsymbol{V}$ is PD.
- However, the math is easier if we pick a Wishart prior for the inverse covariance (precision) matrix $\boldsymbol{\Sigma}^{-1}$.
- If $\boldsymbol{\Sigma}^{-1} \sim \text{Wishart}_P(\boldsymbol{W}, \nu)$, then $\boldsymbol{\Sigma} \sim \text{Inverse-Wishart}_P(\boldsymbol{V}, \nu)$, where $\boldsymbol{V} = \boldsymbol{W}^{-1}$.
- This is the definition of the inverse-Wishart distribution.
- The inverse-Wishart prior for $\boldsymbol{\Sigma}$ has PDF

$$f(\boldsymbol{\Sigma}) = \frac{|\boldsymbol{V}|^{\nu/2}}{2^{\nu P/2}\Gamma_P(\nu/2)} |\boldsymbol{\Sigma}|^{-(\nu+P+1)/2} \exp\left[-\frac{1}{2}\text{Tr}(\boldsymbol{V}\boldsymbol{\Sigma}^{-1})\right].$$

Derivation of the posterior

- The posterior is

$$
\begin{aligned}
p(\boldsymbol{\Sigma}|\mathcal{Y}) &\propto |\boldsymbol{\Sigma}|^{-n/2}\exp\left[-\frac{1}{2}\sum_{i=1}^{n}(\mathbf{Y}_i-\boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{Y}_i-\boldsymbol{\mu})\right] \\
&\qquad \times|\boldsymbol{\Sigma}|^{-(\nu+P+1)/2}\exp\left[-\frac{1}{2}\mathrm{Tr}(\boldsymbol{V}\boldsymbol{\Sigma}^{-1})\right] \\
&= |\boldsymbol{\Sigma}|^{-(n+\nu+P+1)/2}\exp\left[-\frac{1}{2}\mathrm{Tr}\left(\left\{\sum_{i=1}^{n}(\mathbf{Y}_i-\boldsymbol{\mu})(\mathbf{Y}_i-\boldsymbol{\mu})'+\boldsymbol{V}\right\}\boldsymbol{\Sigma}^{-1}\right)\right].
\end{aligned}
$$

- The posterior is

$$
\boldsymbol{\Sigma}|\mathcal{Y}\sim \text{Inverse-Wishart}_P\left(SSE+\boldsymbol{V}, n+\nu\right),
$$

where $SSE=\sum_{i=1}^{n}(\mathbf{Y}_i-\boldsymbol{\mu})(\mathbf{Y}_i-\boldsymbol{\mu})'$.

Shrinkage

- The mean of an Inverse-Wishart$_P(\boldsymbol{V}, \nu)$ distribution only exists if $\nu > P + 1$.
- The prior mean (if it exists) is $\frac{1}{\nu - P - 1}\boldsymbol{V}$.
- The posterior mean of $\boldsymbol{\Sigma}$ is

$$\frac{SSE + \boldsymbol{V}}{n + \nu - P - 1}.$$

- It is common to take ν and all the elements of $\boldsymbol{V}$ to be small to give a non-informative prior.
- For large n and comparatively negligible ν and the elements of $\boldsymbol{V}$, the posterior mean is approximately same as the sample variance SSE/n.

Conjugate prior for a normal precision matrix

- The precision matrix is the inverse covariance matrix, $\boldsymbol{\Omega} = \boldsymbol{\Sigma}^{-1}$.
- If $\boldsymbol{Y}_i$'s have mean vector $\boldsymbol{\mu}$ and precision matrix $\boldsymbol{\Omega}$, the likelihood is proportional to

$$\prod_{i=1}^{n} f(\mathbf{y}_i|\boldsymbol{\Omega}) \propto |\boldsymbol{\Omega}|^{n/2} \exp\left[-\frac{1}{2}\sum_{i=1}^{n}(\mathbf{y}_i - \boldsymbol{\mu})'\boldsymbol{\Omega}(\mathbf{y}_i - \boldsymbol{\mu})\right].$$

- If $\boldsymbol{\Omega} \sim \text{Wishart}_P(\boldsymbol{W}, \nu)$, then its PDF is given by

$$f(\boldsymbol{\Omega}) = \frac{|\boldsymbol{W}|^{-\nu/2}}{2^{\nu P/2}\Gamma_P(\nu/2)}|\boldsymbol{\Omega}|^{(\nu-P-1)/2} \exp\left[-\frac{1}{2}\text{Tr}(\boldsymbol{W}^{-1}\boldsymbol{\Omega})\right].$$

- In that case, we have $\boldsymbol{\Omega}|\mathcal{Y} \sim \text{Wishart}([SSE + \boldsymbol{W}^{-1}]^{-1}, n + \nu)$.
- This is the exact same analysis as the inverse-Wishart prior for covariance.

Thank you!

Lecture 12

Objective priors: Part 1

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Informative versus non-informative priors

- In some cases informative priors are available.
- Potential sources include: literature reviews; pilot studies; expert opinions; etc.
- **Prior elicitation** is the process of converting expert opinions to prior distribution.
- For example, an expert might not comprehend an inverse-gamma PDF, but if they give you a point estimate and a spread, you can back calculate a and b.
- There are tools for this, such as `https://jeremy-oakley.shinyapps.io/SHELF-single/`

Informative versus non-informative priors

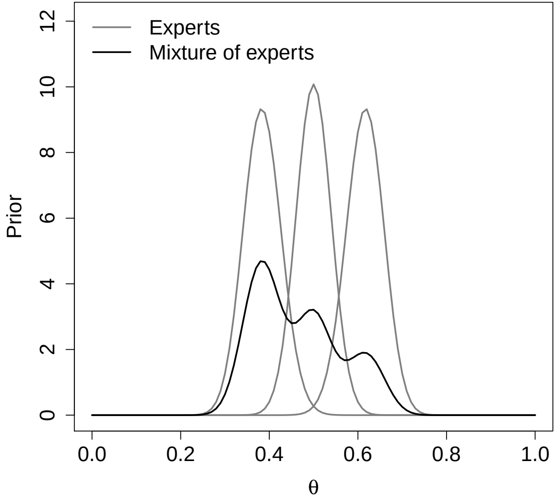
- Strong priors for the parameters of interest can be hard to defend.
- Strong priors for nuisance parameters are more common.
- For example, when you are doing a Bayesian t-test to study the mean μ, you might use an informative prior for the nuisance parameter σ^2.
- Anytime informative priors are used, you should conduct a **sensitivity analysis**.
- That is, compare the posterior for several priors.

Prior elicitation

- Prior information can come from a variety of sources.
- Say that source j (e.g., journal article, expert, pilot study) suggests prior $\pi_j(\theta)$.
- A **mixture of experts** prior combines the J sources into a single prior.
- Say source j is given weight $w_j > 0$ with $\sum_{j=1}^{J} w_j = 1$.
- The mixture of experts prior is

$$\pi(\theta) = \sum_{j=1}^{J} w_j \pi_j(\theta).$$

Mixture of experts prior ($w_j \in \{0.2, 0.3, 0.5\}$)



Mixture of experts prior

- The posterior distribution under the mixture of experts prior is

$$
\begin{aligned}
p(\theta|\mathbf{Y}) &\propto f(\mathbf{Y}|\theta) \times \pi(\theta) \\
&\propto \sum_{j=1}^{J} w_j f(\mathbf{Y}|\theta)\pi_j(\theta) \\
&\propto \sum_{j=1}^{J} Q_j p_j(\theta|\mathbf{Y}).
\end{aligned}
$$

- Here $p_j(\theta|\mathbf{Y}) \propto f(\mathbf{Y}|\theta)\pi_j(\theta)$ is the posterior under prior π_j.
- If each π_j is conjugate, the p_j belong to the same family as that of π_j.
- The posterior is also a mixture distribution over the same parametric family as the prior, and thus the prior is conjugate.

Prior elicitation

- Another powerful idea is to elicit prior information using real-life scenarios.
- “Do you expect more events on hot days than cold days?”
- “Would you expect more events on a rainy day in August or a sunny day in December?”
- Later you can frame these questions using the parameters in your model and back out the prior that best matches the expert responses.

Informative versus non-informative priors

- In most cases prior information is not available and so non-informative priors are used.
- Other names: “vague”, “weak”, “flat”, “diffuse”, etc.
- These all refer to priors with large variance.
- Examples: $\theta \sim \text{Uniform}(0, 1)$ or $\mu \sim \text{Normal}(0, 1000^2)$.
- Non-informative priors can be conjugate or not conjugate.
- The idea is that the likelihood overwhelms the prior.
- You should verify this with a sensitivity analysis.

Improper priors

- Extreme case: $\mu \sim \text{Normal}(0, \tau^2)$ and we set $\tau = \infty$.
- A "prior" that doesn't integrate to one is called an **improper** prior.
- Example: $\pi(\mu) = 1$ for all $\mu \in \mathbb{R}$.
- It's OK to use an improper prior so long as you verify that the posterior integrates to one.
- Subjective Bayesian interpretation is tricky.

Improper prior for the model Normal($\mu, 1$)

- Observations: $Y_1, \ldots, Y_n \overset{IID}{\sim}$ Normal($\mu, 1$). Let $\bar{Y} = (Y_1 + \ldots + Y_n)/n$.
- Likelihood: $f(Y_1, \ldots, Y_n|\mu) \propto \exp\left[-\frac{1}{2}\sum_{i=1}^{n}(Y_i - \mu)^2\right] \propto \exp\left[-\frac{n}{2}(\mu - \bar{Y})^2\right]$.
- Prior: $\pi(\mu) = 1$ for all $\mu \in \mathbb{R}$.
- Posterior: $p(\mu|Y_1, \ldots, Y_n) \propto f(Y_1, \ldots, Y_n|\mu)\pi(\mu) \propto \exp\left[-\frac{n}{2}(\mu - \bar{Y})^2\right]$
- Thus, $\mu|Y_1, \ldots, Y_n \sim$ Normal($\bar{Y}, \frac{1}{n}$).
- Despite the prior being improper, posterior is proper.

Subjective versus objective priors

- A subjective Bayesian picks a prior that corresponds to their current state of knowledge before collecting data.
- Of course, if the readers do not share this prior, then they might not accept the analysis, and so non-informative priors and a sensitivity analysis are common.
- An **objective analysis** is one that requires no subjective decisions by the analyst.
- Subjective decisions include picking the likelihood, treatment of outliers, transformations, ... and prior specification.
- A completely objective analysis may be feasible in tightly controlled experiments, but is impossible in many analyses.

Objective Bayes

- An objective Bayesian attempts to replace the subjective choice of prior with an algorithm that determines the prior.
- There are many approaches: Jeffreys, reference, probability matching, maximum entropy, empirical Bayes, penalized complexity, etc.
- Jeffreys priors are the most common in the literature.
- Many of these priors are **improper** and so you have to check that the posterior is proper.

Objective Bayes

- One subjective decision is the parameterization to use.
- For example, should we select $\sigma \sim \text{Uniform}(0, 10)$ or $\sigma^2 \sim \text{Uniform}(0, 100)$?
- These are quite different; $\sigma \sim \text{Uniform} \rightarrow p(\sigma^2) \propto 1/\sigma$.
- Any objective prior must give the same results, e.g., posterior mean for σ, for any parameterization.
- The Jeffreys prior is invariant to transformations.

Jeffreys prior

- The Jeffreys prior for parameter θ is $p(\theta) \propto \sqrt{I(\theta)}$.
- The Fisher information is $I(\theta) = -\mathrm{E}_{Y|\theta}\left[\frac{d^2}{d\theta^2}\log p(Y|\theta)\right]$.
- Once you have specified the likelihood, the Jeffreys prior is determined with no additional input.
- Therefore you do not have to make a subjective decision about the prior (other than “subjective” decision to use a Jeffreys prior).

Examples of Jeffreys priors

Likelihood	Jeffreys prior
$Y \sim \text{Binomial}(n, \theta)$	$\theta \sim \text{Beta}(1/2, 1/2)$
$Y \sim \text{N}(\mu, 1)$	$p(\mu) \propto 1$
$Y \sim \text{N}(0, \sigma^2)$	$p(\sigma) \propto 1/\sigma$

Jeffreys priors for the model $Y \sim \text{Binomial}(n, \theta)$

- The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = \log \binom{n}{x} + Y \log(\theta) + (n - Y) \log(1 - \theta).$$

- The first derivative is

$$\ell'(\theta) = \frac{Y}{\theta} - \frac{n - Y}{1 - \theta}.$$

- The second derivative is

$$\ell''(\theta) = -\frac{Y}{\theta^2} - \frac{n - Y}{(1 - \theta)^2}.$$

Jeffreys priors for the model $Y \sim \text{Binomial}(n, \theta)$

- Recalling that $E(Y) = n\theta$,

$$
\begin{aligned}
I(\theta) = -E(\ell''(\theta)) &= \frac{n\theta}{\theta^2} + \frac{n - n\theta}{(1-\theta)^2} \\
&= \frac{n}{\theta} + \frac{n}{(1-\theta)} \\
&= \frac{n}{\theta(1-\theta)}.
\end{aligned}
$$

- Finally, the prior is $\pi(\theta) \propto I(\theta)^{1/2} \propto \theta^{-1/2}(1-\theta)^{-1/2} = \theta^{1/2-1}(1-\theta)^{1/2-1}$.

- This the kernel of a Beta(1/2,1/2) PDF.

Jeffreys priors for the model $Y \sim \text{Normal}(\mu, 1)$

- The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2}\log(2\pi) - \frac{1}{2}(Y - \mu)^2.$$

- The first derivative is

$$\ell'(\theta) = Y - \mu.$$

- The second derivative is

$$\ell''(\theta) = -1.$$

- Therefore,

$$I(\theta) = -E(\ell''(\theta)) = 1.$$

- Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} = 1.$$

Jeffreys priors for σ for the model $Y \sim \text{Normal}(0, \sigma^2)$

- The log likelihood is

$$\ell(\theta) = \log[f(Y|\theta)] = -\frac{1}{2}\log(2\pi) - \log(\sigma) - \frac{1}{2}\frac{Y^2}{\sigma^2}.$$

- The first derivative is

$$\ell'(\theta) = -\frac{1}{\sigma} + \frac{Y^2}{\sigma^3}.$$

- The second derivative is

$$\ell''(\theta) = \frac{1}{\sigma^2} - 3\frac{Y^2}{\sigma^4}.$$

- Therefore,

$$I(\theta) = -E(\ell''(\theta)) = -\frac{1}{\sigma^2} + 3\frac{\sigma^2}{\sigma^4} = \frac{2}{\sigma^2}.$$

- Finally, the prior is

$$\pi(\theta) \propto I(\theta)^{1/2} \propto \frac{1}{\sigma}.$$

Thank you!

Lecture 13

Objective priors: Part 2

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Prior elicitation: Recap

- Potential sources of prior information include: literature reviews; pilot studies; expert opinions; etc.

- Prior elicitation is the process of converting expert opinion to prior distribution.

- Prior information can come from a variety of sources and the mixture of experts prior is

$$\pi(\theta) = \sum_{j=1}^{J} w_j \pi_j(\theta).$$

- Another powerful idea is to elicit prior information using real-life scenarios.

Improper priors and objective priors: Recap

- A ‘prior’ that integrates to infinity is called an improper prior.
- Example: $\pi(\mu) = 1$ for all $\mu \in \mathbb{R}$.
- It’s OK to use an improper prior as long as you verify that the posterior integrates to one.
- An objective analysis is one that requires no subjective decisions by the analyst.
- There are many approaches: Jeffreys, reference, probability matching, maximum entropy, empirical Bayes, penalized complexity, etc.

Jeffreys prior: Recap

- The Jeffreys prior (JP) for parameter θ is $p(\theta) \propto \sqrt{I(\theta)}$.
- The Fisher information is $I(\theta) = -\mathrm{E}_{Y|\theta}\left[\frac{d^2}{d\theta^2}\log p(Y|\theta)\right]$.
- Once you have specified the likelihood, the Jeffreys prior is determined with no additional input.
-

Likelihood	Jeffreys prior
$Y \sim \text{Binomial}(n, \theta)$	$\theta \sim \text{Beta}(1/2, 1/2)$
$Y \sim \text{N}(\mu, 1)$	$p(\mu) \propto 1$
$Y \sim \text{N}(0, \sigma^2)$	$p(\sigma) \propto 1/\sigma$

Why Jeffreys prior?

- You use a JP on θ, denoted $\pi_1(\theta)$, and I place a JP on $\gamma = g(\theta)$, denoted $\pi_2(\gamma)$.

- Using the change of variables formula, the prior for γ induced by $\pi_1(\cdot)$ is

$$\pi_3(\gamma) = \pi_1(g^{-1}(\gamma)) \left| \frac{dg^{-1}(\gamma)}{d\gamma} \right|.$$

- It can be shown that $\pi_2(\cdot) = \pi_3(\cdot)$.

- This establishes invariance to reparameterization.

An example

- Likelihood: $Y \sim \text{Binomial}(n, \theta)$. Here, JP is $\theta \sim \text{Beta}(1/2, 1/2)$.
- You choose θ as your parameter and thus $\pi_1(\theta) = \frac{1}{\pi}\theta^{1/2-1}(1-\theta)^{1/2-1}$.
- I choose $\gamma = \theta^2 \in [0, 1]$ as my parameter. Thus, $g(x) = x^2$.
- The prior for γ induced by $\pi_1(\cdot)$ is

$$\pi_3(\gamma) = \frac{1}{2\pi}(\sqrt{\gamma})^{-1}(1-\sqrt{\gamma})^{1/2-1}.$$

- Now, I rewrite the model as $Y \sim \text{Binomial}(n, \sqrt{\gamma})$. Then obtain $\pi_2(\cdot)$.
- After a few steps of boring algebra, we get $\pi_2(\cdot) = \pi_3(\cdot)$.

Reference priors

- These priors try to formally be ‘non-informative’.
- Similar to JP, they are objective.
- They are defined as the prior that gives the maximum expected ‘distance’ between the log-prior and log-posterior (Kullback-Leibler divergence).
- For univariate models, they give JP. For multivariate models, they are different.
- Reference priors are harder to compute than JP.
- The solution of the optimization problem is a PDF and not a scalar or vector.

Probability matching priors (PMP)

- The PMP is designed so that posterior credible intervals have correct frequentist coverage.
- There are only a few cases where this can be done exactly.
- For example, if $Y_1, \ldots, Y_n | \mu \sim \text{Normal}(\mu, 1)$, the PMP is $p(\mu) = 1$.
- The posterior is $\mu | Y_1, \ldots, Y_n \sim \text{Normal}(\bar{Y}, 1/n)$.
- In this case, credible sets have correct frequentist coverage.
- Approximations are available for medium-complexity cases.

Empirical Bayes

- Empirical Bayes is also objective.
- Here you pick the priors based on the data.
- Example: σ^2 has prior mean s^2.
- Example: σ^2 is fixed at s^2 in the Bayesian analysis of μ.
- More formally, you can use marginal maximum likelihood to fix nuisance parameters. The analysis then proceeds as a usual Bayesian analysis.
- Often criticized for 'using the data twice'.
- Despite this concern, EB remains a useful tool to stabilize computing in high-dimensional problems.

Penalized complexity (PC) priors

- A PC prior begins with a simple base model, e.g., linear regression with all the slopes equal zero.
- The full model, e.g., regression with non-zero slopes, is shrunk towards to the base model.
- The ‘distance’ of the full model to the base model has exponential prior to penalize the more complex model from deviating from the base model.
- Requires picking the parameter in the exponential prior and setting priors for the parameters in the base model.
- So technically this is not purely objective.

Maximum entropy priors

- In a maximum entropy prior you fix a few quantities of the prior distribution, e.g., $\text{E}(\theta) = 0.5$.
- The prior is taken to be the distribution with the maximum entropy of these that satisfy the known constraints.
- **Entropy** is a measure of uncertainty, e.g., the entropy of the PMF $f(x)$ is

$$-\sum_{x \in \mathcal{S}} f(x) \log[f(x)]$$

- If θ has support $\mathbb{R}$ and the mean and variance are known, the maximum entropy prior is Gaussian.
- Not purely objective because you have to set the constraints.

Problem discussion

Improper prior example 1

Say $Y|\mu \sim \text{Normal}(0, 1)$ and μ has improper prior $\pi(\mu) = 1$ for all μ. Prove that the posterior distribution of μ is a proper PDF.

- The posterior is

$$\begin{aligned} p(\mu|Y) &\propto f(Y|\mu) \times \pi(\mu) \\ &\propto \exp\left[-\frac{1}{2}(Y-\mu)^2\right] \\ &\propto \exp\left[-\frac{1}{2}(\mu-Y)^2\right]. \end{aligned}$$

- By comparing the kernel of the posterior with that of a normal density, we have

$$\mu|Y \sim \text{Normal}(Y, 1).$$

Improper prior example 2

Suppose that the likelihood $f(Y|\theta) \geq c(Y)$ for all θ. Show that an improper prior on θ will yield an improper posterior.

- For some positive constant $c_0(Y)$ (dependent on Y but not on θ), the posterior is

$$\begin{aligned} p(\theta|Y) &= c_0(Y) \times f(Y|\theta) \times \pi(\theta) \\ &\geq c_0(Y) \times c(Y) \times \pi(\theta). \end{aligned}$$

- Thus, we have

$$\begin{aligned} \int p(\theta|Y)d\theta &\geq c_0(Y) \times f(Y|\theta) \times \int \pi(\theta)d\theta \\ &\geq c_0(Y) \times c(Y) \times \infty \\ &= \infty. \end{aligned}$$

Improper prior example 3

Consider a sample of size n from the mixture of two normal densities given by $f(y|\theta) = 0.5\phi(y) + 0.5\phi(y - \theta)$, where $\phi(z)$ denotes the standard normal density function. Show that any improper prior for leads to an improper posterior.

- For some positive $c_0(\mathbf{Y})$ (dependent on $\mathbf{Y}$ but not on θ), the posterior is

$$\begin{aligned} p(\theta|\mathbf{Y}) &= c_0(\mathbf{Y}) \times f(\mathbf{Y}|\theta) \times \pi(\theta) \\ &= c_0(\mathbf{Y}) \times \prod_{i=1}^{n} \{0.5\phi(y_i) + 0.5\phi(y_i - \theta)\} \times \pi(\theta) \\ &\geq c_0(\mathbf{Y}) \times \prod_{i=1}^{n} \{0.5\phi(y_i)\} \times \pi(\theta) = d_0(\mathbf{Y}) \times \pi(\theta). \end{aligned}$$

- Thus, $\int p(\theta|\mathbf{Y})d\theta \geq d_0(\mathbf{Y}) \times \int \pi(\theta)d\theta = d_0(\mathbf{Y}) \times \infty = \infty$.

Improper prior example 4

Say $Y|\lambda \sim \text{Poisson}(\lambda)$. (a) Derive the Jeffreys' prior for λ. (b) Is this prior proper? (c) Derive the posterior and give conditions on Y to ensure it is proper.

- The log likelihood is
$$\ell(\lambda) = \log[f(Y|\lambda)] = -\lambda + Y\log(\lambda) - \log(Y!).$$
- The first derivative is
$$\ell'(\lambda) = -1 + Y/\lambda.$$
- The second derivative is
$$\ell''(\lambda) = -Y/\lambda^2.$$
- Therefore,
$$I(\lambda) = -E(\ell''(\lambda)) = \lambda/\lambda^2 = 1/\lambda.$$
- Finally, the prior is
$$\pi(\lambda) \propto I(\lambda)^{1/2} = \lambda^{-1/2}.$$

Improper prior example 4 (Contd.)

- $\int \pi(\lambda) d\lambda \propto \int_0^\infty \lambda^{-1/2} d\lambda = \infty$. Thus, improper prior.
- The posterior is $p(\lambda|Y) \propto \exp[-\lambda]\lambda^Y \times \lambda^{-1/2} = \exp[-\lambda]\lambda^{Y+1/2-1}$.
- Thus, $\lambda|Y \sim \text{Gamma}(Y + 1/2, 1)$.
- Thus, the posterior is proper for all Y.

Improper prior example 5

Say $Y|\lambda \sim$ Exponential(λ). (a) Derive the Jeffreys' prior for λ. (b) Is this prior proper? (c) Derive the posterior and give conditions on Y to ensure it is proper.

- The log likelihood is

$$\ell(\lambda) = \log[f(Y|\lambda)] = \log(\lambda) - \lambda Y.$$

- The first derivative is

$$\ell'(\lambda) = 1/\lambda - Y.$$

- The second derivative is

$$\ell''(\lambda) = -1/\lambda^2.$$

- Therefore,

$$I(\lambda) = -E(\ell''(\lambda)) = 1/\lambda^2.$$

- Finally, the prior is

$$\pi(\lambda) \propto I(\lambda)^{1/2} = \lambda^{-1}.$$

Improper prior example 4 (Contd.)

- $\int \pi(\lambda) d\lambda \propto \int_0^\infty \lambda^{-1} d\lambda = \infty$. Thus, improper prior.
- The posterior is $p(\lambda|Y) \propto \lambda \exp[-\lambda Y] \times \lambda^{-1} = \exp[-\lambda Y]$.
- Thus, $\lambda|Y \sim$ Exponential(Y).
- Thus, the posterior is proper for all Y.

Thank you!

1/ $Y_1, \dots, Y_n \mid \mu \overset{iid}{\sim} N(\mu, \sigma^2)$, σ^2 is fixed and $\mu \sim N(0, \sigma^2/m)$

a/ Give a 95% posterior interval for μ.

A: $p(\mu \mid Y_1, \dots, Y_n) \propto f(Y_1, \dots, Y_n \mid \mu) \times \pi(\mu)$

$$\propto \prod_{i=1}^{n} e^{-\frac{1}{2}\frac{(Y_i-\mu)^2}{\sigma^2}} \times e^{-\frac{1}{2}\cdot\frac{m}{\sigma^2}\mu^2}$$

$$\propto e^{-\frac{1}{2\sigma^2}\left[\sum_{i=1}^{n}(Y_i-\mu)^2 + m\mu^2\right]}$$

$$\sum_{i=1}^{n}(Y_i-\mu)^2 + m\mu^2 = \sum_{i=1}^{n}(Y_i^2 - 2Y_i\mu + \mu^2) + m\mu^2$$

$$= (n+m)\mu^2 - 2\mu\sum Y_i + \sum Y_i^2$$

Thus, $p(\mu \mid Y_1, \dots, Y_n) \propto e^{-\frac{1}{2\sigma^2}\left[(n+m)\mu^2 - 2\mu\sum Y_i\right]}$

thus, $\mu \mid Y_1, \dots, Y_n \sim N\left(\frac{\sum Y_i}{n+m}, \frac{\sigma^2}{n+m}\right)$

Thus, a 95% posterior interval for μ can be calculated. For example, the equal-tailed interval would be

$$\left\{ \frac{\sum Y_i}{n+m} \pm Z_{0.975}\frac{\sigma}{\sqrt{n+m}} \right\},$$

where $Z_{0.975}$ is the 97.5%-th percentile of $N(0,1)$ distribution.

b/ select a value of m and argue that the 95% credible region is same as the 95% confidence interval.

A: Here, 95% CI is $\frac{\sum Y_i}{n} \pm Z_{0.975}\frac{\sigma}{\sqrt{n}}$.

Thus, they match when $m \to 0$.

2. Over his career, Reggie Jackson played in 2820 regular-season games and hit 563 home runs (a player can hit 0, 1, 2, .. home runs in a game). He also played in 27 world series games and hit 10 home runs in these games. Assuming non-informative conjugate priors, summarize the posterior dist. of his home-run rate in the regular season and world series.

A: $Y_1, \dots, Y_{2820} \overset{iid}{\sim} \text{Poisson}(\lambda_1)$, $\lambda_1 \sim \text{Gamma}(a, b)$

$Z_1, \dots, Z_{27} \overset{iid}{\sim} \text{Poisson}(\lambda_2)$, $\lambda_2 \sim \text{Gamma}(a, b)$

Choose a and b to be small.

So, following the derivations in the class,

$$p(\lambda_1 | Y_1, .., Y_{2820}) \propto f(Y_1, .., Y_{2820} | \lambda_1) \pi(\lambda_1)$$

$$\prod_{i=1}^{2820} e^{-\lambda_1} \frac{\lambda_1^{Y_i}}{Y_i!} \times \lambda_1^{a-1} e^{-b\lambda_1}$$

$$\propto \lambda_1^{\sum Y_i + a - 1} e^{-(2820 + b)\lambda_1}$$

$$\lambda_1 | Y_1, .., Y_{2820} \sim \text{Gamma}(\sum Y_i + a, b + 2820)$$

Here, $\sum Y_i = 563$.

So, $\lambda_1 | Y_1, .., Y_{2820} \sim \text{Gamma}(a + 563, b + 2820)$

Similarly, $\lambda_2 | Z_1, .., Z_{27} \sim \text{Gamma}(a + 10, b + 27)$ indep

It is okay to pick $a = b = 0.1$ or 0.01.

Q: Is there sufficient evidence to claim that he performs better in the World Series?

A: $P(\lambda_2 > \lambda_1 | Y_1, .., Y_{2820}, Z_1, .., Z_{27})$

$$= \int_0^\infty \int_{\lambda_1}^\infty f(\lambda_1 | Y_1, .., Y_{2820}) f(\lambda_2 | Z_1, .., Z_{27}) \, d\lambda_2 \, d\lambda_1$$

Then, need to do numerically.

3. $Y | \theta \sim \text{Exponential}(\theta)$. Find a conjugate prior for θ and derive the posterior.

A: Done in class.

$\theta \sim \text{Gamma}(a, b)$

$$p(\theta | Y) \propto f(Y|\theta) \pi(\theta) \propto \theta e^{-\theta Y} \times \theta^{a-1} e^{-b\theta}$$

$$\propto \theta^{1+a-1} e^{-\theta(b+Y)}$$

Thus, $\theta | Y \sim \text{Gamma}(1 + a, b + Y)$.

Finding conjugate prior: Likelihood $= \theta^1 e^{-Y \cdot \theta} = \theta^{2-1} e^{-Y\theta}$

Looks like a gamma density with shape 2 and rate Y.

So, the conjugate prior is gamma.

4/ $Y|\theta \sim$ Neg-Binomial(θ, m), and $\theta \sim$ Beta(a, b).

Success prob. No of ~~successes~~ failures (Convention in the class)

$Y = \#$ ~~failures~~ Successes before m ~~successes~~ failures.

Q: Derive the posterior of θ.

A: $$p(\theta|Y) \propto f(Y|\theta)\,\pi(\theta)$$
$$\propto \theta^{Y}(1-\theta)^{m} \times \theta^{a-1}(1-\theta)^{b-1}$$
$$\propto \theta^{Y+a-1}(1-\theta)^{m+b-1}$$

Thus, $\theta|Y \sim \text{Beta}(a+Y, b+m)$.

b) $m=5$, $Y=10$, $a=b=1$. ~~Then op~~ Then plot $\theta|Y$.

A: $\theta|Y \sim \text{Beta}(1+10, 1+5) = \text{Beta}(11, 6)$. For the rest, a software is needed.

In terms of R code, it would be (equal-tailed CI).

(qbeta(0.025, 11, 6), qbeta(0.975, 11, 6)).

5/ In the past 50 years, an average of $\lambda_0 = 75$ large wildfires per year is observed in California. For the next 10 years, you will record the number of large wildfires and then fit a Poisson/gamma model to these data. Let the rate of large wildfires in this future period, λ, have Prior $\lambda \sim \text{Gamma}(a, b)$. Select a and b so that the prior is non-informative and Variance $= 100$ and $\text{Prob}(\lambda > \lambda_0) = 0.5$.

A: So, $\lambda \sim \text{Gamma}(a, b)$

$\Pr(\lambda > 75) = 0.5 \Rightarrow$ median $= 75$.

$\text{Var}(\lambda) = 100 \Rightarrow \dfrac{a}{b^2} = 100 \Rightarrow b = \dfrac{\sqrt{a}}{10}$

$\lambda \sim \text{Gamma}\left(a, \frac{\sqrt{a}}{10}\right)$, and $P(\lambda > 75) = 0.5$ Solve in R.

6/ An algorithm is used. It is Known that the algo is unbiased, So assume that measurements follow normal dist with mean zero, $Y_i \mid \sigma^2 \overset{iid}{\sim} N(0, \sigma^2)$. If $\sigma > c$, then the algorithm must be replaced. We have $n = 20$, $\sum Y_i = -2$, $\sum Y_i^2 = 15$. We conduct a Bayesian analysis with $\sigma^2 \sim$ Inv-Gamma(a, b). Compute the posterior probability that $\sigma > c$.

A: Numerically, this can be done only using a software.

$$\pi(\sigma^2) \propto (\sigma^2)^{-a-1} e^{-b/\sigma^2}$$

$$f(Y_1, .., Y_n \mid \sigma^2) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2} Y_i^2} \propto (\sigma^2)^{-\frac{n}{2}} e^{-\frac{\frac{1}{2}\sum Y_i^2}{\sigma^2}}$$

$$p(\sigma^2 \mid Y_1, .., Y_n) \propto f(Y_1, .., Y_n \mid \sigma^2)\, \pi(\sigma^2)$$

$$\propto (\sigma^2)^{-\frac{n}{2} - a - 1} e^{-\frac{1}{\sigma^2}\left[b + \frac{1}{2}\sum_{i=1}^{n} Y_i^2\right]}$$

$$\propto (\sigma^2)^{-\left(\frac{n}{2} + a\right) - 1} e^{-\frac{1}{\sigma^2}\left[b + \frac{1}{2}\sum_{i=1}^{n} Y_i^2\right]}$$

$$\sigma^2 \mid Y_1, .., Y_n \sim \text{Inv-Gamma}\left(a + \frac{n}{2},\ b + \frac{1}{2}\sum_{i=1}^{n} Y_i^2\right)$$

$$n = 20,\ \sum Y_i^2 = 15 \Rightarrow \sigma^2 \mid Y_1, .., Y_n \sim \text{Inv-Gamma}\left(a + 10,\ b + \frac{15}{2}\right).$$

For the rest, use R.

7/ In a simple logistic regression, the probability of a success is regressed onto a single covariate X as $p(X) = \frac{e^{\alpha + X\beta}}{1 + e^{\alpha + X\beta}}$

Assuming $X \sim N(0, 1)$, $\alpha \sim N(0, c^2)$, $\beta \sim N(0, c^2)$, Find c such that the induced prior on the success prob. $p(X)$ is roughly uniform $(0, 1)$.

A:- This is a purely computational problem. So, do it in R. You will find $c \approx 1.3$. $\alpha, \beta \sim N(0, 1.3^2)$ are not non-informative priors, as the prior variance is ~~very~~ small.

8. The Mayo clinic conducted a study of $n = 50$ patients followed for one year on a new medication and found that 30 patients experienced no ASE, 12 experienced one ASE, 6 experienced two ASE and 2 experienced ten ASE.

a) Derive the posterior of λ: $Y_1, \ldots, Y_n \mid \lambda \overset{iid}{\sim} Poi(\lambda)$, $\lambda \sim Ga(a,b)$.

A: $Y_1, \ldots, Y_{30} = 0$, $\quad Y_{43}, Y_{44}, \ldots, Y_{48} = 2$

$Y_{31}, \ldots, Y_{42} = 1$, $\quad Y_{49}, Y_{50} = 10$

$$\sum_{i=1}^{50} Y_i = 1 \times 12 + 2 \times 6 + 2 \times 10$$

$$= 12 + 12 + 20 = 44.$$

$$p(\lambda \mid Y_1, \ldots, Y_n) \propto f(Y_1, \ldots, Y_n \mid \lambda) \times \lambda^{a-1} e^{-b\lambda}$$

$$\propto \prod_{i=1}^{n} e^{-\lambda} \frac{\lambda^{Y_i}}{Y_i!} \times \lambda^{a-1} e^{-b\lambda}$$

$$\propto e^{-n\lambda} \, \lambda^{\sum_{i=1}^{n} Y_i} \, \lambda^{a-1} e^{-b\lambda}$$

$$\propto e^{-(n+b)\lambda} \, \lambda^{\sum_{i=1}^{n} Y_i + a - 1}$$

$$\lambda \mid Y_1, \ldots, Y_n \sim \text{Gamma}\left(a + \sum_{i=1}^{n} Y_i, \; b + n\right).$$

(b) with $a = b = 0.01$, give the posterior mean.

A: Posterior mean $= \dfrac{a + \sum Y_i}{b + n} = \dfrac{0.01 + 44}{0.01 + 50} = \dfrac{44.01}{50.01}$.

(c) For other a and b, the posterior mean can be similarly calculated.

(d) Plot data versus Poisson($\hat{\lambda}$) PMF. Do in R.

~~(e) The current medication is thought to have around one ASE per year. Say rate $= \lambda_2$, old medication $= \lambda_1$.~~

~~data $Z_1, \ldots, Z_n$, $\sum Z_i =$~~

~~Posterior $\lambda_1 \mid Y_1, \ldots, Y_n \sim \text{Gamma}(a + 44, b + 50)$~~

~~$\lambda_2 \mid Z_1, \ldots, Z_n \sim \text{Gamma}(a + 50, b + 50)$~~

(e) What is the posterior probability that this new medication has a higher side effect?

A: ~~Pr~~ $\lambda | Y_1, \ldots, Y_n \sim \text{Gamma}(a+44, b+50)$.

So, posterior probability $\Pr(\lambda > 1 | Y_1, \ldots, Y_n)$

$= \Pr(\text{Gamma}(a+44, b+50) > 1) = 1 - \text{pgamma}(1, a+44, b+50)$.

Now, we can do sensitivity analysis ~~for~~ by choosing different a and b.

9/ A smoker is trying to quit and their goal is to abstain for one week because studies have shown that this is the most difficult period. Denote Y as the number of days before the smoker relapses, and θ as the probability that they relapse on any given day of the study.

a) Argue that a negative binomial distribution for $Y|\theta$ is reasonable.

A: Here, ~~the~~ the event of smoking again is a failure. And the number of days ~~until~~ until a ~~success~~ failure can be seen as ~~a~~ success. Thus, a negative binomial model is reasonable.

Q: What are the most important assumptions being made and are they valid for this smoking cessation attempt?

A: Here, the events on different days are assumed to be independent. This is unlikely to be true in practice. Another, for each day, the success/failure probability remains the same. This is also unlikely.

b) Assuming a negative binomial likelihood and Unif(0,1) prior for θ. Here, θ = probability that they relapse on any day. that is, θ is ~~score~~ failure probability (~~we~~ in the class, we defined negative binomial model, where θ was the success probability) what is the prior probability ~~tho~~ the smoker will be smoke-free for at least 7 days?

A:- Pr(Smoker will be smoke-free for the first ~~Soe~~ seven days (at least))

$= Pr(\text{No Smoking on day } 1, \ldots, \text{day } 7)$

$= Pr(\text{No Smoking on day } 1) \times \ldots \times$ [with our class convention.

$\theta \sim U(0,1), \quad Y|\theta \sim \text{Neg-Binomal}(1, 1-\theta)$

$Y = \#$ success before first failure.

$Y \geq 7$

$\theta =$ failure prob.

$$Pr(Y \geq 7) = \sum_{y=7}^{\infty} Pr(Y=y)$$

$$= \sum_{y=7}^{\infty} \int Pr(Y=y|\theta) f(\theta)\, d\theta$$

$$Pr(Y=y|\theta) = \binom{1+y-1}{y} \theta^{1} (1-\theta)^{y}$$

$$= \theta (1-\theta)^y$$

Take, $1-\theta = \theta^*$, $\theta^* = 1-\theta$, $\theta(1-\theta)^y = (1-\theta^*)\theta^{*y}$, $d\theta^* = -d\theta$

$$\text{So, } Pr(Y \geq 7) = \sum_{y=7}^{\infty} \int_0^1 \theta (1-\theta)^y\, d\theta = \sum_{y=7}^{\infty} \int_0^1 \theta^{*y}(1-\theta^*)\, d\theta^*$$

$$= \sum_{y=7}^{\infty} \left[\int_0^1 \theta^y\, d\theta - \int_0^1 \theta^{y+1}\, d\theta \right] = \sum_{y=7}^{\infty} \left[\left(\frac{\theta^{y+1}}{y+1}\right)\Bigg|_0^1 - \left(\frac{\theta^{y+2}}{y+2}\right)\Bigg|_0^1 \right]$$

$$= \sum_{y=7}^{\infty} \left[\frac{1}{y+1} - \frac{1}{y+2} \right] = \sum_{y=7}^{\infty} \frac{1}{(y+1)(y+2)}$$

(c) Plot the posterior distribution of θ assuming $Y=5$.

$Y|\theta \sim \text{Neg-Bin}(1, 1-\theta)$

$$\pi(\theta|Y) \propto f(Y|\theta)\, \pi(\theta) \rightarrow \text{Unif}(0,1)$$

$$\propto \theta (1-\theta)^y \times 1 = \theta^{1+1-1}(1-\theta)^{y+1-1}$$

$\theta|Y \sim \text{Beta}(2, Y+1)$, So, $Y=5 \Rightarrow \theta|Y \sim \text{Beta}(2,6)$

(d) Given this attempt resulted in $Y=5$ and that θ is the same in the second attempt, what is the posterior predictive prob. that the second attempt will result in at least 7 smoke-free days?

A:- $Y=5 \Rightarrow \theta | Y \sim \text{Beta}(2, 6)$

$Y^* = \#$ Smoke-free days in second attempt.

$$\Pr(Y^* \geq 7 | Y) = \int \Pr(Y^* \geq 7 | \theta)\, \pi(\theta | Y)\, d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \int \Pr(Y^* \geq 7 | \theta) \times \theta^{2-1} (1-\theta)^{6-1} d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \int \sum_{y^*=7}^{\infty} \Pr(Y^* = y^* | \theta) \times \theta^{2-1} (1-\theta)^{6-1} d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \sum_{y^*=7}^{\infty} \int \theta (1-\theta)^{y^*} \times \theta^{2-1} (1-\theta)^{6-1} d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \sum_{y^*=7}^{\infty} \int_0^1 \theta^{3-1} (1-\theta)^{y^*+6-1} d\theta$$

$$= \frac{\Gamma(8)}{\Gamma(2)\Gamma(6)} \times \sum_{y^*=7}^{\infty} \frac{\Gamma(3)\Gamma(y^*+6)}{\Gamma(y^*+9)}$$

For the rest, we need a software.

Lecture 15

Deterministic computation

Arnab Hazra

(Edited from the lecture notes by Brian J. Reich, NC State)

Brief recap of Chapter 1 and 2

- Probability essentials (PMF, PDF, marginal and conditional distributions)
- Bayes' theorem and Bayes versus frequentist paradigm
- Posterior summarization and posterior predictive distribution
- Conjugate priors
- Prior elicitation
- Improper priors and objective priors (Jeffreys' prior)

Bayesian computing

- Given the prior and data, the posterior is fixed and a Bayesian analysis boils down to summarizing the posterior.
- We need point estimates, credible sets, etc.
- Summarizing a P-dimensional posterior distribution is challenging for large P.
- In the 80's, Bayesian computing was unable to do this for more than a few parameters.
- In the 90's, new algorithms were developed that revolutionized Bayesian statistics.
- Understanding these algorithms is obviously important.

Revolution in Bayesian statistics (JASA, 1990)

Sampling-Based Approaches to Calculating Marginal Densities

ALAN E. GELFAND AND ADRIAN F. M. SMITH*

Stochastic substitution, the Gibbs sampler, and the sampling–importance–resampling algorithm can be viewed as three alternative sampling- (or Monte Carlo-) based approaches to the calculation of numerical estimates of marginal probability distributions. The three approaches will be reviewed, compared, and contrasted in relation to various joint probability structures frequently encountered in applications. In particular, the relevance of the approaches to calculating Bayesian posterior densities for a variety of structured models will be discussed and illustrated.

KEY WORDS: Conditional probability structure; Data augmentation; Gibbs sampler; Hierarchical models; Importance sampling; Missing data; Monte Carlo sampling; Posterior distributions; Stochastic substitution; Variance components.

Approaches to Bayesian computing

Some approaches to dealing with complicated joint posteriors:

- Just use a point estimate (e.g., MAP), ignore uncertainty
- Approximate the posterior as Gaussian
- Numerical integration
- Markov Chain Monte Carlo (MCMC) sampling

Outline of Chapter 3

- Deterministic methods
  - MAP estimation
  - Numerical integration
  - Bayesian Central Limit Theorem
- MCMC algorithms
  - Gibbs sampling
  - Metropolis-Hastings sampling
- Just Another Gibbs Sampler (JAGS)
- Diagnostic and improving convergence
  - Setting initial values
  - Convergence diagnostics
  - Improving convergence
  - Dealing with large datasets

MAP estimation

- Sometimes you don't need an entire posterior distribution and a single point estimate works fine.

- Example: prediction in machine learning.

- The Maximum a Posteriori (MAP) estimate is the posterior mode

$$\hat{\boldsymbol{\theta}}_{MAP} = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\; p(\boldsymbol{\theta}|\mathbf{Y}) = \underset{\boldsymbol{\theta}}{\operatorname{argmax}}\; \log[f(\mathbf{Y}|\boldsymbol{\theta})] + \log[\pi(\boldsymbol{\theta})].$$

- This is similar to the maximum likelihood estimation but includes the prior.

Univariate example 1

Say $Y|\theta \sim \text{Poisson}(N\theta)$ and $\theta \sim \text{Gamma}(a, b)$, find $\hat{\theta}_{MAP}$.

- The likelihood is $f(Y|\theta) \propto \exp[-N\theta]\theta^Y$.
- The log-likelihood is $\ell(\theta|Y) = \log[f(Y|\theta)] = c_f - N\theta + Y\log(\theta)$.
- The prior is $\pi(\theta) \propto \theta^{a-1}\exp[-b\theta]$.
- The log-prior is $\log[\pi(\theta)] = c_\pi + (a-1)\log(\theta) - b\theta$.
- Thus, the MAP estimate is

$$\hat{\theta}_{MAP} = \underset{\theta}{\text{argmax}}\ p(\theta|Y) = \underset{\theta}{\text{argmax}}\ \{\log[f(Y|\theta)] + \log[\pi(\theta)]\}.$$

Univariate example 1 (contd.)

Say $Y|\theta \sim \text{Poisson}(N\theta)$ and $\theta \sim \text{Gamma}(a, b)$, find $\hat{\theta}_{MAP}$.

- Thus, the MAP estimate is

$$\hat{\theta}_{MAP} = \operatorname*{argmax}_{\theta} \{-(N+b)\theta + (Y+a-1)\log(\theta)\}.$$

- Taking the derivative and setting to zero gives

$$-(N+b) + (Y+a-1)\frac{1}{\theta} = 0.$$

- The solution is

$$\theta = \frac{Y+a-1}{N+b}.$$

Univariate example 2

Say $Y|\theta \sim \text{Binomial}(N, \theta)$ and $\theta \sim \text{Beta}(a, b)$, find $\hat{\theta}_{MAP}$.

- The likelihood is $f(Y|\theta) \propto \theta^Y(1-\theta)^{N-Y}$.
- The log-likelihood is $\ell(\theta|Y) = \log[f(Y|\theta)] = c_f + Y\log(\theta) + (N-Y)\log(1-\theta)$.
- The prior is $\pi(\theta) \propto \theta^{a-1}(1-\theta)^{b-1}$.
- The log-prior is $\log[\pi(\theta)] = c_\pi + (a-1)\log(\theta) + (b-1)\log(1-\theta)$.
- Thus, the MAP estimate is

$$\hat{\theta}_{MAP} = \underset{\theta}{\operatorname{argmax}}\ p(\theta|Y) = \underset{\theta}{\operatorname{argmax}}\ \{\log[f(Y|\theta)] + \log[\pi(\theta)]\}.$$

Univariate example 2 (contd.)

Say $Y|\theta \sim \text{Binomial}(N, \theta)$ and $\theta \sim \text{Beta}(a, b)$, find $\hat{\theta}_{MAP}$.

- Thus, the MAP estimate is

$$\hat{\theta}_{MAP} = \underset{\theta}{\text{argmax}} \{(Y + a - 1)\log(\theta) + (N - Y + b - 1)\log(1 - \theta)\}.$$

- Taking the derivative and setting to zero gives

$$\frac{Y + a - 1}{\theta} - \frac{N - Y + b - 1}{1 - \theta} = 0.$$

- The solution is

$$\theta = \frac{Y + a - 1}{Y + a - 1 + N - Y + b - 1} = \frac{Y + a - 1}{N + a + b - 2}.$$

Multivariate example

Say $\mathbf{Y}|\boldsymbol{\mu} \sim \text{Normal}_P(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\boldsymbol{\mu} \sim \text{Normal}_P(\boldsymbol{\mu}_0, m^{-1}\boldsymbol{\Sigma})$. Here, $\boldsymbol{\Sigma}$ is known.

- The likelihood is $f(\mathbf{Y}|\boldsymbol{\mu}) = |2\pi\boldsymbol{\Sigma}|^{-1/2} \exp\left[-\frac{1}{2}(\mathbf{Y} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})\right]$.
- The log-likelihood is $\ell(\boldsymbol{\mu}|\mathbf{Y}) = \log[f(\mathbf{Y}|\boldsymbol{\mu})] = c_f - \frac{1}{2}(\mathbf{Y} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$.
- The prior is $\pi(\boldsymbol{\mu}) \propto \exp\left[-\frac{m}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)\right]$.
- The log-prior is $\log[\pi(\boldsymbol{\mu})] = c_\pi - \frac{m}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)$.
- Thus, the MAP estimate is

$$\hat{\boldsymbol{\mu}}_{MAP} = \underset{\boldsymbol{\mu}}{\text{argmax}}\ p(\boldsymbol{\mu}|\mathbf{Y}) = \underset{\boldsymbol{\mu}}{\text{argmax}}\ \{\log[f(\mathbf{Y}|\boldsymbol{\mu})] + \log[\pi(\boldsymbol{\mu})]\}.$$

Multivariate example (contd.)

Say $\mathbf{Y}|\boldsymbol{\mu} \sim \text{Normal}_P(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\boldsymbol{\mu} \sim \text{Normal}_P(\boldsymbol{\mu}_0, m^{-1}\boldsymbol{\Sigma})$. Here, $\boldsymbol{\Sigma}$ is known.

- Thus, the MAP estimate is

$$\hat{\mu}_{MAP} = \underset{\boldsymbol{\mu}}{\text{argmax}} \left\{ -\frac{1}{2}(\mathbf{Y} - \boldsymbol{\mu})'\boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) - \frac{m}{2}(\boldsymbol{\mu} - \boldsymbol{\mu}_0)'\boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \boldsymbol{\mu}_0) \right\}.$$

- Taking the multivariate derivative and setting to zero gives

$$(1 + m)\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \boldsymbol{\Sigma}^{-1}(\mathbf{Y} + m\boldsymbol{\mu}_0) = \mathbf{0}.$$

- The solution is

$$\boldsymbol{\mu} = \frac{\mathbf{Y} + m\boldsymbol{\mu}_0}{1 + m}.$$

Bayesian central limit theorem

- Another simplification is to approximate the posterior as Gaussian.
- Berstein-Von Mises Theorem: As the sample size grows the posterior doesn't depend on the prior.
- Frequentist result: As the sample size grows the likelihood function is approximately normal.
- Bayesian CLT: For large n and some other conditions, $\theta|\mathbf{Y}_1, \ldots, \mathbf{Y}_n \approx$ Normal.

Bayesian central limit theorem

- Bayesian CLT: For large n and some other conditions

$$\theta|\mathbf{Y}_1, \ldots, \mathbf{Y}_n \sim \text{Normal}[\hat{\theta}_{MAP}, \boldsymbol{H}(\hat{\theta}_{MAP})^{-1}].$$

- $\boldsymbol{H}$ is the negative hessian matrix.
- The (j, k) element of $\boldsymbol{H}$ is

$$-\frac{\partial^2}{\partial\theta_j\partial\theta_k}\log[p(\theta|\mathbf{Y})]$$

evaluated at $\hat{\theta}_{MAP}$.

- We have marginal and conditional means, standard deviations and intervals for the normal distribution.

Univariate example

Say $Y|\theta \sim \text{Binomial}(n, \theta)$ and $\theta \sim \text{Beta}(a = 0.5, b = 0.5)$, find the Gaussian approximation for $p(\theta|\mathbf{Y})$.

- We have seen that

$$\theta = \frac{Y + a - 1}{n + a + b - 2} = \frac{Y - 0.5}{n - 1}.$$

- After some boring algebra,

$$H(\theta) = \frac{Y + a - 1}{\theta^2} + \frac{n - Y + b - 1}{(1 - \theta)^2}.$$

- Thus,

$$\theta|Y \quad \sim \quad \text{Normal}[\hat{\theta}_{MAP}, H(\hat{\theta}_{MAP})^{-1}]$$

Illustration of the Bayesian CLT

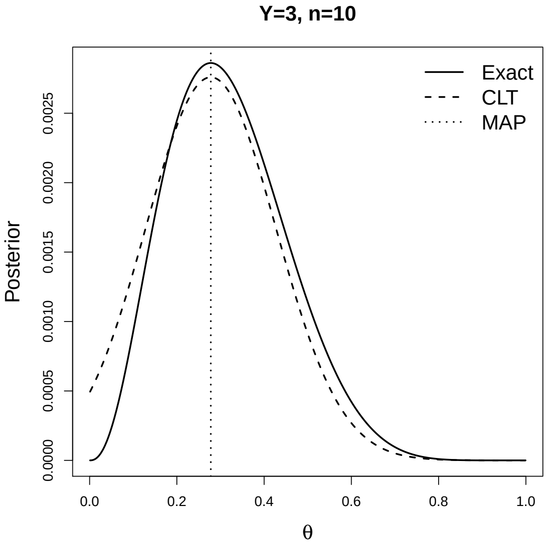


Illustration of the Bayesian CLT

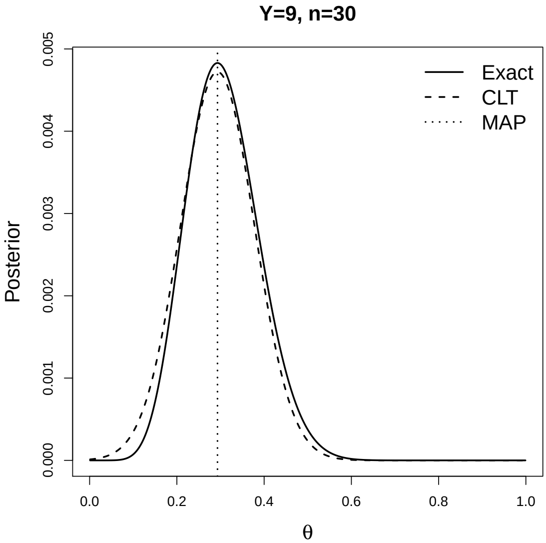
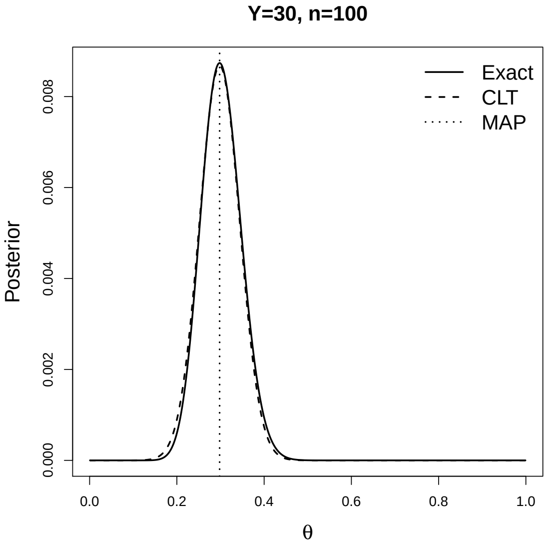


Illustration of the Bayesian CLT



Bayesian central limit theorem

- For large datasets with a small number of parameters evoking the Bayes CLT is probably the best approach.
- The approximate posterior can be computing using standard software (e.g., `glm` in `R`).
- The numerical values (e.g., intervals) will equal the frequentist values, but the interpretation remains Bayesian.
- Why not just do a frequentist analysis? Well, why not just do a Bayesian analysis?

Numerical integration

- Many posterior summaries of interest are integrals over the posterior.
- Ex: $\mathrm{E}(\theta_j|\mathbf{Y}) = \int \theta_j p(\boldsymbol{\theta}) d\boldsymbol{\theta}$.
- Ex: $\mathrm{V}(\theta_j|\mathbf{Y}) = \int [\theta_j - \mathrm{E}(\theta|\mathbf{Y})]^2 p(\boldsymbol{\theta}) d\boldsymbol{\theta}$.
- These are p dimensional integrals that we usually can't solve analytically
- A grid approximation is a crude approach.
- Gaussian quadrature is better.

Numerical integration

- Numerical integration is only feasible for small p.
- The Iteratively Nested Laplace Approximation (INLA) is an even more sophisticated method.
- INLA combines Gaussian approximations with numerical integration.
- This works well if most of the parameters are approximately normal and only a few are non-Gaussian and require numerical integration.

Revolution in high-dimensional Bayesian statistics (JRSS-B, 2009)

Journal of the Royal Statistical Society
SERIES B Statistical Methodology

J. R. Statist. Soc. B (2009)
71, *Part* 2, *pp.* 319–392

Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations

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[*Read before* The Royal Statistical Society *at a meeting organized by the* Research Section *on Wednesday*, *October 15th, 2008*, Professor I. L. Dryden *in the Chair*]

Summary. Structured additive regression models are perhaps the most commonly used class of models in statistical applications. It includes, among others, (generalized) linear models, (generalized) additive models, smoothing spline models, state space models, semiparametric regression, spatial and spatiotemporal models, log-Gaussian Cox processes and geostatistical and geoadditive models. We consider approximate Bayesian inference in a popular subset of structured additive regression models, *latent Gaussian models*, where the latent field is Gaussian, controlled by a few hyperparameters and with non-Gaussian response variables. The posterior marginals are not available in closed form owing to the non-Gaussian response variables. For such models, Markov chain Monte Carlo methods can be implemented, but they are not without problems, in terms of both convergence and computational time. In some practical applications, the extent of these problems is such that Markov chain Monte Carlo sampling is simply not an appropriate tool for routine analysis. We show that, by using an integrated nested Laplace approximation and its simplified version, we can directly compute very accurate approximations to the posterior marginals. The main benefit of these approximations is computational: where Markov chain Monte Carlo algorithms need hours or days to run, our approximations provide more precise estimates in seconds or minutes. Another advantage with our approach is its generality, which makes it possible to perform Bayesian analysis in an automatic, streamlined way, and to compute model comparison criteria and various predictive measures so that models can be compared and the model under study can be challenged.

Thank you!